About this document

Product	Bellomassi Order-to-Delivery Platform (frozen & chilled food distribution)
Document set	Complete Application Workflow Documentation
Version	1.0 · prepared from the source code of `bellomassi-premium-ui` (application data version 8)
Audience	Business users, client stakeholders, trainers, testers and developers

This folder documents how the application **actually works today**. Every workflow, status, rule and message was taken from the source code (`src/config`, `src/services`, `src/pages`, `src/routes`). Nothing in these documents changes the application — no code, route, UI or LocalStorage structure was modified to produce them.

How to read these documents

Each chapter has two layers:

- **Business explanation** — plain English: who does what, in which order, and why.
- **Technical reference** — the route, page component, service function, permission key and LocalStorage key behind each step, so a developer or tester can find it in the code.

Implementation status legend

Every feature is classified with one of these labels. The label describes what the code really does, not what a menu item suggests.

Label	Meaning
Implemented	Works end to end in the browser; data is saved to LocalStorage and the rules are enforced by a service.
Partially Implemented	The main flow works but part of the business process is not built (the gap is named).
Simulated	Behaves realistically for a demo but no external system is involved (e.g. DEMO GPS, demo OTP, demo email).
UI Only	A screen or control exists but it does not change any data.
Coming Soon	Shown in the app as a placeholder for a later phase.
Not Implemented	The business step does not exist in the application.
Not Verified	Exists in code but its behaviour could not be confirmed with a live provider (e.g. backend gateway mode).

Chapters

#	File	What it covers
01	Application Overview	Purpose, apps, roles, modules, technology, data storage, simulations
02	Overall Workflow	End-to-end business flow with the master flowchart and swimlane

#	File	What it covers
03	Login-Wise Workflow	Sign-in pages, demo accounts, landing pages, session, lockout, password reset, sign-out
04	Role-Wise Workflow	All 13 roles: dashboard, menus, KPIs, actions, daily routine, limits, hand-offs
05	Module-Wise Workflow	Every module with the 17 standard headings
06	Order-to-Delivery Workflow	The core process from order capture to delivery note and invoice
07	Approval Workflow	Order approval, return and rejection; scrap and replenishment approvals; document review
08	Driver Workflow	Driver app: today, start, arrive, OTP, proof of delivery, failure
09	Customer Workflow	Customer app: catalog, cart, order, tracking, OTP, notes, invoices
10	Inventory Workflow	Receipts, transfers, adjustments, movements, low stock, scrap, replenishment
11	Dispatch & VMS Workflow	Dispatch planning, assignment, confirmation, tracking, exceptions, vehicle/driver compliance
12	Finance & Invoice Workflow	Invoice templates, numbering, generation, cancellation, credit
13	Communication & Documents Workflow	Email, WhatsApp, automation, call history, notifications, documents and OCR
14	Reports & Dashboard Workflow	Role dashboards, period filter, Reports Center (118 reports), AI & Jarvis
15	Exception Workflow	Every exception path and how it is resolved
16	Role Responsibility Matrix	Who creates, approves, views each record; permission matrix
17	Status Lifecycle	Every status set with its exact values and state diagrams
18	Screen-Wise User Guide	Every screen in simple English
19	Glossary	Business and technical terms

Master flowcharts

The twelve master flowcharts are located as follows:

#	Flowchart	Chapter
1	Complete application workflow	02
2	Login-wise workflow	03
3	Order-to-delivery workflow	06
4	Approval workflow	07
5	Driver workflow	08
6	Customer workflow	09
7	Inventory workflow	10
8	Dispatch / VMS workflow	11
9	Finance / invoice workflow	12
10	Communication workflow	13
11	Reports workflow	14

#	Flowchart	Chapter
12	Exception workflow	15

The swimlane diagram for Sales, Sales Manager, Warehouse/Customer Service, Dispatch, Driver, Customer and Finance is in chapter 02 (section 2.3) and repeated for the order lifecycle in chapter 06.

Consolidated editions

The same content is published as one document in three formats, in `docs/workflow/dist/`:

- `Bellomassi-Workflow-Documentation.html` — single page, diagrams rendered, printable
- `Bellomassi-Workflow-Documentation.pdf` — cover page, table of contents, headers, footers and page numbers
- `Bellomassi-Workflow-Documentation.docx` — Word edition with the same structure

The diagrams are written in Mermaid. They render directly on GitHub, GitLab and in VS Code's Markdown preview, and are pre-rendered as images in the consolidated editions.

01 · Application Overview

1.1 What Bellomassi is

Bellomassi is an **order-to-delivery management system** for a frozen and chilled food distributor. It covers the full chain from a salesman (or the customer) placing an order, through sales approval, warehouse preparation, dispatch planning and the driver's doorstep delivery, to the delivery note and the customer invoice. Around this core it provides inventory control, vehicle and driver compliance, document management with OCR, communication (email, WhatsApp, call log), reporting, forecasting and an AI assistant.

The application in this repository is a **client-demo prototype**:

- It is a single React web application. There is **no backend server and no database**.
- All data lives in the **browser's LocalStorage** (application data version 8). Each browser has its own copy of the data; clearing site data resets it to the seeded demo data.
- Sign-in is a **browser-session simulation** with published demo passwords. It is not production authentication.
- Integrations (SMTP email, OpenWA WhatsApp, OCR, AI providers) have a **demo mode** that is active by default and an optional **backend gateway mode** that is only used when `VITE_INTEGRATION_GATEWAY_URL` is configured. Gateway mode is **Not Verified** in this document because no gateway was available.

1.2 The three user-facing apps

One code base serves three experiences. They share the same data and the same accounts.

App	Entry	Who uses it	Layout
Office web app	`/login` → `/dashboard`	Super Admin, Administrator, Sales Executive, Sales Manager, Customer Service, Dispatcher, Finance, Warehouse Manager, Inventory Executive, QA Manager, Supplier	Sidebar + header (`components/layout/AppLayout`)
Driver app (mobile-first)	`/driver/login` → `/driver/dashboard`	Driver	Bottom tab bar (`DriverLayout`)
Customer app (mobile-first)	`/customer/login` → `/customer/dashboard`	Customer (portal user)	Bottom tab bar (`CustomerLayout`)

1.3 Roles at a glance

Thirteen system roles are defined in `src/config/roles.js`. Administrators can also create **custom roles** (Role Management) by choosing permissions from the catalog in `src/config/rbacCatalog.js`.

Role code	Label	Workspace	Data scope (what records they see)
SUPER_ADMIN	Super Admin	Admin Panel	All records; can grant the Super Admin role
ADMIN	Administrator	Admin Panel	All records

Role code	Label	Workspace	Data scope (what records they see)
SALESMAN	Sales Executive	Order Taking App	Own orders, own customers' documents and deliveries
SALES_MANAGER	Sales Manager	Order Taking App	All non-draft orders; customer and order documents
CUSTOMER_SERVICE	Customer Service	Order Taking App	Orders from approval onwards; all documents
DISPATCHER	Dispatcher	Order Taking App	Orders from consolidation onwards; fleet and hub documents
DRIVER	Driver	Drivers App	Own dispatches, deliveries and documents
FINANCE	Finance Executive	Notifications & Reports	All non-draft orders; invoices, customers, vendor documents
WAREHOUSE_MANAGER	Warehouse Manager	Warehouse	Dispatch & warehouse scope
INVENTORY_EXECUTIVE	Inventory Executive	Warehouse	Dispatch & warehouse scope
QA_MANAGER	QA Manager	Quality & Cold Chain	Dispatch & warehouse scope
CUSTOMER	Customer (Portal)	Customer Portal	One linked customer's orders, deliveries and invoices
SUPPLIER	Supplier (Portal)	Supplier Portal	One linked supplier's approved replenishment

> Note: the role description of CUSTOMER in `roles.js` still says "read-only portal". In the current code the customer can **place orders** from the customer app (permission `customer.order`). This document follows the code.

1.4 Modules and implementation status

Module	Main routes	Status
Authentication & session	`/login`, `/driver/login`, `/customer/login`, `/forgot-password`, `/reset-password`, `/logout`	Implemented (simulated security)
Role dashboards + period filter	`/dashboard`, `/driver/dashboard`, `/customer/dashboard`	Implemented
Master data — products, customers, hubs, suppliers, zones	`/products`, `/customers`, `/hubs`, `/suppliers`, `/dispatch/zones`	Implemented
Customer credit	`/customers/credit`	Partially Implemented (balance is maintained on the customer record; no payment entry)
Order management	`/orders/*`	Implemented
Order approval	`/orders/approval`	Implemented
Order consolidation	`/orders/consolidation`	Implemented
Inventory — stock, receipts, transfers, adjustments, movements, low stock	`/inventory/*`	Implemented

Module	Main routes	Status
Scrap management	`/inventory/scrap/*`	Implemented
Warehouse dashboard / QA & cold chain	`/warehouse`, `/quality`	Implemented (temperature monitoring: Not Implemented — "No sensor integration")
Dispatch planning, assignment, confirmation	`/dispatch/*`	Implemented
Delivery tracking & live tracking	`/dispatch/delivery-tracking`, `/dispatch/live-tracking`	Simulated (DEMO GPS)
Driver app	`/driver/*`	Implemented (GPS and OTP simulated)
Delivery OTP	driver confirm screen, customer tracking	Simulated (no SMS)
Proof of delivery (photo / signature)	driver confirm screen	Implemented (stored in browser)
Delivery notes	`/dispatch/delivery-notes/:id`	Implemented
Customer app	`/customer/*`	Implemented (payments: Not Implemented)
Vehicle Management System (VMS)	`/vms/*`	Implemented
Documents & OCR	`/documents/*`	Implemented; OCR is Simulated in demo mode
Invoices & templates	`/invoices/*`	Implemented (demo tax, not GST-compliant)
Payments, credit notes, refunds, expenses	—	Not Implemented
Email	`/communication/email/*`	Simulated (demo mode); gateway mode Not Verified
WhatsApp (OpenWA)	`/communication/whatsapp/*`	Simulated (mock session); gateway mode Not Verified
Automated notifications	`/communication/automation`	Implemented (sends through the simulated channels)
Call history	`/communication/calls/*`	Implemented (manual call log; no telephony)
In-app notifications	`/notifications`	Implemented
Reports Center	`/reports/center`	Implemented — 118 reports, 24 have no source data in the prototype
AI Assistant, Jarvis, Insights	`/ai`, `/ai/assistant`, `/ai/jarvis`, `/ai/insights`	Simulated (rule-based intent engine; external AI provider Not Verified)
Forecasting, predictive inventory, model management	`/ai/forecasting`, `/ai/predictive-inventory`, `/ai/models`	Implemented (in-browser statistical models)
Replenishment (purchase requests)	`/ai/replenishment`	Implemented (stops at goods receipt; supplier invoicing Not Implemented)
Administration — users, roles, permissions, audit, settings, organization	`/admin/*`	Implemented

1.5 Technology

Layer	Technology
UI	React 19, React Router 7, Tailwind CSS v4, Lucide icons, Recharts 3, sonner (toasts)
Build	Vite 8
Dates	dayjs with UTC and time-zone plugins (organisation time zone from Application Settings)
Persistence	`src/services/storageService.js` — the only module that reads or writes browser storage
Access control	`src/config/permissions.js` (`can(user, key)`), `src/services/rbacService.js`, `src/routes/ProtectedRoute.jsx`
Navigation	`src/config/navigation.js` — one list of 206 entries that drives routes, menus, breadcrumbs, search and access

Architecture: **page → hook → service → storageService → LocalStorage.** Pages never write storage directly; every business rule (who may do what, which status may move where) lives in a service function that returns `{ ok: true, record }` or `{ ok: false, error, errors }`.

1.6 Data storage (LocalStorage)

All keys are prefixed `bellomassi_`. The full list from `src/utils/constants.js` (`STORAGE_KEYS`):

Area	Keys
App & session	`app_version`, `demo_users`, `session`, `settings`, `demo_data`, `activities`, `theme`, `notifications`, `notification_preferences`, `auth`
Master data	`customers`, `products`, `hubs`, `dispatch_zones`, `drivers`, `vehicles`, `company`, `organization`
Inventory	`inventory`, `stock_receipts`, `stock_transfers`, `stock_adjustments`, `stock_movements`, `scrap_requests`
Orders	`orders`, `order_sequence`, `approvals`, `consolidations`, `order_timeline`
Dispatch & delivery	`dispatches`, `dispatch_sequence`, `deliveries`, `delivery_sequence`, `delivery_otps`, `delivery_notes`, `dispatch_timeline`, `driver_tracking`, `delivery_pod`
VMS & documents	`vms_settings`, `document_types`, `documents`, `ocr_results`
Invoices	`invoice_templates`, `invoices`, `invoice_settings`
Communication	`email_templates`, `email_drafts`, `email_history`, `email_inbox`, `email_settings`, `whatsapp_templates`, `whatsapp_messages`, `openwa_settings`, `communication_settings`, `notification_templates`, `notification_history`, `call_history`
AI & planning	`ai_settings`, `ai_conversations`, `ai_messages`, `ai_insights`, `ai_activity`, `jarvis_settings`, `jarvis_history`, `forecasts`, `forecast_settings`, `inventory_predictions`, `reorder_recommendations`, `ai_models`, `replenishment_drafts`, `sales_history`
Administration	`roles`, `permissions`, `audit_logs`
Customer app & reports	`customer_carts`, `report_prefs`

Suppliers are stored inside `bellomassi_forecast_settings` (`suppliers` array), not in a key of their own.

1.7 What is simulated

These behaviours look real in a demo but no external system is involved. The UI labels each of them.

Behaviour	Where	How it is simulated
GPS tracking	Driver app, customer tracking, Live Tracking	`utils/demoGeo` + `trackingService` move a marker along a straight-line route over time. Badge: **DEMO GPS**.
Delivery OTP	Driver confirm, customer tracking	`otpService` generates a 6-digit code stored in `bellomassi_delivery_otps`; it is shown in the customer app. **No SMS is sent.**
Email	Communication › Email, automation	Demo mode records `QUEUED_DEMO` → `SENT_DEMO` ("Sent · demo (not sent)").
WhatsApp	Communication › WhatsApp, automation	Mock OpenWA session; messages recorded as "Sent · demo".
OCR	Documents › OCR Queue	MOCK provider pre-fills fields from the linked record, form values and file name; results are labelled "Simulated OCR".
AI Assistant / Jarvis	AI & Intelligence	A demo intent engine maps questions to permission-checked tools; voice uses the browser's speech API.
Phone calls	Call History	A manual call log — nothing is dialled.
Cold-chain temperatures	QA dashboard	No readings: "No sensor integration".
Password reset code	Forgot password	The 6-digit code is shown on screen (no email is sent).

1.8 Coming Soon and placeholder items

Item	Where	What the user sees	Classification
Credit history	Customer details (`/customers/:id`)	Panel with a "Later phase" badge: "Credit movements (invoices, payments and limit changes over time) arrive with invoicing."	Coming Soon
Credit limits / Invoices / Order reports panels	Finance dashboard	Three information panels with a "Later phase" badge. Each has a working button to an implemented module (Customer Credit, Invoices, Order Reports); the badge text is older than the invoice module.	UI Only (links work)
Coming Soon page	`src/pages/ComingSoon.jsx`	Generic page for modules of later phases. No route in `navigation.js` currently uses it.	Not used
Warehouse settings — picking strategy (FEFO), putaway confirmation, QA hold on receipt	Application Settings › Warehouse	Saved, but no workflow reads them	UI Only

1.9 How the documentation was verified

- Roles, menus, permissions, routes and status values were exported directly from the source (`navigation.js`, `roles.js`, `permissions.js`, `rbacCatalog.js`, `orderConfig.js`, `dispatchConfig.`

`js`, `reportService.js`) with a small read-only script (`scripts/dump-inventory.mjs`), so lists in this document match the code exactly.

- Business rules and messages were read from the service functions named in each chapter.
- Behaviour of the main flows was previously exercised with browser test suites (order-to-delivery, approval, dispatch, driver app, customer app, scrap, reports, dashboards for all 13 roles).

02 · Overall Workflow

2.1 The business in one paragraph

A **Sales Executive** (or the **customer** in the customer app) raises an order for a customer and a fulfilment hub. The **Sales Manager** checks credit, stock and data quality and approves, returns or rejects it. **Customer Service** verifies the approved order against the warehouse and consolidates it, then marks it ready for dispatch. The **Dispatcher** groups ready orders into a dispatch, assigns a compliant driver and vehicle, and confirms the dispatch — at that moment hub stock is deducted and a delivery record is created for each order. The **Driver** starts each stop in sequence, records arrival, collects the customer's OTP, captures quantities and proof of delivery, and completes or fails the stop. Completing a stop generates a **delivery note**. **Finance** issues the invoice from the order (on ordered or delivered quantities). Failed and partial deliveries go back to the Dispatcher for review. Around this flow, the warehouse roles keep stock accurate (receipts, transfers, adjustments, scrap), the QA Manager verifies documents and watches expiry, and every step raises in-app notifications and optional email / WhatsApp messages.

2.2 Master flowchart 1 — Complete application workflow

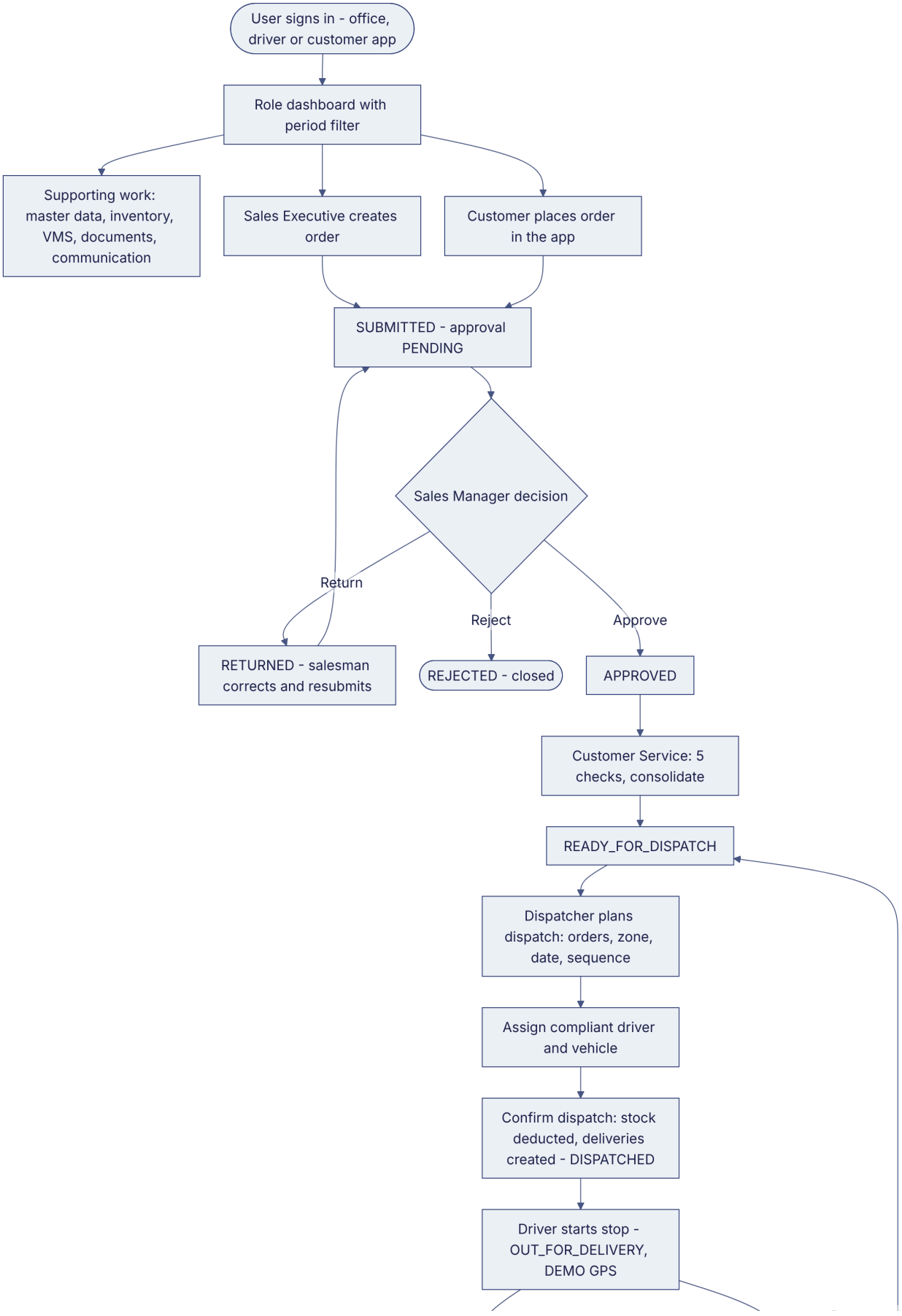


Diagram continued (1 of 2)

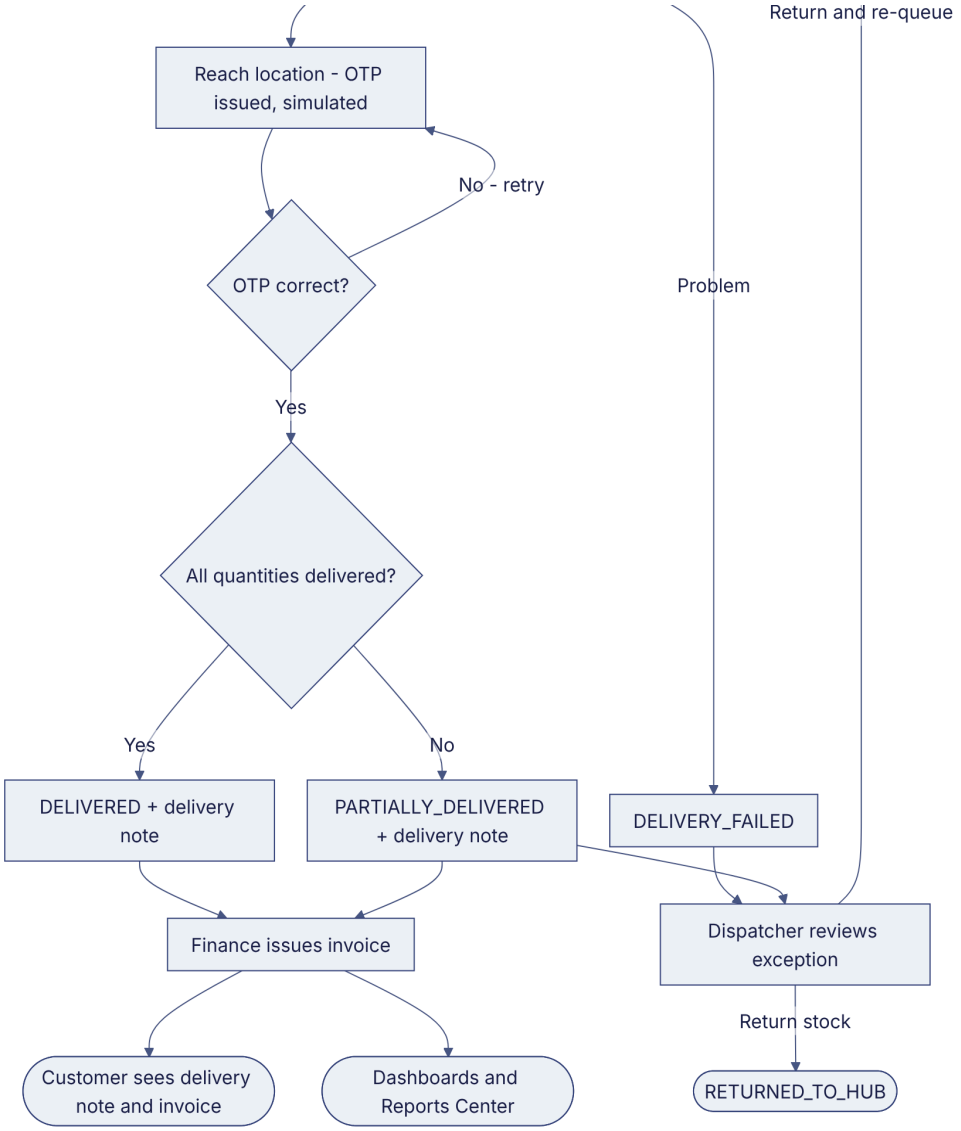
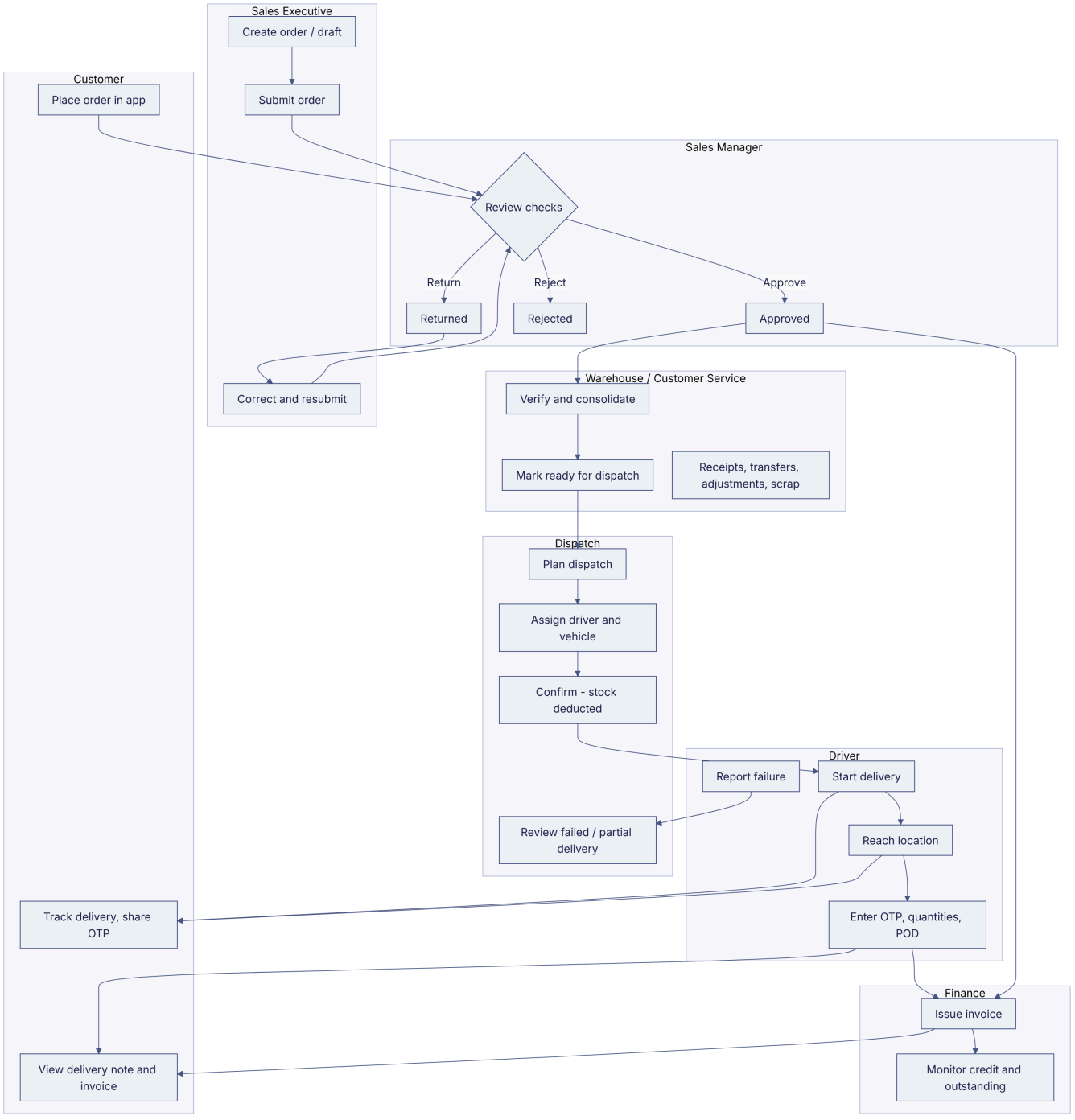


Diagram continued (2 of 2)

2.3 Swimlane — who does what



2.4 Stages, owners and system effects

#	Stage	Owner role	Route	Service function	Resulting order status	Notifications
1	Create / save draft	Sales Executive, Administrator	`/orders/create`	`orderService.createOrder`	DRAFT	—

#	Stage	Owner role	Route	Service function	Resulting order status	Notifications
2	Submit	Sales Executive, Administrator, Customer (app)	`/orders/create`, `/customer/cart`	`orderService.submitOrder`, `customerAppService.placeCustomerOrder`	SUBMITTED (approval PENDING)	Sales Manager; event ORDER_CREATED
3a	Approve	Sales Manager	`/orders/approval/:id`	`approvalService.approveOrder`	APPROVED (consolidation PENDING)	Customer Service, salesman; event ORDER_APPROVED
3b	Return for correction	Sales Manager	`/orders/approval/:id`	`approvalService.returnOrderForCorrection`	RETURNED	Salesman (link to edit)
3c	Reject	Sales Manager	`/orders/approval/:id`	`approvalService.rejectOrder`	REJECTED	Salesman; event ORDER_REJECTED
4	Consolidate	Customer Service	`/orders/consolidation/:id`	`consolidationService.consolidateOrder`	CONSOLIDATED	Dispatcher
5	Ready for dispatch	Customer Service	`/orders/consolidation/:id`	`consolidationService.markReadyForDispatch`	READY_FOR_DISPATCH	Dispatcher; event ORDER_READY
6	Plan dispatch	Dispatcher, Administrator	`/dispatch/create`	`dispatchService.createDispatch`	(order linked to dispatch)	—
7	Assign driver & vehicle	Dispatcher, Administrator	`/dispatch/assignments`, `/dispatch/:id`	`dispatchService.assignDispatch`	—	Driver; event DISPATCH_ASSIGNED
8	Confirm dispatch	Dispatcher, Administrator	`/dispatch/:id`	`dispatchService.confirmDispatch`	DISPATCHED	Driver, Administrator
9	Start delivery	Driver	`/driver/deliveries/:id`	`deliveryService.startDelivery`	OUT_FOR_DELIVERY	Customer, Dispatcher; event DELIVERY_STARTED
10	Reach location	Driver	`/driver/deliveries/:id`	`deliveryService.markArrived`	OUT_FOR_DELIVERY (arrivedAt set, OTP issued)	Customer sees OTP
11a	Complete	Driver	`/driver/deliveries/:id/confirm`	`deliveryService.confirmDelivery`	DELIVERED or PARTIALLY_DELIVERED	Customer, Dispatcher; event DELIVERY_COMPLETED
11b	Fail	Driver	`/driver/deliveries/:id`	`deliveryService.failDelivery`	DELIVERY_FAILED	Administrator, Dispatcher, customer; event DELIVERY_FAILED
12	Review exception	Dispatcher, Administrator	`/dispatch/deliveries/:id`	`deliveryService.reviewDelivery`	RETURNED_TO_HUB or READY_FOR_DISPATCH (re-queue)	—
13	Invoice	Finance, Administrator	`/invoices/create`	`invoiceService.generateInvoice`	(order unchanged; invoice ISSUED)	event INVOICE_GENERATED

2.5 Supporting workflows

Workflow	Owner	Chapter
Stock receipt, transfer, adjustment, movement ledger	Warehouse Manager, Inventory Executive, QA Manager (adjustments)	10
Scrap request → approval → write-off	Warehouse roles / Dispatcher (request); Warehouse Manager / Administrator (approve)	07, 10
Replenishment draft → approval → goods receipt	Warehouse / Inventory / Dispatcher (draft); Administrator (approve)	07, 10
Vehicle and driver compliance (VMS)	Administrator; Dispatcher views	11
Documents and OCR review	Uploaders; Customer Service, QA Manager, Administrator review	13
Email, WhatsApp, automated notifications, call log	Sales, CS, Dispatch, Finance, Administrator	13
Dashboards, reports, forecasting, AI assistant	All office roles (by permission)	14
Users, roles, permissions, settings, audit	Super Admin, Administrator	04, 05

2.6 Key rules that hold across the whole flow

- **One step at a time.** Each service refuses an action when the record is not in the expected status (for example, a dispatch can only be confirmed when it is `ASSIGNED`; a delivery can only be completed when it is `OUT_FOR_DELIVERY`).
- **Permissions are checked twice** — the route (`ProtectedRoute` + `navigation.js` permission) and the service (`can(user, key)`), so a hidden button cannot be bypassed.
- **Stock moves exactly once.** Dispatch confirmation deducts stock once (`stockDeducted` flag); cancellation reverses it once (`stockReversed`); a reviewed exception returns stock once (`returnedToHub`); a scrap request writes off once.
- **Everything is traceable.** Orders have a timeline (`bellomassi_order_timeline`), dispatches have a timeline (`bellomassi_dispatch_timeline`), administrative changes go to the audit log (`bellomassi_audit_logs`) and user activity to `bellomassi_activities`.
- **Nothing is deleted after submission.** Only draft orders can be deleted; submitted orders are kept for the audit trail.

03 · Login-Wise Workflow

Status: Implemented as a **browser-session simulation**. Passwords are stored as a demo FNV-1a hash in `bellomassi_demo_users`; this is not production authentication.

3.1 Sign-in pages

Page	Route	Component	Identifier accepted	Intended for
Office sign-in	`/login`	`pages/auth/Login.jsx`	Email or username	All office roles (any account works)
Driver app sign-in	`/driver/login`	`pages/auth/AppLogin.jsx` (driver variant)	Mobile number, email or username	Driver
Customer app sign-in	`/customer/login`	`pages/auth/AppLogin.jsx` (customer variant)	Mobile number, email or username	Customer
Forgot password	`/forgot-password`	`pages/auth/ForgotPassword.jsx`	Email or username	Everyone
Reset password	`/reset-password`	`pages/auth/ResetPassword.jsx`	6-digit demo code	Everyone
Sign out	`/logout`	`logoutConfirm` page	—	Everyone
Unauthorized	`/unauthorized`	`Unauthorized`	—	Shown when a role opens a route it may not use

If a non-driver signs in on `/driver/login` (or a non-customer on `/customer/login`), sign-in still succeeds; a toast says *"This account is not a driver account — opening your own workspace"* and the user goes to their own landing page.

3.2 Demo accounts

All accounts are listed on the office login screen. Email = `<username>@bellomassi.demo`.

Role	Username	Password	Landing page after sign-in
Super Admin	superadmin	Super@123	`/dashboard` (Admin dashboard)
Administrator	admin	Admin@123	`/dashboard` (Admin dashboard)
Sales Manager	manager	Manager@123	`/dashboard` (Sales Manager dashboard)
Sales Executive	salesman	Sales@123	`/dashboard` (Salesman dashboard)
Customer Service	cs	Cs@123	`/dashboard` (Customer Service dashboard)
Warehouse Manager	warehouse	Warehouse@123	`/dashboard` (Warehouse dashboard)
Inventory Executive	inventory	Inventory@123	`/dashboard` (Inventory variant of Warehouse dashboard)
Dispatcher	dispatcher	Dispatch@123	`/dashboard` (Dispatcher dashboard)
Driver	driver	Driver@123 (or mobile `+91 98400 11206`)	`/driver/dashboard`

Role	Username	Password	Landing page after sign-in
Finance Executive	finance	Finance@123	`/dashboard` (Finance dashboard)
QA Manager	qa	Quality@123	`/dashboard` (QA dashboard)
Customer (portal)	customer	Customer@123 (or mobile `+91 98410 22310`)	`/customer/dashboard`
Supplier (portal)	supplier	Supplier@123	`/dashboard` (Supplier portal dashboard)

Landing pages come from the role record (`landingPath` in `bellomassi_roles`, seeded by `services/seed/adminSeed.js`: DRIVER → `/driver/dashboard`, CUSTOMER → `/customer/dashboard`, all others → `/dashboard`). An Administrator can change a role's landing page in Role Management to one of: Dashboard, Driver app — Today, Order list, Order approval, Dispatch dashboard, Inventory dashboard, Notification center, Customer app — Home, Reports center (`LANDING_PAGES` in `rbacCatalog.js`). A customer who opens `/dashboard` is redirected to `/customer/dashboard`.

3.3 Master flowchart 2 — Login-wise workflow

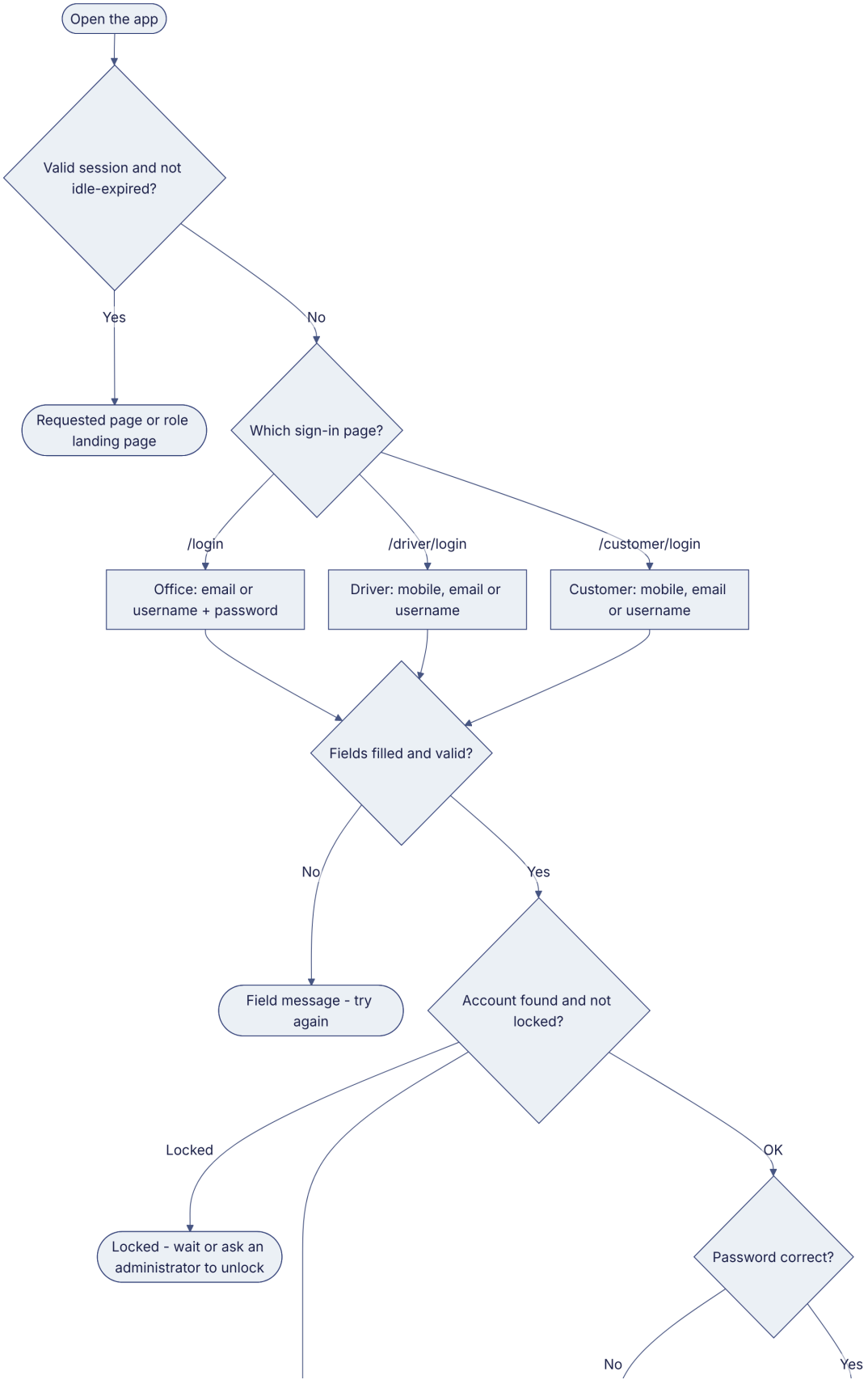


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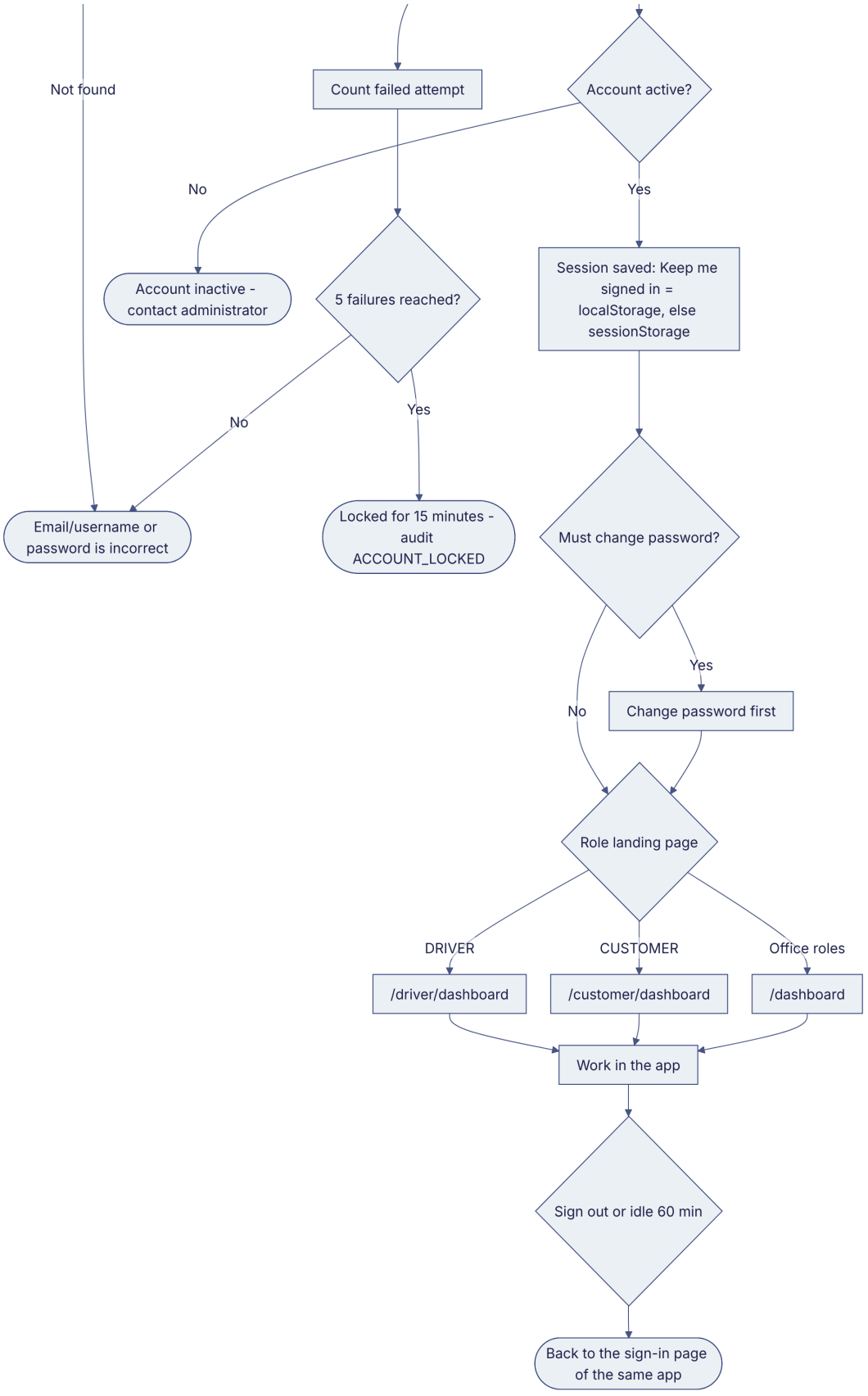


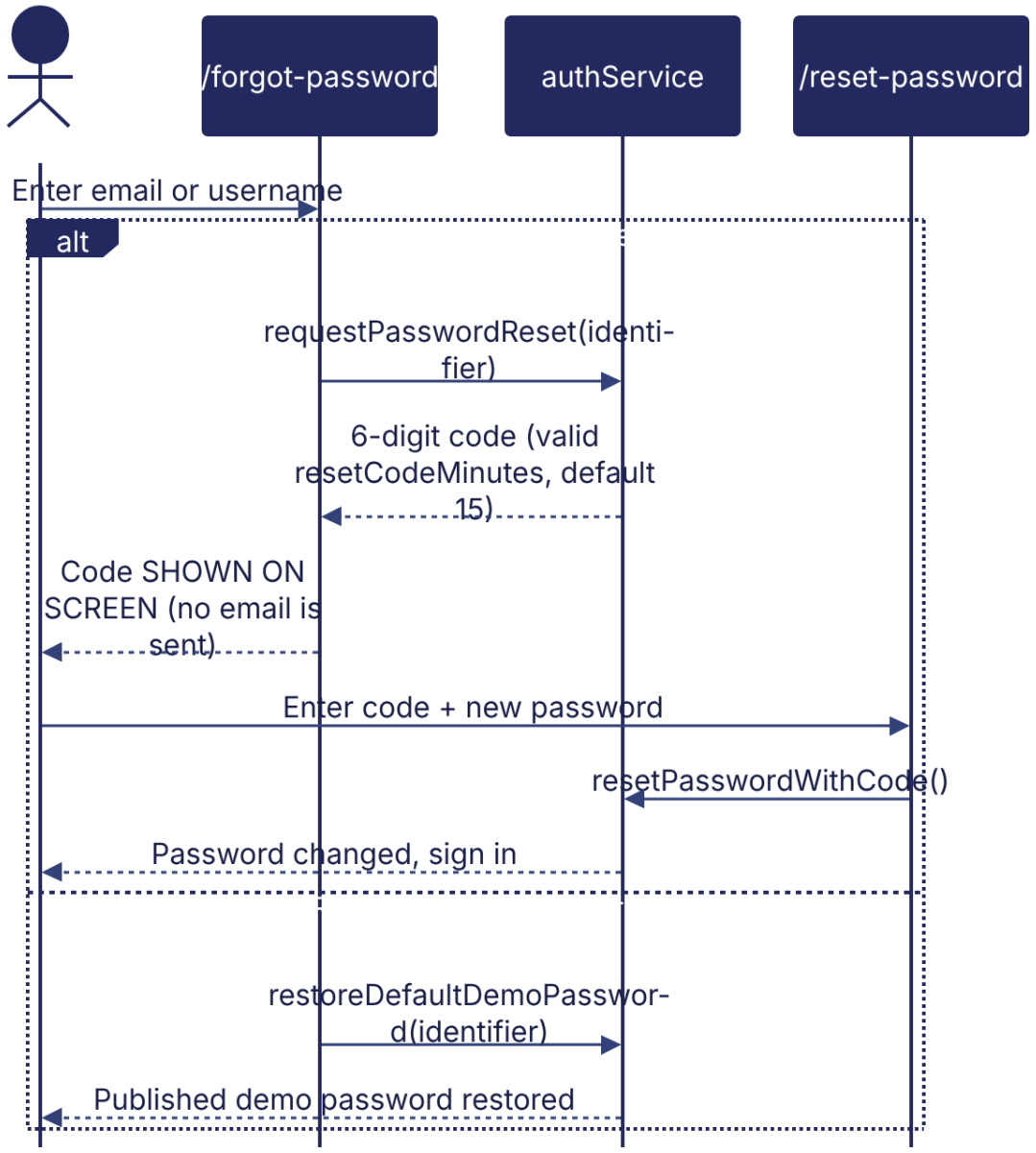
Diagram continued (2 of 2)

3.4 Sign-in rules (from `services/authService.js`)

Rule	Behaviour	Message
Empty identifier	Blocked	"Enter your email address or username."
Malformed email	Blocked	"Enter a valid email address."
Empty password	Blocked	"Enter your password."
Unknown account / wrong password	Blocked; failed attempt counted	"The email/username or password is incorrect. Use one of the demo accounts below."
Too many failures	Account locked (`maxFailedAttempts` 5, `lockoutMinutes` 15 by default — Application Settings › Security)	"Too many failed attempts. This account is locked for N minutes."
Locked account	Blocked until the lock expires or an administrator unlocks the user (`userService.unlockUser`)	"This account is temporarily locked after too many failed attempts…"
Inactive account	Blocked	"This account is inactive. Contact your administrator to reactivate it."
Forced password change	After an administrator reset (`forcePasswordChangeAfterReset`) the user must change password before any other page	Redirect to `/account/change-password?required=1`
Idle timeout	`components/layout/SessionWatcher.jsx` signs out after `sessionTimeoutMinutes` (default 60) of inactivity	Signed out (reason "timeout")
Remember me	"Keep me signed in" stores the session in localStorage; otherwise sessionStorage (closes with the tab). The last username can be remembered.	—

Session data is stored under `bellomassi_session`; failed-attempt counters under `bellomassi_auth`.

3.5 Forgot / reset password (Simulated)



Errors: “No demo account uses that email address or username.”, “This account is inactive. Contact your administrator.”, “Enter the 6-digit code.”, “This code is not valid. Request a new code.”, “This code has expired. Request a new code.”, and for accounts created inside the app: “This account was created in the app and has no published demo password. Use a reset code instead.”

New passwords must follow the policy in Application Settings › Security (minimum length 8, mixed characters by default; `authService.passwordPolicy`).

3.6 Sign-out

1. The user chooses **Sign Out** (header menu, sidebar, driver profile or customer profile) → `/logout` confirmation.
2. On confirm, the session is cleared and the user returns to the sign-in page of the app they were using (`signOutTarget`: driver pages → `/driver/login`, customer pages → `/customer/login`, otherwise `/login`).

3.7 Route protection

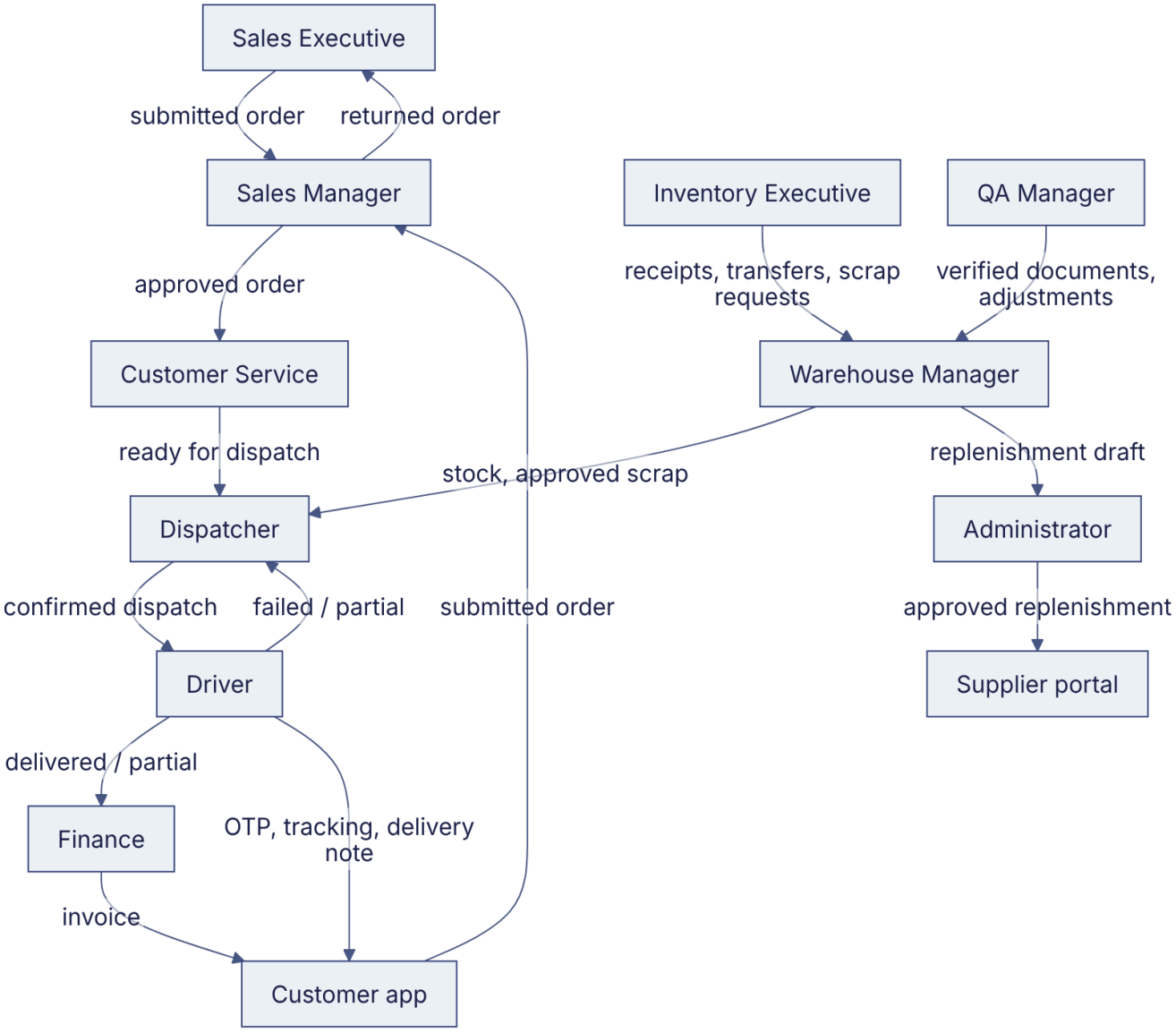
- `src/routes/ProtectedRoute.jsx` sends signed-out visitors to the right sign-in page and checks the page's permission from `navigation.js`. A signed-in user without the permission sees `/unauthorized`.
- Menus only show items the user's role may open (`SidebarNav` filters by permission), so most users never see the Unauthorized page.

04 · Role-Wise Workflow

This chapter describes each of the 13 system roles. Menus were generated from `src/config/navigation.js` filtered by each role's default permissions in `src/config/permissions.js` — they are exactly what the sidebar shows for the seeded role. (An Administrator can change any non-locked role's permissions in Role Management; the menus then change too.)

Every office role also has: **Dashboard**, and under **My Account** — My Profile (`/profile`), Notification Center (`/notifications`), Change Password (`/account/change-password`), Preferences (`/settings`) and Sign Out (`/logout`). Every office dashboard has the **period filter** (chapter 14).

The flowchart below shows how work passes between roles.



4.1 Super Admin

Topic	Details
Purpose	Owner of the platform: every module, every role and setting. The only role that can grant, change, reset or delete Super Admin accounts.
Login	`/login` · **superadmin / Super@123**
Dashboard	`/dashboard` → `AdminDashboard` (Operations overview)
Menus	Everything the Administrator sees (4.2). Because Super Admin holds every permission key, the **My Deliveries** (driver app) and **My Account (Customer)** groups also appear; those pages show “No driver profile linked” / “No customer account linked” unless the user is linked to a driver or customer record.
KPIs	Total orders, Pending approval, Approved, Rejected, Consolidated, Ready for dispatch, Order pipeline; Operations overview: Sales (delivered, 30 days), Delivery (7 days), Inventory value, Finance — invoiced, Purchase (replenishment), AI insights
CRUD	All master data, users, roles, permissions, settings, organization, templates
Approvals	Orders, scrap, replenishment, documents (all permissions)
Reports	All 118 reports in the Reports Center; every module report
Notifications	Administrative and operational notifications addressed to ADMIN / SUPER_ADMIN
Daily workflow	Review dashboard → user and role administration → audit log → settings
Cannot do	Delete or deactivate their **own** account; remove their own access to user management
Next responsible user	Administrator (day-to-day administration)
Logout	Header menu › Sign Out → `/logout` → `/login`

4.2 Administrator

Topic	Details
Purpose	Runs the system day to day: administers users, roles and settings and can perform most business steps (create orders, plan and confirm dispatches, issue invoices, approve scrap and replenishment).
Login	`/login` · **admin / Admin@123**
Dashboard	`/dashboard` → `AdminDashboard` (same KPIs as Super Admin)
Menus	**Order Management:** Create Order, My Orders, Order List, Order Approval, Order Consolidation, Returned Orders, Rejected Orders, Order Reports · **Customers:** Customer List, Customer Credit · Products, Suppliers, Hubs · **Inventory:** Inventory Dashboard, Warehouse Dashboard, QA & Cold Chain, Stock Overview, Low Stock, Stock Receipts, Stock Transfers, Stock Adjustments, Stock Movements, Scrap Management · **Dispatch:** Dispatch Dashboard, Ready for Dispatch, Dispatch Planning, Driver Assignment, Delivery Tracking, Delivery History, Dispatch Zones, Drivers, Vehicles, Delivery Reports, Live Tracking (Demo GPS) · **Vehicle Management:** VMS Dashboard, Vehicles, Driver Documents, Document Expiry, VMS Reports, Compliance Rules · **Documents:** Document Repository, Upload Documents, OCR Queue, Verified, Rejected, Document Types, Document Reports · **Invoices:** Invoice Dashboard, Invoices, New Invoice, Invoice Templates, Invoice Reports · **Communication:** Communication Center, Email, WhatsApp, Automated Notifications, Call History, Call Follow-ups · **Administration:** User Management, Role Management, Permission Matrix, User Role Assignment, Audit Log, Application Settings · **Organization:** Company, Branches, Business Units, Departments, Designations, Employee Master, Warehouse Master · **AI & Intelligence:** AI Overview, AI Assistant, Jarvis, Forecasting, Predictive Inventory, Replenishment, AI Insights, Model Management, AI Settings · **Reports:** Reports Center

Topic	Details
CRUD	Customers, products, suppliers, hubs, zones, drivers, vehicles, document types, invoice templates, email/WhatsApp templates, users, roles, organization records
Approvals	Scrap (`scrap.approve`), replenishment (`procurement.approve` / `procurement.reject` — **only** Administrator and Super Admin hold these), documents (`documents.review` , `documents.reject`). **Not** order approval or consolidation: the Administrator can open the Order Approval and Consolidation queues but `orders.approve` , `orders.reject` and `orders.consolidate` are not granted by default (Super Admin holds them).
Reports	All report categories
Notifications	Failed deliveries, dispatch confirmations, document expiry, low stock, purchase approvals
Daily workflow	Check dashboard → approve replenishment drafts → handle user requests (unlock, reset password) → review audit log → adjust settings
Cannot do	Change or delete a Super Admin account; change the Administrator role unless they are Administrator or Super Admin
Next responsible user	Depends on the module (e.g. approved replenishment → Supplier / warehouse goods receipt)
Logout	`/logout` → `/login`

4.3 Sales Executive (SALESMAN)

Topic	Details
Purpose	Takes orders for the customers assigned to them and keeps them moving through approval.
Login	`/login` · **salesman / Sales@123**
Dashboard	`SalesmanDashboard` : My orders, Draft orders, Pending approval, Approved orders, Returned orders, Rejected orders, Needs your action, My order trend, My recent orders
Menus	**Order Management:** Create Order, My Orders, Order List, Returned Orders, Rejected Orders · **Customers:** Customer List · Products · **Inventory:** Stock Overview · **Documents:** Repository, Upload, OCR Queue, Verified, Rejected · **Communication:** Communication Center, Email, WhatsApp, Call History, Call Follow-ups · **AI:** AI Overview, AI Assistant, Jarvis · Reports Center
Data scope	Own records only (orders where `salesmanId` or `createdBy` is the user)
CRUD	Create, edit and delete **draft** orders; duplicate an order; edit and resubmit **returned** orders; log, edit and delete own calls; upload documents for own records
Approvals	None (submits for approval)
Reports	Reports Center — only categories their permissions allow (Customer reports via `customers.view` , Communication reports via `communication.calls`)
Notifications	Order submitted / approved / returned (with link to edit) / rejected / ready for dispatch
Daily workflow	Dashboard “Needs your action” → fix returned orders → create today’s orders (or from a call) → submit → follow up calls → check approvals
Cannot do	Approve or reject orders; consolidate; dispatch; invoice; see other salesmen’s orders; delete a submitted order (“Only draft orders can be deleted”)
Next responsible user	**Sales Manager** (approval)

Topic	Details
Logout	`/logout` → `/login`

4.4 Sales Manager

Topic	Details
Purpose	Approves, returns or rejects submitted orders after checking credit, stock and data quality; watches sales performance.
Login	`/login` · **manager / Manager@123**
Dashboard	`SalesManagerDashboard` : Pending approvals, Returned orders, Recent pending orders (oldest first), Active accounts and credit position
Menus	**Order Management:** Order List, Order Approval, Returned Orders, Rejected Orders, Order Reports · **Customers:** Customer List, Customer Credit · Products, Suppliers, Hubs · **Inventory:** Stock Overview, Low Stock · **Dispatch:** Dispatch Dashboard, Dispatch Planning, Delivery Tracking, Delivery History · **Documents** (5 pages) · **Invoices:** Invoice Dashboard, Invoices · **Communication:** Center, Email, WhatsApp, Call History, Call Follow-ups · **AI:** AI Overview, AI Assistant, Jarvis, Forecasting, Predictive Inventory, Replenishment, AI Insights, Model Management · Reports Center
Data scope	All submitted and later orders (not other users' drafts)
CRUD	Sales targets (`sales.targets.manage`); calls; documents upload; forecasts (generate / export)
Approvals	**Order approval** — Approve (`orders.approve`), Return for correction and Reject (`orders.reject`)
Reports	Sales, Inventory, Procurement, Dispatch & Delivery (history), Customer, Supplier, Communication, AI categories
Notifications	"New order for approval", "Order resubmitted" (with credit / stock warning flags)
Daily workflow	Order Approval queue (oldest first) → open each order → read approval checks → approve / return / reject → check returned orders → review credit position and reports
Cannot do	Create orders; consolidate; plan or confirm dispatches; issue invoices; manage master data
Next responsible user	**Customer Service** (approved) or **Sales Executive** (returned)
Logout	`/logout` → `/login`

4.5 Customer Service

Topic	Details
Purpose	Verifies approved orders against the warehouse, consolidates them and releases them to dispatch; reviews documents.
Login	`/login` · **cs / Cs@123**
Dashboard	`CustomerServiceDashboard` : Approved orders, Awaiting consolidation, Consolidated orders, Ready for dispatch, Approved orders by stage, Queue by salesman, Consolidation queue (oldest approval first), Recent approved orders

Topic	Details
Menus	**Order Management:** Order List, Order Consolidation · Customer List · Products, Hubs · **Inventory:** Stock Overview, Low Stock · **Dispatch:** Dispatch Dashboard, Ready for Dispatch, Dispatch Planning, Delivery Tracking, Delivery History · **Documents:** Repository, Upload, OCR Queue, Verified, Rejected, Document Reports · **Communication:** Center, Email, WhatsApp, Call History, Call Follow-ups · **AI:** AI Overview, AI Assistant, Jarvis, AI Insights · Reports Center
Data scope	Orders from approval onwards; all documents
CRUD	Consolidation records; calls; documents (upload, review, reject); VMS document upload
Approvals	Consolidation (5-point verification); **document review** (verify / reject)
Reports	Customer, Communication, Dispatch & Delivery (where permitted)
Notifications	“Order awaiting consolidation” after each approval
Daily workflow	Consolidation queue → open order → tick the 5 checks, review hub stock → Consolidate → Mark ready for dispatch → OCR queue / document review → customer calls
Cannot do	Approve or reject orders; create dispatches; assign drivers; confirm dispatch; issue invoices
Next responsible user	**Dispatcher**
Logout	`/logout` → `/login`

4.6 Dispatcher

Topic	Details
Purpose	Plans dispatches from ready orders, assigns compliant drivers and vehicles, confirms loading (stock deduction), follows deliveries and resolves exceptions.
Login	`/login` · **dispatcher / Dispatch@123**
Dashboard	`DispatcherDashboard` / `DispatchOverview`: Ready for dispatch, Planned dispatches, Assigned dispatches, Confirmed dispatches, In transit, Completed deliveries, Failed deliveries, Partial deliveries, Active drivers, Available vehicles; charts: Dispatch trend, Delivery completion trend, Delivery status distribution, Driver performance, Zone delivery performance; tables: Recent dispatches, In-transit deliveries
Menus	Order List · Products, Suppliers, Hubs · **Inventory:** Inventory Dashboard, Stock Overview, Low Stock, Stock Receipts, Stock Transfers, Stock Adjustments, Stock Movements, Scrap Management · **Dispatch:** all 11 pages including Live Tracking (Demo GPS) · **VMS:** Dashboard, Vehicles, Driver Documents, Document Expiry, VMS Reports · **Documents** (5) · **Communication:** Center, Email, WhatsApp · **AI:** AI Overview, Assistant, Jarvis, Forecasting, Predictive Inventory, Replenishment, AI Insights, Model Management · Reports Center
Data scope	Orders from consolidation onwards; fleet and hub documents
CRUD	Dispatches (create, edit while DRAFT/PLANNED/ASSIGNED, cancel), delivery sequence; scrap requests (create); replenishment drafts (create); documents upload. Drivers, vehicles and zones are **view-only** for the Dispatcher (`fleet.manage` is Administrator only).
Approvals	Confirms dispatch (loading check); reviews failed / partial deliveries; can inspect delivery OTPs (`dispatch.otp.inspect`)
Reports	Dispatch & Delivery, Fleet, Inventory-related (where permitted), Procurement, AI
Notifications	“Ready for dispatch: …”, “Delivery failed: … Review required”, dispatch confirmations, document expiry

Topic	Details
Daily workflow	Ready for Dispatch → New Dispatch (orders, zone, date, sequence) → assign driver & vehicle → confirm (tick review) → Live Tracking → review failed / partial stops → re-queue or return stock
Cannot do	Approve orders; consolidate; issue invoices; create or edit drivers / vehicles / zones; approve scrap or replenishment
Next responsible user	**Driver**; for exceptions, the Dispatcher again
Logout	`/logout` → `/login`

4.7 Driver

Topic	Details
Purpose	Delivers the stops of their confirmed dispatch in sequence and records proof of delivery.
Login	`/driver/login` · **driver / Driver@123** or mobile **+91 98400 11206** (also works on `/login`)
Dashboard	`/driver/dashboard` (Today) — period filter defaults to **Today**; Pending / Completed stops, Today / Tomorrow plan ("Published by the Dispatcher"), next stop, driver performance
Menus (bottom bar)	Today, My Deliveries, Route, Completed, My Delivery Notes, Driver Profile, My Documents
Data scope	Own dispatches, deliveries and documents only — another driver's delivery shows "Access denied"
CRUD	Upload own documents (`documents.upload`); delivery notes are generated automatically
Approvals	None
Actions	Start delivery → Reach location → Enter OTP & complete (quantities, receiver, photo/signature) → or Report issue / mark failed
Reports	Driver performance on the Today screen only
Notifications	New dispatch assigned, dispatch confirmed ("Start your first delivery when you leave the hub"), sequence updated, removed from dispatch, dispatch cancelled
Daily workflow	See chapter 08
Cannot do	Start a stop out of sequence, start a future-dated dispatch, start before the Dispatcher confirms, complete without the correct OTP, restart a closed stop, see office modules
Next responsible user	**Customer** (receives goods), **Finance** (invoice), **Dispatcher** (exceptions)
Logout	Driver Profile › Sign Out → `/driver/login`

4.8 Finance Executive

Topic	Details
Purpose	Issues and cancels customer invoices, manages invoice templates and numbering, watches credit exposure.
Login	`/login` · **finance / Finance@123**

Topic	Details
Dashboard	`FinanceDashboard`: Order value (submitted), Approved order value, Delivered to invoice, Invoices issued, Collections, Outstanding (customer balances), Credit exposure, Credit utilisation, Over credit limit, Order value overview
Menus	Order List, Order Reports · Customer Credit · Products, Suppliers · Inventory Dashboard, Stock Overview · **Dispatch:** Dispatch Dashboard, Dispatch Planning, Delivery History, Delivery Reports · **Documents** (5) · **Invoices:** Invoice Dashboard, Invoices, New Invoice, Invoice Templates, Invoice Reports · **Communication:** Center, Email, WhatsApp · **AI:** AI Overview, Assistant, Jarvis, Forecasting, Predictive Inventory, Replenishment, AI Insights, Model Management · Reports Center
Data scope	All non-draft orders; invoices, customers, vendor documents
CRUD	Invoices (issue, cancel), invoice templates, invoice numbering settings; email invoice (demo)
Approvals	None in the order flow; replenishment approval is **not** granted to Finance by default
Reports	Finance, Sales, Dispatch & Delivery, Inventory, Procurement, Supplier, AI
Notifications	Invoice generated; purchase approval information
Daily workflow	Invoice Dashboard → New Invoice for delivered / approved orders → print or email → review outstanding and credit utilisation → reports
Cannot do	Record payments, credit notes or refunds (**Not Implemented**); change a customer's credit balance (Customer master is Administrator only); approve orders
Next responsible user	**Customer** (receives invoice)
Logout	`/logout` → `/login`

4.9 Warehouse Manager

Topic	Details
Purpose	Owns hub stock: receipts, transfers, adjustments, scrap approval, picking & loading view, document expiry.
Login	`/login` · **warehouse / Warehouse@123**
Dashboard	`WarehouseDashboard`: Stock value, Low / out of stock, Damaged / held, Expiry, Inventory alerts, Picking & loading (consolidated and ready orders at the hub), Current stock (lowest available first), vehicle & driver documents expired or due
Menus	Order List · Products, Suppliers, Hubs · **Inventory:** all 10 pages incl. Warehouse Dashboard and QA & Cold Chain · **Dispatch:** Dashboard, Ready for Dispatch, Dispatch Planning, Delivery History · **Documents** (5) · **AI:** Overview, Assistant, Jarvis, Forecasting, Predictive Inventory, Replenishment, AI Insights, Model Management · Reports Center
CRUD	Stock receipts, transfers, adjustments, min/max/reorder levels, scrap requests, replenishment drafts
Approvals	**Scrap requests** (approve / reject / post write-off) — not their own requests
Reports	Inventory, Procurement, Scrap, Dispatch & Delivery history, Supplier, AI
Notifications	Low stock alert, scrap requests pending approval, purchase approval information
Daily workflow	Inventory alerts → receive goods (Stock Receipt, incl. replenishment receipts) → transfers between hubs → approve scrap → check picking list for today's dispatches

Topic	Details
Cannot do	Approve orders; dispatch; approve replenishment (Administrator only); manage products or hubs master data
Next responsible user	**Dispatcher** (stock ready for loading); **Administrator** (replenishment approval)
Logout	`/logout` → `/login`

4.10 Inventory Executive

Topic	Details
Purpose	Day-to-day stock transactions at the hub.
Login	`/login` · **inventory / Inventory@123**
Dashboard	Warehouse dashboard in **inventory** variant (`InventoryDashboardForRole`)
Menus	Products, Hubs · **Inventory:** Inventory Dashboard, Warehouse Dashboard, Stock Overview, Low Stock, Stock Receipts, Stock Transfers, Stock Adjustments (view), Stock Movements, Scrap Management · **Documents** (5) · **AI:** AI Overview, AI Assistant, Replenishment · Reports Center
CRUD	Stock receipts, stock transfers, scrap requests, replenishment drafts. **No** stock adjustments (`inventory.adjustments.manage` not granted).
Approvals	None
Reports	Reports Center categories allowed by permissions (Procurement, Scrap)
Daily workflow	Receive goods → complete transfers → raise scrap requests with photos → draft replenishment from low stock
Cannot do	Approve scrap; post adjustments; approve replenishment; see orders or dispatch
Next responsible user	**Warehouse Manager** (scrap approval), **Administrator** (replenishment)
Logout	`/logout` → `/login`

4.11 QA Manager

Topic	Details
Purpose	Quality and cold chain: document verification, expiry, quarantine (damaged stock), adjustments.
Login	`/login` · **qa / Quality@123**
Dashboard	`QaDashboard` : Inspections pending, Inspection queue (documents waiting for OCR review and verification), Quarantine (damaged stock), Expired documents, Expiry (vehicle & driver documents), **Cold chain & temperature alerts — "No sensor integration", "No live readings"**
Menus	Products, Hubs · **Inventory:** Warehouse Dashboard, QA & Cold Chain, Stock Overview, Low Stock, Stock Receipts, Stock Transfers, Stock Adjustments, Stock Movements, Scrap Management · **VMS:** Dashboard, Vehicles, Driver Documents, Document Expiry, VMS Reports · **Documents:** 5 pages + Document Reports · **AI:** Overview, Assistant, AI Insights · Reports Center
CRUD	Stock adjustments (e.g. damaged, expired), scrap requests, documents (upload, review, reject)

Topic	Details
Approvals	Document verification / rejection
Reports	Scrap, AI, Fleet (via `vms.view`)
Daily workflow	Inspection queue → verify / reject documents → check expiries → adjust damaged stock → raise scrap
Cannot do	Approve scrap (unless granted); create receipts or transfers; dispatch; invoice
Next responsible user	**Warehouse Manager** (scrap approval)
Logout	`/logout` → `/login`

4.12 Customer (portal / customer app)

Topic	Details
Purpose	A customer's staff orders products, tracks deliveries, shares the OTP and downloads delivery notes and invoices.
Login	`/customer/login` · **customer / Customer@123** or mobile **+91 98410 22310**
Dashboard	`/customer/dashboard` (Home): Active orders, Pending approval, Delivered, Outstanding (current); "Your delivery is on the way" card with DEMO GPS; recent orders, delivery notes, notifications
Menus (bottom bar)	Home, Catalog, Cart, My Orders, Delivery Notes, Invoices, Payments, Profile
Data scope	One linked customer (`user.customerId`)
CRUD	Cart (add, change quantity, remove); place order; re-order to cart
Approvals	None — orders go to the Sales Manager
Reports	None
Notifications	Order placed, approved, out for delivery, delivered, delivery not completed
Daily workflow	See chapter 09
Cannot do	Cancel a placed order (Not Implemented); pay online or see payment history (Not Implemented — "No payments recorded in this prototype"); see other customers' data
Next responsible user	**Sales Manager**
Logout	Profile › Sign Out → `/customer/login`

4.13 Supplier (portal)

Topic	Details
Purpose	Shows a supplier the replenishment (purchase) requests Bellomassi has approved for them and the goods receipts posted.
Login	`/login` · **supplier / Supplier@123**
Dashboard	`SupplierPortalDashboard`: Open purchase requests, Approved replenishment (newest first), Lines requested, Lead time, Goods receipts

Topic	Details
Menus	Dashboard and My Account only (permission `dashboard.view` only)
Data scope	One linked supplier
CRUD / approvals	None — **read-only**
Cannot do	Accept or reject requests, submit invoices, update delivery dates (**Not Implemented**)
Next responsible user	Warehouse Manager / Inventory Executive (goods receipt)
Logout	`/logout` → `/login`

4.14 Custom roles

Administrators can create roles in **Role Management** (`/admin/roles/create`, `rbacService`). A custom role gets a name, data scope, landing page and any set of permission keys. Its dashboard is the permission-driven **workspace launcher** (`GenericDashboard` — “Your workspace” / “No modules assigned yet”), never another role’s KPIs. Status: **Implemented**.

05 · Module-Wise Workflow

Each module is described with the same 17 headings: **Purpose · Roles · Entry point · Screens · Preconditions · Steps · Actions · System behaviour · Validation · Decisions · Approval · Rejection · Status transitions · Related modules · Output · Exceptions · Flowchart.** "—" means the heading does not apply to that module.

Status labels follow the legend in the README.

Module index

#	Module	Status
5.1	Authentication & session	Implemented (simulated security)
5.2	Role dashboards	Implemented
5.3	Products	Implemented
5.4	Customers & credit	Implemented / credit Partially Implemented
5.5	Hubs	Implemented
5.6	Suppliers	Implemented
5.7	Order capture (create, draft, submit, resubmit)	Implemented
5.8	Order approval	Implemented
5.9	Order consolidation	Implemented
5.10	Inventory (stock, receipts, transfers, adjustments, movements, low stock)	Implemented
5.11	Scrap management	Implemented
5.12	Warehouse dashboard and QA & cold chain	Implemented (temperature monitoring Not Implemented)
5.13	Dispatch planning, assignment and confirmation	Implemented
5.14	Delivery tracking, live tracking and exception review	Implemented / GPS Simulated
5.15	Driver app	Implemented (GPS and OTP Simulated)
5.16	Delivery notes, OTP and proof of delivery	Implemented / OTP Simulated
5.17	Customer app	Implemented (payments Not Implemented)
5.18	Fleet master — zones, drivers, vehicles	Implemented
5.19	Vehicle Management System (VMS)	Implemented
5.20	Documents and OCR	Implemented / OCR Simulated
5.21	Invoices and invoice templates	Implemented (demo tax)
5.22	Email	Simulated (gateway Not Verified)
5.23	WhatsApp (OpenWA)	Simulated (gateway Not Verified)
5.24	Automated notifications	Implemented over simulated channels
5.25	Call history	Implemented (no telephony)

#	Module	Status
5.26	Notification center	Implemented
5.27	Reports Center	Implemented (24 of 118 reports have no source data)
5.28	AI Assistant and Jarvis	Simulated
5.29	Forecasting, predictive inventory and model management	Implemented (in-browser models)
5.30	Replenishment	Implemented (ends at goods receipt)
5.31	AI insights	Implemented (rule-based)
5.32	Users, roles and permissions	Implemented
5.33	Audit log	Implemented
5.34	Application settings	Implemented
5.35	Organization	Implemented
5.36	My account (profile, password, preferences)	Implemented

5.1 Authentication & session

Heading	Details
Purpose	Identify the user, open their workspace and protect pages by permission.
Roles	All
Entry point	`/login`, `/driver/login`, `/customer/login`
Screens	Login, Driver login, Customer login, Forgot password, Reset password, Change password, Sign out, Unauthorized
Preconditions	Account exists in `bellomassi_demo_users` and is active
Steps	Enter identifier and password → sign in → (change password if required) → landing page
Actions	Sign in, Keep me signed in, Remember username, Forgot password, Reset with code, Restore demo password, Sign out
System behaviour	`authService.login` checks lock, password hash, status; writes `bellomassi_session`; `SessionWatcher` signs out after idle timeout
Validation	Identifier and password required; valid email format; password policy on change
Decisions	Locked? Active? Must change password? Which landing page?
Approval	—
Rejection	Wrong credentials, locked, inactive (messages in chapter 03)
Status transitions	Signed out → Signed in → (Idle-expired / Signed out)
Related modules	Users (unlock, reset password), Application settings › Security, Audit log (ACCOUNT_LOCKED)
Output	Session; landing page
Exceptions	Lockout after 5 failures for 15 minutes (defaults)

Heading	Details
Flowchart	Chapter 03, master flowchart 2

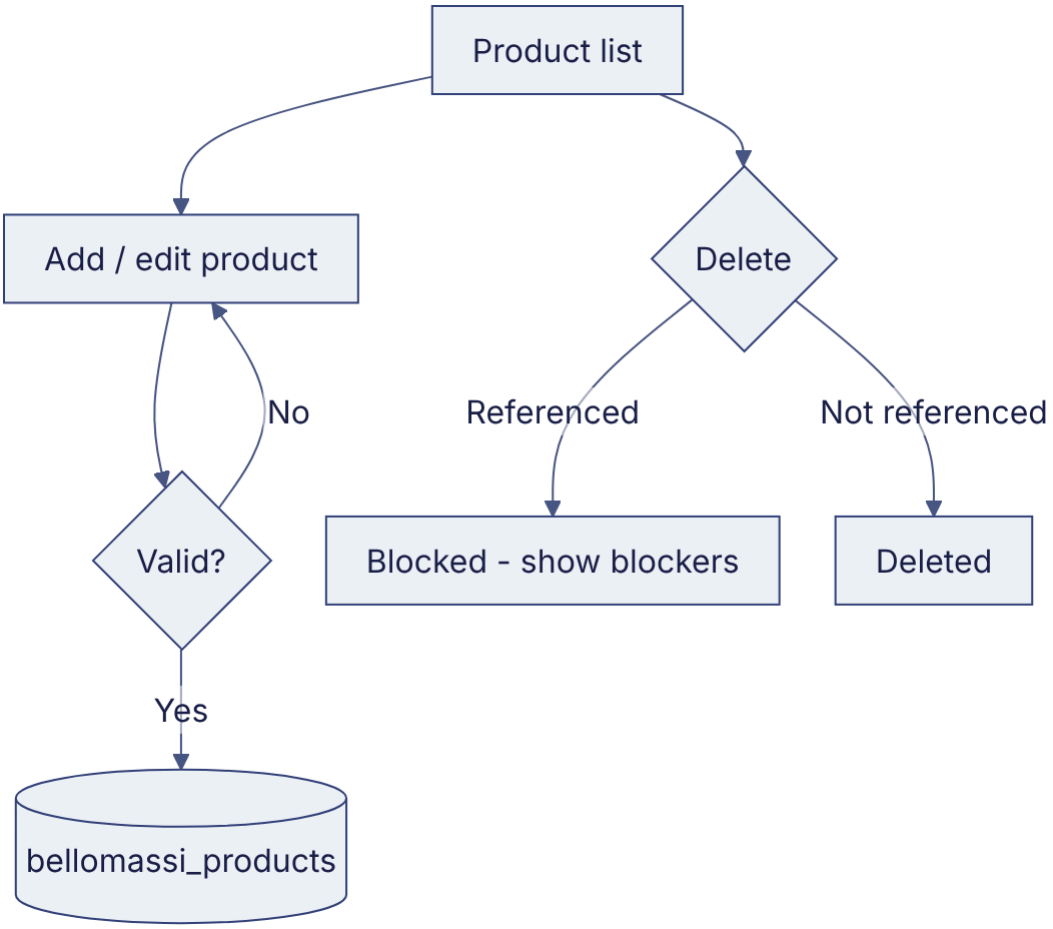
5.2 Role dashboards

Heading	Details
Purpose	One screen per role with the KPIs and queues that role acts on.
Roles	All (`dashboard.view`)
Entry point	`/dashboard` (`pages/dashboard/Dashboard.jsx` → `RoleDashboard.jsx`), `/driver/dashboard` , `/customer/dashboard`
Screens	Admin, Salesman, Sales Manager, Customer Service, Dispatcher, Finance, Warehouse (and Inventory variant), QA, Customer portal, Supplier portal, Driver Today, Customer Home, Generic workspace launcher
Preconditions	Signed in
Steps	Open dashboard → choose period → (optional) Compare → click a KPI or queue row to drill into the module
Actions	Period select, custom date range, specific period, compare toggle, reset
System behaviour	Dated collections are narrowed to the period on their business date (`utils/dashboardFilter.js`) and passed to the existing calculations; current positions are labelled **Current**
Validation	Custom range: start after end / missing / invalid dates show a message
Decisions	—
Approval / Rejection	—
Status transitions	—
Related modules	Every module (drill-down links)
Output	KPIs, charts, queues
Exceptions	Custom roles without dated data show the launcher without a filter
Flowchart	Chapter 14

5.3 Products

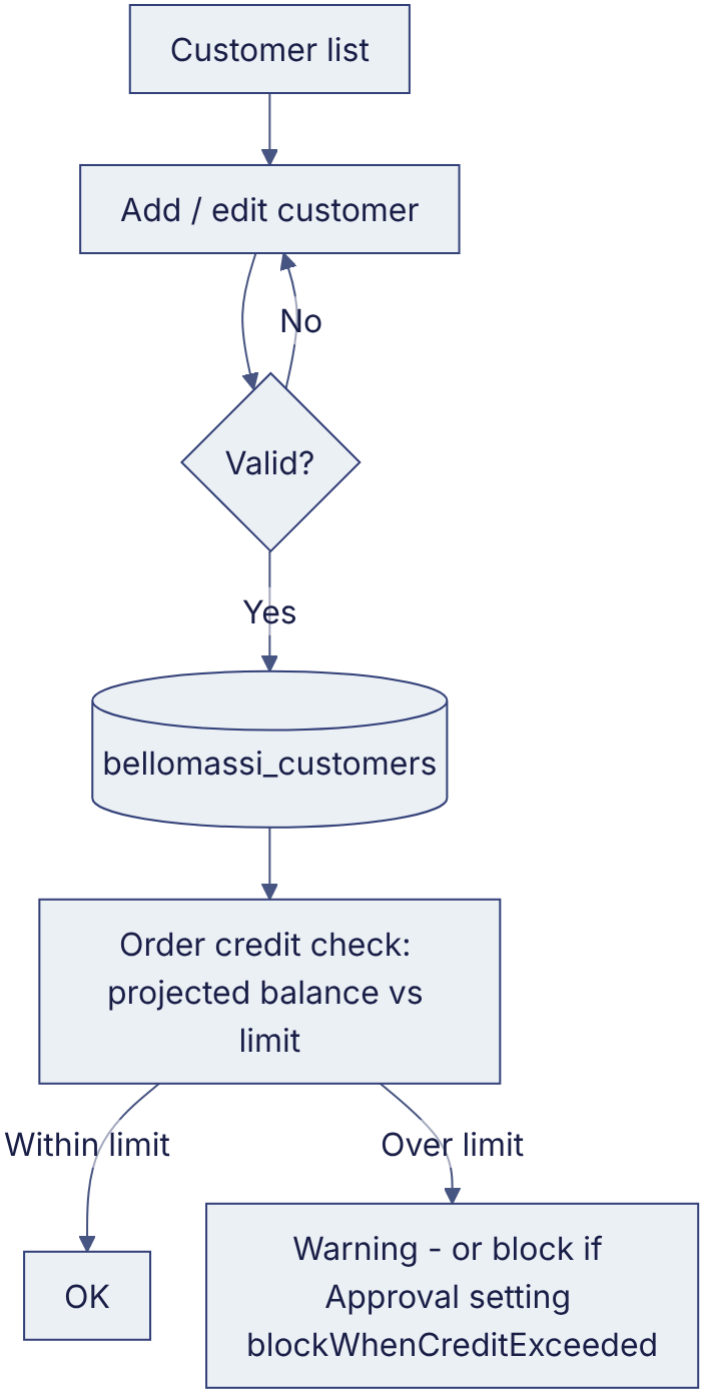
Heading	Details
Purpose	Product master: code, SKU, barcode, category, unit, price, tax, discount limit, storage condition (Frozen / Chilled / Ambient), stock levels, batch control.
Roles	View: most office roles (`products.view`); manage: Administrator (`products.manage`)
Entry point	`/products`
Screens	Product list, Add product (`/products/create`), Product details (`/products/:id`), Edit product
Preconditions	—
Steps	Add product → fill sections → save; edit or delete later

Heading	Details
Actions	Create, edit, delete, search, filter, export
System behaviour	`productService.createProduct/updateProduct/deleteProduct` → `bellomassi_products`
Validation	Code 3–20 letters/numbers/hyphens, unique; unique SKU and barcode (8–14 digits); category, status, unit, product type required; tax 0–100; max discount 0–100%; maximum stock greater than minimum
Decisions	Track inventory? Allow negative stock? Batch-controlled?
Approval / Rejection	—
Status transitions	Active ↔ Inactive (inactive products block order approval: “Products active” check)
Related modules	Orders, inventory, invoices, customer catalog, forecasting
Output	Product record
Exceptions	Delete is blocked when the product is referenced (`getProductDeleteBlockers`)
Flowchart	see below



5.4 Customers & credit

Heading	Details
Purpose	Customer master (type, contacts, address, GST/PAN, hub, zone, sales rep, VIP) and credit settings (enabled, limit, balance, days).
Roles	View: Administrator, Sales Executive (own), Sales Manager, Customer Service; manage: Administrator; Customer Credit: Administrator, Sales Manager, Finance
Entry point	`/customers`, `/customers/credit`
Screens	Customer list, Add / Edit customer, Customer details (orders, documents, credit), Customer Credit
Preconditions	Hub and zone exist
Steps	Add customer → save → (optional) link a portal user to the customer in User Management
Actions	Create, edit, delete, search, view credit position
System behaviour	`customerService` → `bellomassi_customers`; `getOrderCreditPosition` calculates projected balance for each order
Validation	Code 3–20 characters, unique; type and status required; 10-digit Indian mobile; email; PAN `AAAAA9999A`; GSTIN 15 characters containing the PAN; 6-digit pincode; customer-since not in future; credit days whole number; credit balance non-negative
Decisions	Credit enabled? Credit limit exceeded (warning or block per settings)?
Approval / Rejection	—
Status transitions	Active / Inactive / other statuses from `CUSTOMER_STATUSES`; only **Active** customers pass order approval
Related modules	Orders, approval (credit check), invoices, customer app, reports
Output	Customer record, credit position
Exceptions	**Partially Implemented:** the credit balance is a field maintained on the customer record; issuing invoices and recording deliveries do **not** change it, and payments cannot be recorded
Flowchart	see below



5.5 Hubs

Heading	Details
Purpose	Fulfilment locations (Main, Regional, Branch, Distribution hub, Warehouse) that hold stock and run dispatches.
Roles	View: most office roles; manage: Administrator
Entry point	`/hubs` (also `/admin/organization/hubs`)
Screens	Hub list, Add / Edit hub, Hub details (stock, zones, vehicles)
Preconditions	—

Heading	Details
Steps	Add hub → save
Actions	Create, edit, delete
System behaviour	`hubService` → `bellomassi_hubs`
Validation	Code 2–6 letters/numbers, unique; unique name; type and status required; phone, email, 6-digit pincode
Decisions	Active or Inactive
Approval / Rejection	—
Status transitions	Active ↔ Inactive (inactive hubs cannot receive stock and block order approval)
Related modules	Inventory, orders, dispatch, zones, drivers, vehicles
Output	Hub record
Exceptions	Delete blocked when referenced (`getHubDeleteBlockers`)
Flowchart	`Hub list → Add/Edit → validate → bellomassi_hubs` (same pattern as 5.3)

5.6 Suppliers

Heading	Details
Purpose	Suppliers used in replenishment (lead time, products) and the supplier portal.
Roles	View: Administrator, Sales Manager, Dispatcher, Finance, Warehouse Manager; manage: Administrator
Entry point	`/suppliers`
Screens	Supplier list (create / edit in place)
Preconditions	—
Steps	Add or edit supplier → save
Actions	Create, edit
System behaviour	`supplierService.saveSupplier` → stored inside `bellomassi_forecast_settings.suppliers`
Validation	Required fields on the supplier form
Decisions / Approval / Rejection	—
Status transitions	—
Related modules	Replenishment, supplier portal, procurement reports
Output	Supplier record
Exceptions	Supplier invoices and payments are **Not Implemented**
Flowchart	`Supplier list → edit → bellomassi_forecast_settings`

5.7 Order capture (create, draft, submit, resubmit)

Heading	Details
Purpose	Record what a customer wants, priced with tax and discounts, and send it for approval.
Roles	Sales Executive, Administrator (`orders.create`); Customer via customer app
Entry point	`/orders/create` ; `/orders/:id/edit` ; `/orders/my-orders` ; `/orders/returned`
Screens	Create / Edit order form (customer, hub, type, priority, payment terms, dates, address, lines), My Orders, Order List, Order Details (timeline), Returned Orders, Rejected Orders
Preconditions	Active customer, active hub, active products
Steps	Choose customer (address and hub default from customer) → add lines → review totals and warnings → **Save draft** or **Submit**
Actions	Save draft, submit, edit draft, delete draft, duplicate, resubmit after return
System behaviour	`orderService.createOrder` (DRAFT), `submitOrder` (number assigned from `bellomassi_order_sequence` , SUBMITTED + approval PENDING, credit/stock warning flags, timeline event, notifications to Sales Manager and salesman, event ORDER_CREATED); `resubmitOrder` after a return
Validation	“Select a customer.”, “Assign a salesman.”, “Select the fulfilment hub.”, order type, priority, payment terms; order date valid and not in future; requested delivery date required and not before order date; address, city, state required; 6-digit pincode; “Add at least one product.”; “Order total must be greater than zero.”
Decisions	Save as draft or submit; credit / stock warnings are shown but do not block submission
Approval	Goes to 5.8
Rejection	— (see 5.8)
Status transitions	(new) → DRAFT → SUBMITTED; RETURNED → SUBMITTED (resubmit)
Related modules	Customers, products, inventory (stock check), approval, call history (link order to call), customer app
Output	Order record in `bellomassi_orders` , timeline in `bellomassi_order_timeline`
Exceptions	“Only draft orders can be deleted. Submitted orders are kept for the audit trail.”; “You can only delete your own drafts.” Cancelling a submitted order is **Not Implemented**
Flowchart	see chapter 06

5.8 Order approval

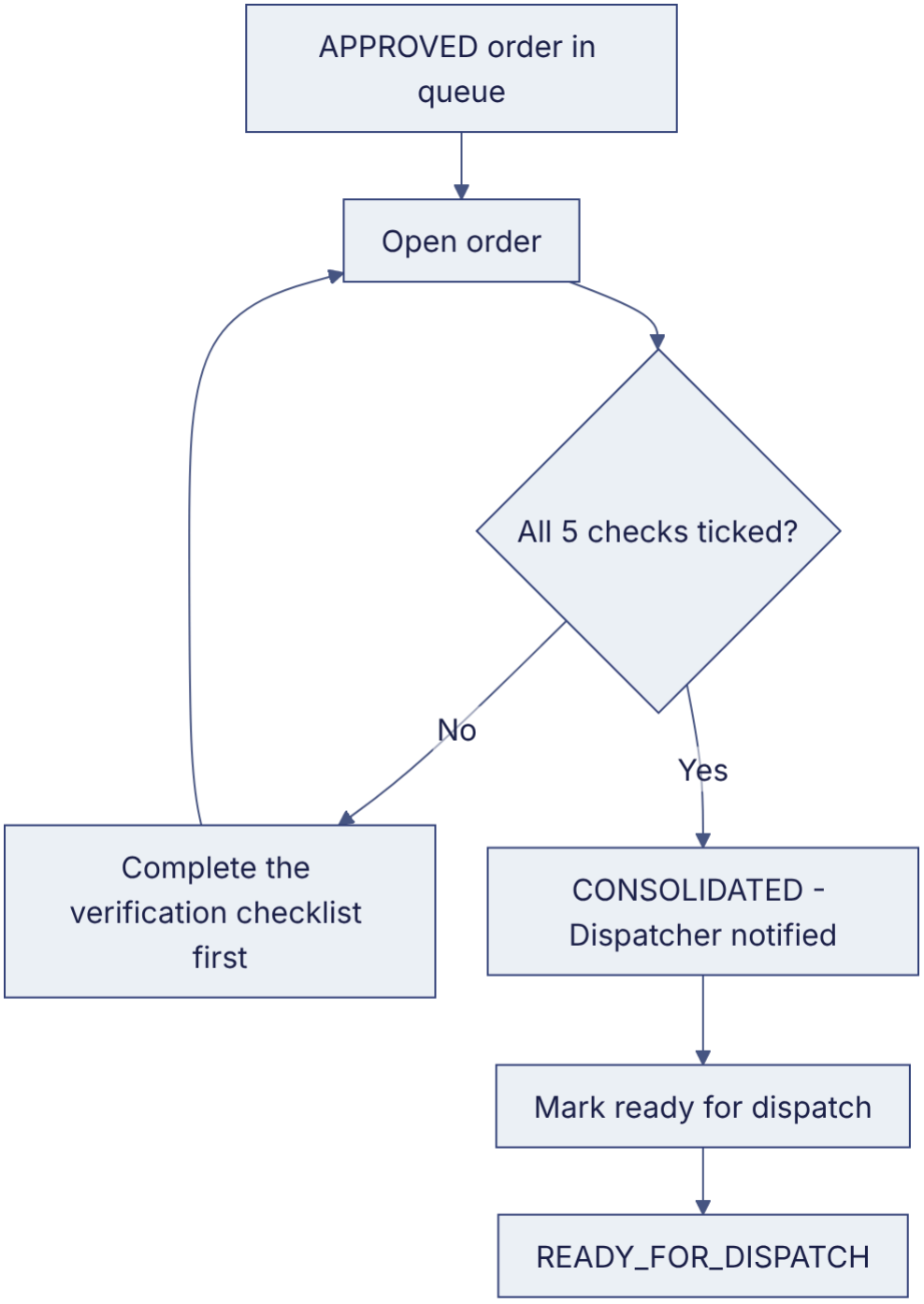
Heading	Details
Purpose	Sales Manager decides each submitted order.
Roles	Sales Manager (`orders.approve` , `orders.reject`); Super Admin (all keys). Administrator can view the queue (`orders.approval.view`) but not decide by default
Entry point	`/orders/approval` → `/orders/approval/:id`
Screens	Approval queue (oldest first), Review order (checks, lines, credit, history, decision panel)
Preconditions	Order SUBMITTED with approval PENDING
Steps	Open order → read the six checks → approve (with remarks and warning acknowledgement), or return / reject with reason and remarks

Heading	Details
Actions	Approve, Return for correction, Reject
System behaviour	`approvalService.approveOrder / returnOrderForCorrection / rejectOrder` write the order, an approval record (`bellomassi_approvals`) and a timeline event in one transaction; notifications and automation events
Validation	Blocking checks (customer active, hub active, order total valid, products active, credit when `blockWhenCreditExceeded` is on); warnings (stock, credit) need acknowledgement; remarks required if setting `requireApprovalRemarks` ; reject / return need a reason from the list and remarks
Decisions	Approve / Return / Reject
Approval	APPROVED, consolidation PENDING
Rejection	REJECTED (final) or RETURNED (back to the salesman)
Status transitions	SUBMITTED → APPROVED / RETURNED / REJECTED
Related modules	Order capture, consolidation, customers (credit), inventory (stock)
Output	Decision record, notifications
Exceptions	Draft orders cannot be decided; already-decided orders show “has already been approved / rejected / returned. No change was made.”
Flowchart	Chapter 07, master flowchart 4

5.9 Order consolidation

Heading	Details
Purpose	Customer Service verifies an approved order against the hub and releases it to dispatch.
Roles	Customer Service (`orders.consolidate`); view queue: Administrator, Customer Service
Entry point	`/orders/consolidation` → `/orders/consolidation/:id`
Screens	Consolidation queue, Consolidate order (verification checklist, stock review, notes)
Preconditions	Order APPROVED and not yet consolidated
Steps	Tick five checks — customer and delivery address verified; fulfilment hub confirmed; products and units checked; quantities match the approved order; hub stock availability reviewed → Consolidate → Mark ready for dispatch
Actions	Consolidate, Mark ready for dispatch
System behaviour	`consolidateOrder` writes `bellomassi_consolidations` , order CONSOLIDATED (consolidation COMPLETED), notifies Dispatcher; `markReadyForDispatch` → READY_FOR_DISPATCH, notifies Dispatcher and salesman, event ORDER_READY. The stock review is information only — nothing is deducted.
Validation	“Confirm every check before consolidating (N left).”
Decisions	All checks confirmed?
Approval	—
Rejection	— (Customer Service cannot reject; an issue must be resolved outside the system or by the Administrator)

Heading	Details
Status transitions	APPROVED → CONSOLIDATED → READY_FOR_DISPATCH
Related modules	Approval, inventory, dispatch
Output	Consolidation record
Exceptions	“Only approved orders can be consolidated.”, “has already been consolidated. No change was made.”, “Only consolidated orders can be marked ready for dispatch.”
Flowchart	see below



5.10 Inventory

Heading	Details
Purpose	Keep stock per product per hub accurate; every change is a stock movement.
Roles	Receipts & transfers: Administrator, Warehouse Manager, Inventory Executive; adjustments: Administrator, Warehouse Manager, QA Manager; view: Dispatcher, Finance, Sales roles (stock overview)
Entry point	`/inventory`, `/inventory/stock`, `/inventory/low-stock`, `/inventory/stock-receipts`, `/inventory/stock-transfers`, `/inventory/stock-adjustments`, `/inventory/movements`
Screens	Inventory dashboard, Stock overview, Stock details (min / reorder / max levels), Low stock, Receipt list / New receipt, Transfer list / New transfer, Adjustment list / New adjustment, Stock movements
Preconditions	Product tracks inventory; hub active (for stock coming in)
Steps	See chapter 10
Actions	Create receipt; create transfer (draft or complete), complete or cancel draft transfer; create adjustment (Increase, Decrease, Damaged, Lost, Correction); update stock levels
System behaviour	`inventoryService` updates `bellomassi_inventory` and appends `bellomassi_stock_movements` (Stock Receipt, Transfer In/Out, Adjustment Increase/Decrease, Damage, Lost Stock, Stock Dispatch, Dispatch Reversal, Delivery Return, Scrap Write-off)
Validation	Hub selected and active; product valid; quantity valid; date valid; reference ≤ 40 characters; transfer hubs must differ and source must hold the quantity; adjustment needs type, reason (notes for “Other”), cannot remove more than available unless the product allows negative stock; minimum, reorder, maximum non-negative and reorder ≤ maximum
Decisions	Transfer: save draft or complete now
Approval / Rejection	— (adjustments post immediately; no approval step)
Status transitions	Transfer: Draft → Completed / Cancelled
Related modules	Dispatch (deduction / reversal / return), scrap, replenishment, orders (stock warnings), reports
Output	Stock rows and movement ledger
Exceptions	“…is already completed — stock was not changed again.”; “Only draft transfers can be cancelled.”
Flowchart	Chapter 10, master flowchart 7

5.11 Scrap management

Heading	Details
Purpose	Write off expired, damaged or rejected stock with approval and evidence.
Roles	Create: Administrator, Dispatcher, Warehouse Manager, Inventory Executive, QA Manager (`scrap.create`); approve / reject / write off: Administrator, Warehouse Manager (`scrap.approve` , `scrap.reject`)
Entry point	`/inventory/scrap`
Screens	Scrap dashboard, Scrap requests, New / Edit request, Scrap request detail, Scrap note (print)
Preconditions	Stock exists at the hub (available or damaged source)

Heading	Details
Steps	Create draft → submit → approver approves or rejects → approver posts write-off
Actions	Save draft, edit draft, submit, approve, reject, post write-off, cancel, print scrap note
System behaviour	`scrapService` → `bellomassi_scrap_requests`; write-off posts a **Scrap Write-off** movement once (`inventoryService.postScrapWriteOff`)
Validation	Reason from list (Expired, Damaged, Quality Rejected, Temperature Excursion, Contamination, Packaging Damage, Other); quantity ≤ eligible stock (minus other open requests); no duplicate open request; photo proof required for Damaged, Packaging Damage, Contamination; batch number required for batch-controlled products; rejection reason ≥ 5 characters
Decisions	Approve or reject
Approval	Approver cannot be the requester (“You cannot approve your own scrap request.”)
Rejection	REJECTED with reason
Status transitions	DRAFT → PENDING_APPROVAL → APPROVED → COMPLETED; PENDING_APPROVAL → REJECTED; DRAFT / PENDING_APPROVAL / APPROVED → CANCELLED
Related modules	Inventory, QA, reports (Scrap category)
Output	Scrap note; stock reduced
Exceptions	“Only an approved scrap request can create a stock write-off.”, “The write-off would make stock negative.”
Flowchart	Chapter 07 §7.3

5.12 Warehouse dashboard and QA & cold chain

Heading	Details
Purpose	Warehouse view of picking/loading, alerts and expiry; QA view of inspections, quarantine and document expiry.
Roles	Warehouse dashboard: Administrator, Warehouse Manager, Inventory Executive, QA Manager; QA & Cold Chain: Administrator, Warehouse Manager, QA Manager
Entry point	`/warehouse`, `/quality`
Screens	Warehouse dashboard, QA dashboard
Preconditions	—
Steps	Review alerts → open the linked record (stock line, document, order)
Actions	Navigation only
System behaviour	Read-only aggregation of inventory, orders, documents, vehicles and drivers
Validation / Decisions / Approval / Rejection	—
Status transitions	—
Related modules	Inventory, documents, VMS, dispatch
Output	Alerts and queues

Heading	Details
Exceptions	**Cold-chain temperature monitoring is Not Implemented** — the card shows the configured ranges (frozen −25 to −18 °C, chilled 0 to 4 °C) with “No sensor integration” / “No live readings”. Picking strategy (FEFO) and QA hold on receipt are settings only — **UI Only**.
Flowchart	`Dashboard → alert → module record`

5.13 Dispatch planning, assignment and confirmation

Heading	Details
Purpose	Turn ready orders into a vehicle run with a driver, a stop sequence and a stock deduction.
Roles	Manage: Administrator, Dispatcher (`dispatch.manage`); view: Sales Manager, Customer Service, Finance, Warehouse Manager
Entry point	`/dispatch` , `/dispatch/ready-orders` , `/dispatch/planning` , `/dispatch/create` , `/dispatch/assignments` , `/dispatch/:id`
Screens	Dispatch dashboard, Ready for dispatch, Dispatch planning, New / Edit dispatch (orders, zone, date, sequence editor), Driver assignment, Dispatch details (timeline, confirmation review, cancel)
Preconditions	Orders READY_FOR_DISPATCH with a complete delivery address, not on another active dispatch
Steps	Select orders → set hub, zone, date → sequence (suggested or manual) → save as draft or plan → assign driver and vehicle → review and confirm
Actions	Create, edit (DRAFT / PLANNED / ASSIGNED), assign / reassign, change sequence, confirm, cancel
System behaviour	`dispatchService` → `bellomassi_dispatches` , `bellomassi_dispatch_timeline` ; confirmation deducts stock once (`stockDeducted`), creates one delivery per order (`bellomassi_deliveries`), sets orders DISPATCHED, notifies driver and Administrator
Validation	Order eligibility; neighbouring-zone warning; driver issues (inactive, busy on another active dispatch, offline, other hub, licence expired / missing, required documents); vehicle issues (inactive, maintenance, busy, other hub, load over capacity, insurance / RC / fitness documents); confirmation tick required
Decisions	Which orders, sequence, driver, vehicle; cancel reason (Vehicle breakdown, Driver unavailable, Customer requested reschedule, Stock issue at hub, Planning error, Other)
Approval	Dispatcher’s own confirmation
Rejection	Cancellation (reason + remarks) — orders return to READY_FOR_DISPATCH; if confirmed, stock is reversed once
Status transitions	DRAFT → PLANNED → ASSIGNED → CONFIRMED → IN_TRANSIT → COMPLETED / PARTIALLY_DELIVERED (“Closed with exceptions”); → CANCELLED
Related modules	Consolidation, inventory, fleet, VMS, driver app, delivery tracking
Output	Dispatch, deliveries, stock movements (Stock Dispatch)
Exceptions	“Assign a driver and vehicle before confirming.”, “…is already confirmed. Stock was deducted once and was not deducted again.”, “Invalid delivery sequence: every stop must appear exactly once.”
Flowchart	Chapter 11, master flowchart 8

5.14 Delivery tracking, live tracking and exception review

Heading	Details
Purpose	Follow every stop and resolve failed and partial deliveries.
Roles	Delivery tracking: Administrator, Sales Manager, Customer Service, Dispatcher; Live tracking: Administrator, Dispatcher (`tracking.fleet`); review: Administrator, Dispatcher
Entry point	`/dispatch/delivery-tracking` , `/dispatch/live-tracking` , `/dispatch/delivery-history` , `/dispatch/deliveries/:id`
Screens	Delivery tracking, Live Tracking (Demo GPS map), Delivery history, Delivery details (items, OTP inspector, POD, review panel), Delivery reports
Preconditions	Dispatch confirmed
Steps	Monitor → open failed / partial delivery → review with remarks → optionally return goods to hub → optionally re-queue (failed only)
Actions	Review, return to hub, re-queue, inspect OTP
System behaviour	`deliveryService.reviewDelivery` : review status REVIEWED; return to hub books undelivered (not damaged) quantity back once (Delivery Return movement); failed + returned → delivery RETURNED, order RETURNED_TO_HUB; re-queue → order READY_FOR_DISPATCH
Validation	“Add review remarks.”; re-queue only for a failed delivery whose goods are returned; goods can be returned only once
Decisions	Return stock? Re-queue?
Approval / Rejection	—
Status transitions	Delivery FAILED → RETURNED; order DELIVERY_FAILED → RETURNED_TO_HUB or READY_FOR_DISPATCH
Related modules	Driver app, dispatch, inventory
Output	Reviewed delivery, return movements
Exceptions	“Only failed or partial deliveries need review.”, “was already reviewed... No change was made.”
Flowchart	Chapter 15

5.15 Driver app

Heading	Details
Purpose	Mobile workflow for the driver’s stops.
Roles	Driver (`driver.app`)
Entry point	`/driver/login` → `/driver/dashboard`
Screens	Today, My deliveries, Delivery (stop detail), Confirm delivery, Route, Completed, My delivery notes, Delivery note, Driver profile, My documents, Document
Preconditions	User linked to a driver record (`drivers.userId`); dispatch CONFIRMED or IN_TRANSIT; dispatch date not in the future
Steps	Chapter 08

Heading	Details
Actions	Start delivery, Reach location, Enter OTP & complete, Report issue / mark failed, Call / WhatsApp / navigate (contact actions), view route, view notes, upload documents
System behaviour	`deliveryService.startDelivery / markArrived / confirmDelivery / failDelivery`, `trackingService` (DEMO GPS), `otpService`
Validation	Sequence order, one stop out at a time, OTP, quantities, receiver name, POD size limit (350 KB)
Decisions	Deliver all, part, or fail
Approval / Rejection	—
Status transitions	Delivery DISPATCHED → OUT_FOR_DELIVERY → DELIVERED / PARTIALLY_DELIVERED / FAILED
Related modules	Dispatch, customer app, delivery notes, invoices
Output	Delivery note, POD, tracking session
Exceptions	Chapter 15
Flowchart	Chapter 08, master flowchart 5

5.16 Delivery notes, OTP and proof of delivery

Heading	Details
Purpose	Evidence that goods were handed over.
Roles	Generated by the driver's completion; viewed by Administrator, Sales roles, Customer Service, Dispatcher, Finance, Warehouse Manager (`deliveryNotes.view`), driver and customer
Entry point	`/dispatch/delivery-notes/:id`, `/driver/delivery-notes/:id`, `/customer/delivery-notes/:id`
Screens	Delivery note (printable)
Preconditions	Delivery completed (full or partial)
Steps	Driver completes → note created automatically
Actions	View, print
System behaviour	`deliveryNoteService` builds BM-DN numbered notes in `bellomassi_delivery_notes`; OTP records in `bellomassi_delivery_otps` (6 digits, default 10-minute expiry, 5 attempts, statuses ACTIVE / VERIFIED / EXPIRED / INVALIDATED / LOCKED); POD in `bellomassi_delivery_pod`
Validation	OTP must match, be active and unexpired; attempts limited
Decisions / Approval / Rejection	—
Status transitions	OTP: ACTIVE → VERIFIED / EXPIRED / LOCKED / INVALIDATED (regenerated)
Related modules	Driver app, customer app, invoices (delivered-quantity basis)
Output	Delivery note
Exceptions	**Simulated:** no SMS is sent; the customer reads the OTP in the customer app
Flowchart	Chapter 08

5.17 Customer app

Heading	Details
Purpose	Self-service ordering and visibility for a customer.
Roles	Customer (`customer.app` , `customer.order`)
Entry point	`/customer/login` → `/customer/dashboard`
Screens	Home, Catalog, Cart, My orders, Order, Track delivery, Delivery notes, Delivery note, Invoices, Invoice, Payments, Profile
Preconditions	User linked to a customer (`user.customerId`)
Steps	Chapter 09
Actions	Add to cart, change quantity, remove, place order, re-order, track, view OTP, view / print notes and invoices
System behaviour	`customerAppService` (cart in `bellomassi_customer_carts` , `placeCustomerOrder` → normal submit with source customer app)
Validation	Cart not empty; delivery address, city, state, 6-digit pincode; requested date (tomorrow onwards in the picker)
Decisions	Delivery address (registered or other), preferred time slot
Approval / Rejection	Handled by the Sales Manager
Status transitions	Same as order statuses; customer labels: Pending approval, Under review (returned), Approved, Rejected
Related modules	Orders, approval, delivery, invoices
Output	Submitted order
Exceptions	Payments: **Not Implemented** (“No payments recorded in this prototype”); cancelling an order: **Not Implemented**
Flowchart	Chapter 09, master flowchart 6

5.18 Fleet master — zones, drivers, vehicles

Heading	Details
Purpose	Delivery zones (pincode groups per hub), driver master (licence, hub, zone, vehicle, availability) and vehicle master (type, capacity, hub, documents).
Roles	View: Administrator, Dispatcher (`fleet.view`); manage: Administrator (`fleet.manage`)
Entry point	`/dispatch/zones` , `/dispatch/drivers` , `/dispatch/vehicles`
Screens	Zone list / form, Driver list / form / details (performance), Vehicle list / form
Preconditions	Hub exists
Steps	Create zone → create vehicle → create driver (link to hub, zone, vehicle, and a user for the driver app)
Actions	Create, edit, delete
System behaviour	`zoneService` , `driverService` , `vehicleService` ; driver availability and vehicle status are recomputed from active dispatches (`syncFleet`)

Heading	Details
Validation	Zone code 2–12 characters, unique, name unique per hub, at least one pincode; driver code 3–12 characters, unique employee number and licence, 10-digit mobile, zone must belong to the driver's hub, vehicle must operate from that hub; vehicle number in Indian format (e.g. TN 01 BX 4521), unique, capacity > 0, type, unit, hub, fuel type, category, ownership
Decisions	Active / Inactive; availability (Available, Assigned, On Delivery, Offline); vehicle status (Available, Assigned, In Transit, Maintenance, Inactive)
Approval / Rejection	—
Status transitions	Vehicle: Available ↔ Assigned ↔ In Transit (derived), Maintenance / Inactive (manual); driver availability derived from dispatches
Related modules	Dispatch (assignment checks), VMS, driver app
Output	Master records
Exceptions	Delete blocked when referenced (`getDriverDeleteBlockers` , `getVehicleDeleteBlockers`)
Flowchart	`Zone → Vehicle → Driver (+ user link) → available for assignment`

5.19 Vehicle Management System (VMS)

Heading	Details
Purpose	Compliance of vehicles and drivers: document expiry tracking and assignment blocking.
Roles	View: Administrator, Dispatcher, QA Manager (`vms.view`); manage vehicles: Administrator (`vms.manage`); upload VMS documents: Administrator, Customer Service, Dispatcher; rules: Administrator (`vms.settings`)
Entry point	`/vms`
Screens	VMS dashboard, Vehicles, Vehicle details / documents, Driver documents, Document expiry register, VMS reports, Compliance rules
Preconditions	Vehicles and drivers exist
Steps	Record expiry dates and upload documents → register shows expiry state → rules decide whether assignment is blocked
Actions	Upload / replace documents, edit expiry dates, change compliance rules, scan expiries
System behaviour	`complianceService.vehicleCompliance / driverCompliance / expiryRegister` ; opening the expiry page runs `automationService.scanDocumentExpiries` (events DOCUMENT_EXPIRING / DOCUMENT_EXPIRED)
Validation	Rule ranges on the settings form
Decisions	Default rules (`DEFAULT_VMS_SETTINGS`): required vehicle documents RC, Insurance, Fitness; required driver document Licence; expired required document **always blocks**; block on warning (within 7 days) off; block on missing off
Approval / Rejection	Document verification (5.20)
Status transitions	Expiry state: Valid → Within 30 days → Within 7 days → Expires today → Expired; Not on record

Heading	Details
Related modules	Dispatch assignment, documents, automation, QA dashboard
Output	Compliance state per vehicle and driver
Exceptions	Assignment blocked with the specific message (e.g. "...'s driving licence expired on ...")
Flowchart	Chapter 11 §11.6

5.20 Documents and OCR

Heading	Details
Purpose	Central repository of business documents (licences, RC, insurance, customer KYC, invoices, etc.) with OCR field capture and review.
Roles	View / upload: most office roles and Driver (upload own); review & reject: Administrator, Customer Service, QA Manager; document types: Administrator
Entry point	`/documents`
Screens	Repository, Upload, OCR queue, OCR review, Document details, Verified, Rejected, Document types / form, Document reports
Preconditions	Active document type; user may upload for the chosen record
Steps	Upload (type, entity, record, file, fields) → Run OCR → review / correct fields → Verify or Reject
Actions	Upload, replace file, run / re-run OCR, save review, verify, reject
System behaviour	`documentService`, `documentStorageService` (file in browser storage), `ocrService` (providers MOCK / BROWSER / BACKEND) → `bellomassi_documents`, `bellomassi_ocr_results`
Validation	"Choose an active document type.", "Choose what this document belongs to.", "Select the related record.", "Choose a file to upload.", "You can only upload documents for your own records."; verify requires OCR or manually saved fields; reject needs a reason
Decisions	Verify or reject
Approval	VERIFIED (locked — replace the file to start a new review)
Rejection	REJECTED with reason for the uploader
Status transitions	PENDING_OCR → OCR_COMPLETED → PENDING_REVIEW → VERIFIED / REJECTED; EXPIRED (derived from expiry date)
Related modules	VMS, customers, drivers, QA dashboard, automation
Output	Verified document with extracted fields
Exceptions	MOCK OCR does not read the image (Simulated); BROWSER provider reports itself unavailable; BACKEND requires the gateway (Not Verified); "The file is not stored in this browser, so it cannot be sent for OCR."
Flowchart	Chapter 13 §13.6

5.21 Invoices and invoice templates

Heading	Details
Purpose	Issue customer invoices from orders using a branded template and numbering scheme.
Roles	View: Administrator, Sales Manager, Finance; issue / cancel: Administrator, Finance (`invoices.manage`); templates: Administrator, Finance
Entry point	`/invoices/dashboard` , `/invoices` , `/invoices/create` , `/invoices/templates`
Screens	Invoice dashboard, Invoice list, New invoice (live preview), Invoice detail, Print invoice, Templates list / form / preview, Invoice reports
Preconditions	Order in an invoiceable status (APPROVED … RETURNED_TO_HUB); no active invoice for the order; an active template
Steps	Chapter 12
Actions	Generate, preview, print, email (demo), cancel, manage templates and numbering
System behaviour	`invoiceService.generateInvoice` snapshots order, customer, hub, delivery note and template into `bellomassi_invoices` ; number from `bellomassi_invoice_settings` (prefix BM-INV, year, 5 digits); event INVOICE_GENERATED
Validation	Eligibility messages; template active; option validation; “The invoice total must be greater than zero (check delivered quantities and discounts).”; cancel requires a reason
Decisions	Quantity basis: ORDERED or DELIVERED (delivered requires a delivery note); discounts and charges
Approval	—
Rejection	Cancellation → CANCELLED (then a new invoice can be issued)
Status transitions	ISSUED → CANCELLED
Related modules	Orders, delivery notes, customers, email, customer app, finance reports
Output	Invoice (printable)
Exceptions	Tax is a configurable **demo** calculation — not GST-compliant e-invoicing; payments, credit notes and refunds are **Not Implemented**
Flowchart	Chapter 12, master flowchart 9

5.22 Email

Heading	Details
Purpose	Compose and track emails (orders, invoices, documents) and read the mailbox.
Roles	View / send: Administrator, Sales roles, Customer Service, Dispatcher, Finance; inbox: Administrator, Sales Manager, Customer Service; templates and settings: Administrator
Entry point	`/communication/email`
Screens	Email home, Compose, Inbox, Sent, Drafts, History, Templates (list / form), Settings
Preconditions	Demo mode (default) or gateway configured
Steps	Compose (to, subject, body, template, attachments) → send → status recorded
Actions	Send, save draft, use template, sync inbox

Heading	Details
System behaviour	`emailService` : demo mode records QUEUED_DEMO → SENT_DEMO (“Sent · demo (not sent)”); backend mode posts to the gateway (SENT = accepted by SMTP, DELIVERED only if the provider reports it). Inbox: seeded demo mailbox; sync may add a simulated message
Validation	Recipient and subject required; settings validated
Decisions	Template or free text
Approval / Rejection	—
Status transitions	DRAFT → QUEUED → SENDING → SENT → DELIVERED / FAILED; demo: QUEUED_DEMO → SENT_DEMO / FAILED_DEMO
Related modules	Automation, invoices, documents
Output	Email history (`bellomassi_email_history`)
Exceptions	**Simulated** in demo mode; gateway mode **Not Verified**
Flowchart	Chapter 13

5.23 WhatsApp (OpenWA)

Heading	Details
Purpose	Send WhatsApp messages to customers and drivers.
Roles	View / send: Administrator, Sales roles, Customer Service, Dispatcher, Finance; templates / settings: Administrator
Entry point	`/communication/whatsapp`
Screens	WhatsApp home, Compose, History, Templates, OpenWA settings (session, QR)
Preconditions	Mock session (demo) or OpenWA session `ready` through the gateway
Steps	Compose → send → history
Actions	Send, use template, start / stop session (settings)
System behaviour	`openWAService` / `whatsappMessageService` : Mock provider records “Sent · demo”; gateway proxies OpenWA REST (201 = handed to WhatsApp, not delivered)
Validation	Recipient number with country code (default 91); message body
Decisions / Approval / Rejection	—
Status transitions	As email (demo statuses)
Related modules	Automation, call history
Output	`bellomassi_whatsapp_messages`
Exceptions	**Simulated**; live **Not Verified**
Flowchart	Chapter 13

5.24 Automated notifications

Heading	Details
Purpose	Send email / WhatsApp automatically when workflow events happen.
Roles	Administrator (`communication.automation`)
Entry point	`/communication/automation`
Screens	Automated notifications (rules per event, master switch, history)
Preconditions	Automation switched on; an enabled rule with a template for the event
Steps	Event fires in a service → rule found → recipients resolved → messages sent → history logged
Actions	Enable / disable automation, edit rule (channel, recipient rule, templates)
System behaviour	`automationService.fireEvent` with de-duplication; events ORDER_CREATED, ORDER_APPROVED, ORDER_REJECTED, ORDER_READY, DISPATCH_ASSIGNED, DELIVERY_STARTED, DELIVERY_COMPLETED, DELIVERY_FAILED, INVOICE_GENERATED, DOCUMENT_EXPIRING, DOCUMENT_EXPIRED; channels Email, WhatsApp, Both; recipients Customer contact, Order salesman, Operations team (Admin / Dispatcher), Document owner
Validation	“Only the Admin can switch automation.”
Decisions	Skipped when switched off, no enabled rule, missing template, or missing address
Approval / Rejection	—
Status transitions	History status: SKIPPED / SENT_DEMO / SENT / PARTIAL / FAILED
Related modules	Email, WhatsApp, every workflow service
Output	`bellomassi_notification_history`
Exceptions	Messages go through the simulated channels in demo mode
Flowchart	Chapter 13, master flowchart 10

5.25 Call history

Heading	Details
Purpose	Manual log of customer calls with outcomes and follow-ups.
Roles	View: Administrator, Sales Executive (own), Sales Manager, Customer Service (`communication.calls`); manage: same (`calls.manage`)
Entry point	`/communication/calls`
Screens	Call dashboard, All calls, Log a call, Edit call, Call details, Follow-ups
Preconditions	Customer exists
Steps	Log call (direction, status, outcome, duration, notes) → schedule follow-up → complete follow-up → link an order
Actions	Create, edit, delete, add note, schedule / complete follow-up, link order
System behaviour	`callService` → `bellomassi_call_history` ; salesmen see only their calls

Heading	Details
Validation	“Choose the follow-up date.”, “Follow-up cannot be before the call.”, note ≤ 500 characters
Decisions	Directions: Incoming, Outgoing; statuses: Connected, Missed, Busy, No Answer, Voicemail; outcomes: Order Received, Interested, Follow-up Required, No Response, Busy, Not Interested, Complaint, Payment Follow-up, Other
Approval / Rejection	—
Status transitions	Follow-up: open → completed (buckets overdue / today / upcoming)
Related modules	Orders, customers, communication reports
Output	Call record
Exceptions	No telephony — nothing is dialled (**Simulated / manual**)
Flowchart	Chapter 13

5.26 Notification center

Heading	Details
Purpose	In-app inbox of notifications for the signed-in user.
Roles	All
Entry point	`/notifications`; bell icon in the header
Screens	Notification center (filter by category, read / unread), preferences
Preconditions	—
Steps	Open → read → follow link
Actions	Mark read, mark all read, mute categories, toggle toasts
System behaviour	`notificationService` (`bellomassi_notifications`, preferences in `bellomassi_notification_preferences`); notifications target roles, users or a driver / customer
Validation / Decisions / Approval / Rejection	—
Status transitions	Unread → Read (per user)
Related modules	All workflow services
Output	—
Exceptions	Browser push notifications are not used (in-app only)
Flowchart	`Service event → pushNotification → bell / Notification center → link to record`

5.27 Reports Center

Heading	Details
Purpose	One catalogue of 118 reports in 11 categories with filters, export and favourites.
Roles	Office roles with `reports.center`; categories filtered by permission

Heading	Details
Entry point	`/reports/center` → `/reports/view/:id`
Screens	Reports Center, Report viewer
Steps	Chapter 14
Preconditions	Category permission
Actions	Search, filter, favourite, run, export CSV, print
System behaviour	`reportService.runReport` builds rows from LocalStorage with the user's data scope; favourites / recent in `bellomassi_report_prefs`
Validation	—
Decisions	—
Approval / Rejection	—
Status transitions	—
Related modules	All
Output	Tables, CSV
Exceptions	24 reports show "no source data" with the reason (payments, credit notes, expenses, batches, cycle counts, fuel, supplier invoices, returns)
Flowchart	Chapter 14, master flowchart 11

5.28 AI Assistant and Jarvis

Heading	Details
Purpose	Ask business questions in natural language (typed or voice) and get answers built from the user's own data.
Roles	View / use: most office roles (`ai.assistant.*` , `jarvis.*`); settings: Administrator
Entry point	`/ai` , `/ai/assistant` , `/ai/jarvis` , `/ai/activity` , `/ai/settings`
Screens	AI overview, Assistant (conversations), Jarvis (voice / wake word), AI activity, AI settings
Preconditions	—
Steps	Ask → intent engine picks a permission-checked tool → result blocks (KPIs, table, chart) → write actions shown as a confirmation card → confirm or cancel
Actions	Ask, confirm / cancel proposal, rename / delete conversation, clear Jarvis history
System behaviour	`aiChatService` , `aiIntentService` , `aiToolRegistry` ; conversations private per user (`bellomassi_ai_conversations` , `bellomassi_ai_messages` , `bellomassi_jarvis_history`); activity in `bellomassi_ai_activity`
Validation	Tools respect the user's permissions and data scope
Decisions	Confirm a proposed write action
Approval / Rejection	User confirmation of proposals
Status transitions	—

Heading	Details
Related modules	All data modules
Output	Answers, executed actions
Exceptions	Provider DEMO is the default (**Simulated**); OpenJarvis, Ollama, Gemini, OpenAI-compatible and custom providers are configurable but **Not Verified**
Flowchart	Chapter 14

5.29 Forecasting, predictive inventory and model management

Heading	Details
Purpose	Demand forecasts and reorder recommendations computed in the browser.
Roles	View: Administrator, Sales Manager, Dispatcher, Finance, Warehouse Manager; generate: Administrator, Sales Manager; planning parameters: Administrator
Entry point	`/ai/forecasting`, `/ai/predictive-inventory`, `/ai/models`
Screens	Forecast dashboard, New forecast, History, Compare, Forecast detail / print; Predictive dashboard, Reorder recommendations, Risk analysis, Demand forecasting, Planning parameters, Prediction detail; Model management
Preconditions	Sales history (`bellomassi_sales_history`)
Steps	Choose target, scope and horizon → generate → review accuracy → export CSV
Actions	Generate, delete, compare, export, enable / disable models
System behaviour	`forecastService`, `services/ml/*` — models Auto (best backtest), Moving average, Weighted moving average, Exponential smoothing, Holt, Seasonal
Validation	Permission checks (“Only an Administrator can change inventory planning parameters.”)
Decisions	Model choice
Approval / Rejection	—
Status transitions	Recommendation: OPEN → IN_DRAFT (added to replenishment) / DISMISSED
Related modules	Replenishment, inventory
Output	`bellomassi_forecasts`, `bellomassi_inventory_predictions`, `bellomassi_reorder_recommendations`
Exceptions	Backend model registration requires a gateway (**Not Verified**)
Flowchart	Chapter 10 §10.8

5.30 Replenishment

Heading	Details
Purpose	Purchase requests to suppliers, from draft to goods receipt.

Heading	Details
Roles	Create: Administrator, Dispatcher, Warehouse Manager, Inventory Executive; approve / reject / convert: Administrator (`procurement.approve` , `procurement.reject`); view: roles with `procurement.view` ; Supplier portal (read-only)
Entry point	`/ai/replenishment`
Screens	Replenishment list, New draft, Draft detail
Preconditions	Hub, products, supplier
Steps	Draft → submit → approve or reject → convert (“Awaiting goods receipt”) → receive via Stock Receipt links → Received
Actions	Create, edit (creator / admin), submit, approve, reject, convert, cancel, open receipt link
System behaviour	`replenishmentService` → `bellomassi_replenishment_drafts` ; RECEIVED is derived when stock receipts referencing the draft cover all lines
Validation	Field validation; “A reason is required.” to reject
Decisions	Approve / reject
Approval	APPROVED → CONVERTED
Rejection	REJECTED
Status transitions	DRAFT → SUBMITTED → APPROVED → CONVERTED → RECEIVED; SUBMITTED → REJECTED; DRAFT / SUBMITTED / APPROVED → CANCELLED
Related modules	Predictive inventory, stock receipts, supplier portal
Output	Purchase request; stock receipts
Exceptions	Supplier invoices, purchase returns and supplier payments **Not Implemented**
Flowchart	Chapter 07 §7.4

5.31 AI insights

Heading	Details
Purpose	Automatically generated observations (stock-out risk, demand spikes, delivery issues).
Roles	View: Administrator, Sales Manager, Customer Service, Dispatcher, Finance, Warehouse Manager, QA Manager; manage: Administrator, Sales Manager, Dispatcher
Entry point	`/ai/insights`
Screens	AI insights
Steps	Refresh → review → mark Reviewed / Resolved
Actions	Refresh, change status
System behaviour	`insightService.refreshInsights` (rule-based `ml/insightsEngine`) → `bellomassi_ai_insights`
Preconditions / Validation / Decisions / Approval / Rejection	—
Status transitions	NEW → REVIEWED → RESOLVED

Heading	Details
Related modules	Dashboards (OpsOverview), inventory, dispatch
Output	Insight records
Exceptions	Rule-based, not a trained model
Flowchart	`Refresh → NEW → REVIEWED → RESOLVED`

5.32 Users, roles and permissions

Heading	Details
Purpose	Who can sign in, with which role(s), hub and data scope.
Roles	Super Admin, Administrator (`users.*` , `roles.*`)
Entry point	`/admin/users` , `/admin/roles` , `/admin/permissions` , `/admin/role-assignment`
Screens	User list / form / details, Role list / form / details, Permission matrix, User role assignment
Preconditions	—
Steps	Create user (role, hub, customer / driver link) → user signs in; create custom role → assign permissions → assign users
Actions	Create, edit, activate / deactivate, delete, reset password, unlock, assign roles, assign hub; create / edit / clone / delete roles
System behaviour	`userService` , `rbacService` → `bellomassi_demo_users` , `bellomassi_roles` ; audit entries
Validation	Passwords match; “You cannot deactivate your own account.”; “Only a Super Admin can change a Super Admin account.”; “Drivers need a home hub.”; “This change would remove your own access to user management.”; role landing page required
Decisions	Role, data scope, landing page
Approval / Rejection	—
Status transitions	User active ↔ inactive; locked → unlocked
Related modules	Authentication, every module
Output	Users, roles
Exceptions	Protected / locked system roles cannot be deleted
Flowchart	`Create role → set permissions → create user → assign role and hub → user signs in`

5.33 Audit log

Heading	Details
Purpose	History of administrative and security events (who changed what, when).
Roles	Administrator, Super Admin (`audit.view`)

Heading	Details
Entry point	`/admin/audit-logs`
Screens	Audit log (filters, details)
System behaviour	`auditService.logAudit` → `bellomassi_audit_logs` (user, role, settings, account lock, etc.)
Preconditions / Steps / Actions	View and filter
Validation / Decisions / Approval / Rejection / Status transitions	—
Related modules	Users, roles, settings, authentication
Output	Audit trail
Exceptions	Business record history is in the order and dispatch timelines, not in the audit log
Flowchart	`Admin action → logAudit → Audit log`

5.34 Application settings

Heading	Details
Purpose	Company-wide configuration.
Roles	View: Administrator (`settings.view`); change: Administrator (`settings.manage`)
Entry point	`/admin/settings`
Screens	Tabs including Localization / numbering, Security (session timeout 60, failed attempts 5, lockout 15, password length 8, mixed password, reset code 15 min, force change after reset), Approval (require remarks, block when credit exceeded, rejection remarks minimum 5), Inventory defaults, Warehouse (FEFO, putaway, QA hold), Cold Chain ranges
Steps	Edit tab → save
System behaviour	`appSettingsService` → `bellomassi_settings` ; values feed `runtimeConfig`
Validation	Ranges (e.g. session timeout 5–1440, failed attempts 3–20, lockout 1–1440)
Preconditions / Decisions / Approval / Rejection / Status transitions	—
Related modules	Authentication, approval, inventory, QA
Output	Settings
Exceptions	Warehouse picking strategy and QA hold are stored but not enforced (**UI Only**)
Flowchart	`Settings tab → validate → save → used by services`

5.35 Organization

Heading	Details
Purpose	Company profile and organisation structure.
Roles	View: Administrator (`organization.view`); manage: Administrator

Heading	Details
Entry point	`/admin/organization/company`
Screens	Company, Branches, Business units, Departments, Designations, Employee master, Hub master, Warehouse master
System behaviour	`organizationService` → `bellomassi_company`, `bellomassi_organization`
Steps / Actions	Create, edit records
Preconditions / Validation / Decisions / Approval / Rejection / Status transitions	Standard form validation
Related modules	Users (employee master), invoices (company details), hubs
Output	Organisation records
Exceptions	—
Flowchart	`Company → Branches → Business units → Departments → Designations → Employees`

5.36 My account

Heading	Details
Purpose	The user’s own profile, password, notification and display preferences.
Roles	All
Entry point	`/profile`, `/account/change-password`, `/settings`, `/driver/profile`, `/customer/profile`
Screens	My profile, Change password, Preferences (theme light / dark, notifications)
System behaviour	`authService.updateUserProfile`, `changeDemoPassword`; theme in `bellomassi_theme`
Validation	Name 2–60 characters; phone format; password policy
Steps / Actions	Edit and save
Preconditions / Decisions / Approval / Rejection / Status transitions	—
Related modules	Authentication
Output	Updated profile
Exceptions	—
Flowchart	`Profile → edit → validate → save`

06 · Order-to-Delivery Workflow

Status: Implemented end to end. GPS tracking and the delivery OTP are **Simulated** (no GPS device, no SMS).

6.1 Business summary

1. An order is raised by a **Sales Executive** (`/orders/create`) or by the **customer** (`/customer/cart`).
2. The **Sales Manager** approves it (or returns / rejects it).
3. **Customer Service** consolidates it after a five-point verification and marks it ready for dispatch.
4. The **Dispatcher** puts it on a dispatch with other orders for the same hub and date, assigns a compliant driver and vehicle, and confirms the dispatch. Stock leaves the hub at this moment.
5. The **Driver** delivers it in sequence: start → reach location → OTP → quantities and proof → complete (or fail).
6. A **delivery note** is generated; **Finance** issues the **invoice**; the **customer** sees both in the customer app.

6.2 Master flowchart 3 — Order-to-delivery

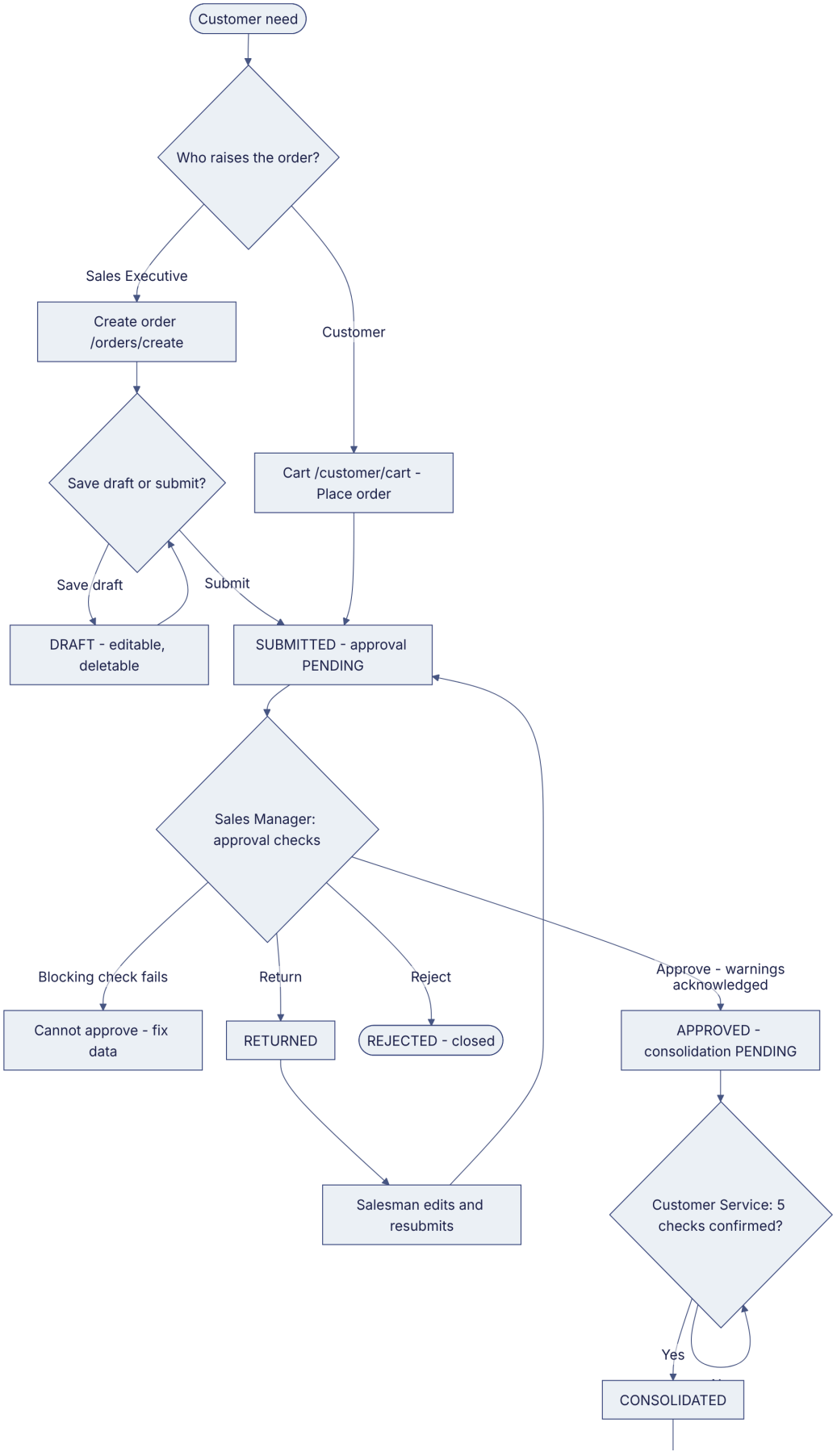


Diagram continued (1 of 3)

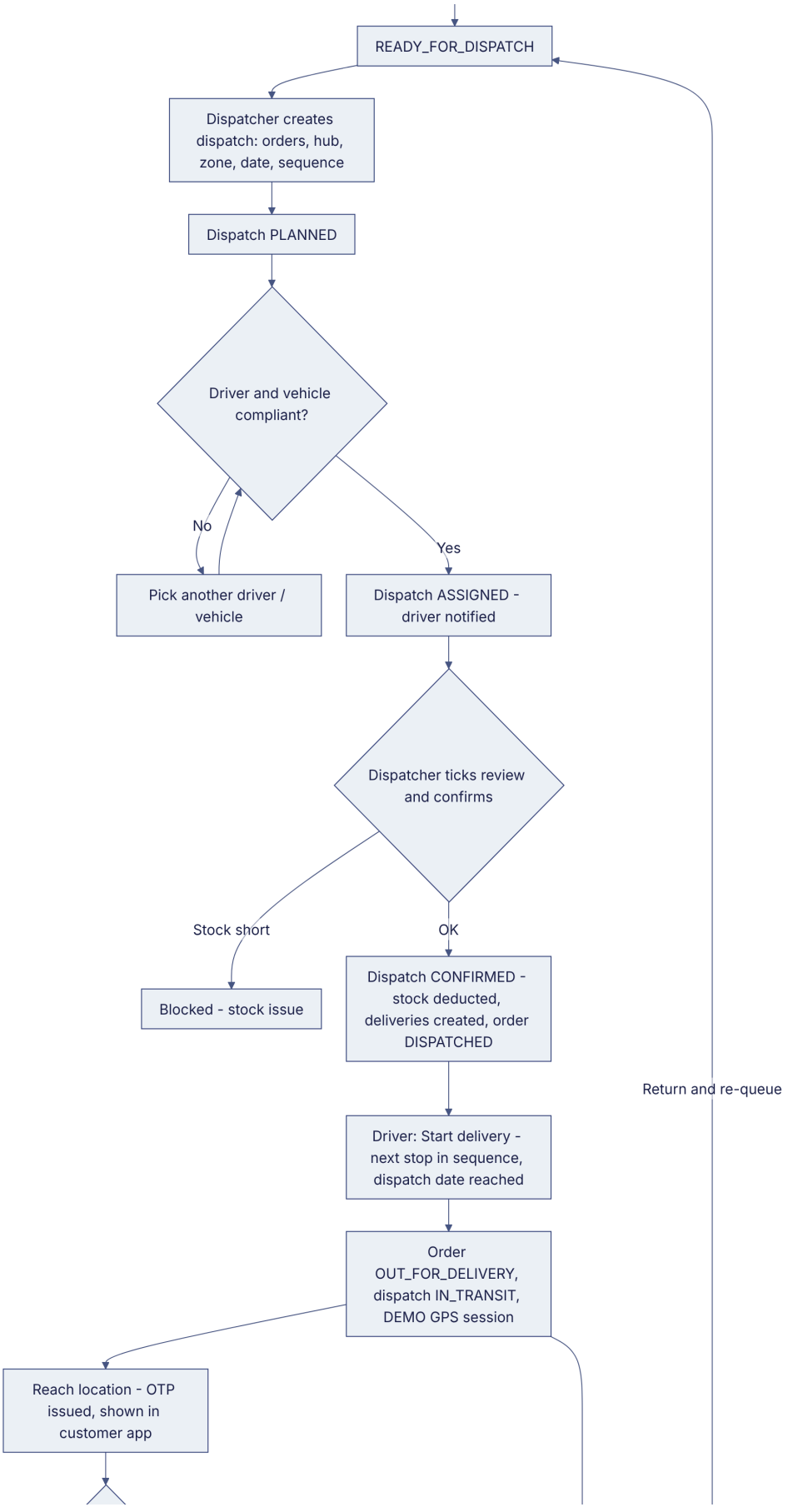


Diagram continued (2 of 3)

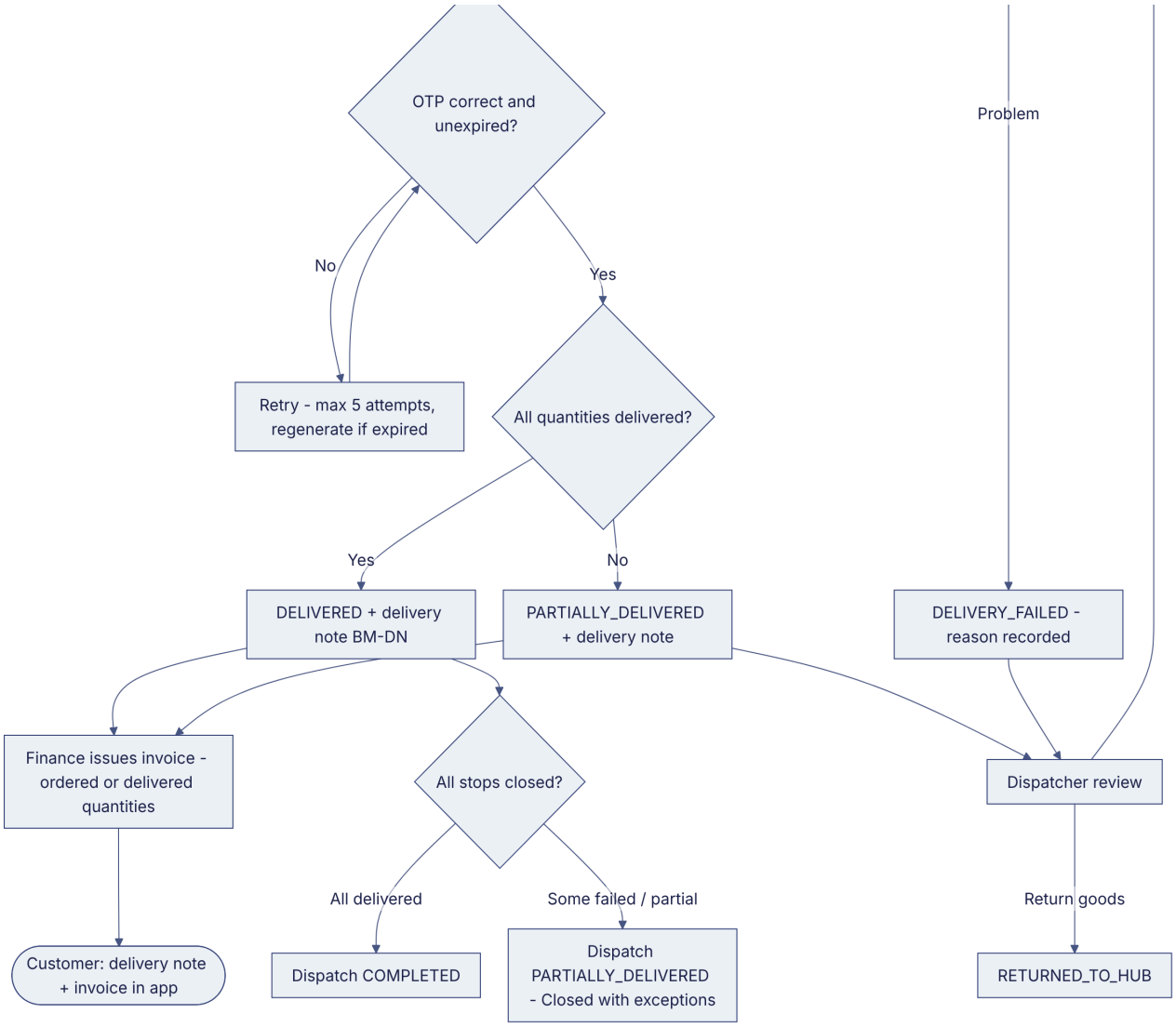
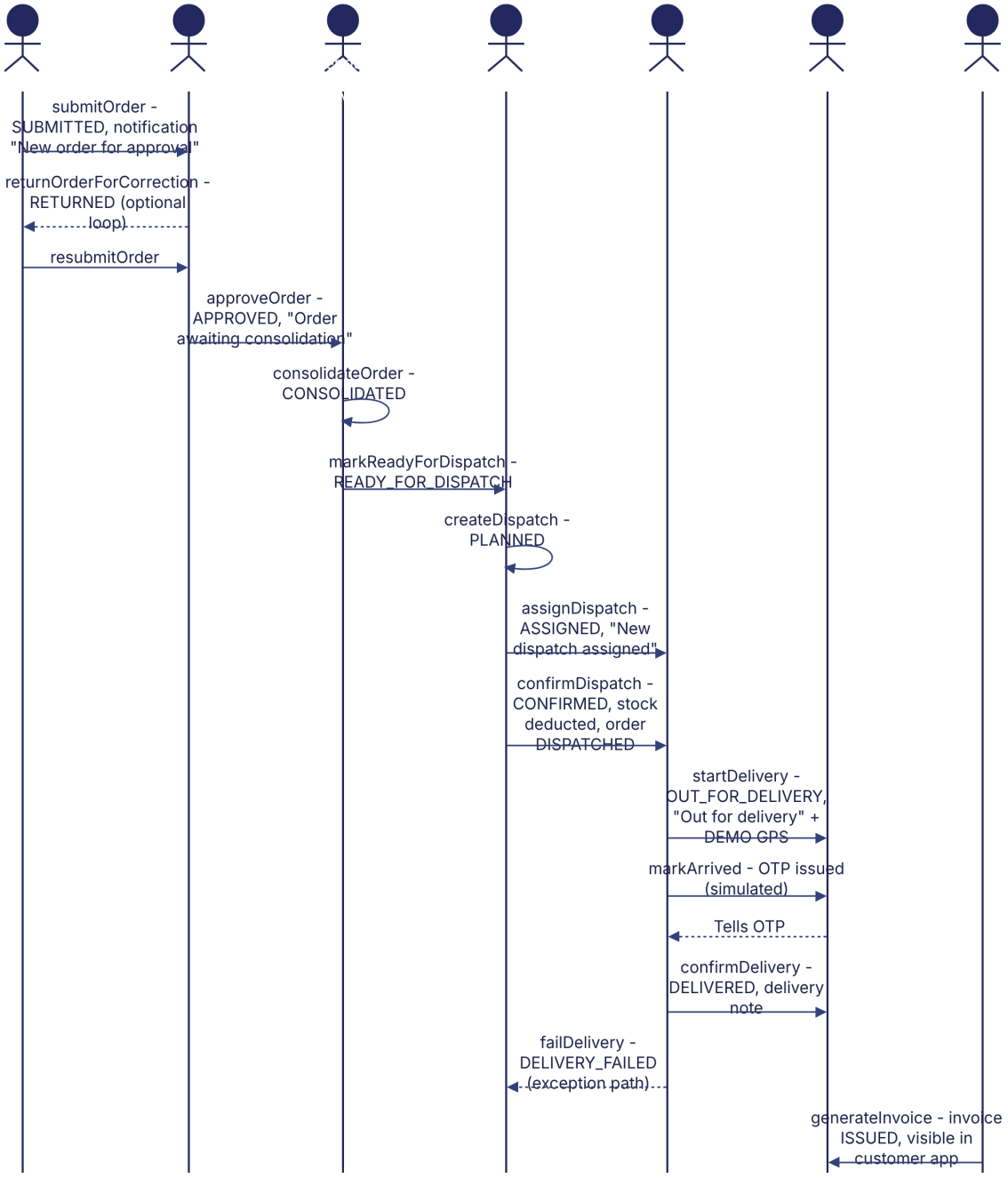


Diagram continued (3 of 3)

6.3 Swimlane — order lifecycle



6.4 Step reference

Step	Screen / route	Component	Service function	Permission	LocalStorage written
Create / draft	`/orders/create`, `/orders/:id/edit`	`orderForm`	`createOrder`, `updateOrder`	`orders.create`	`bellomassi_orders`
Submit	same	`orderForm`	`submitOrder`	`orders.create`	`orders`, `order_sequence`, `order_timeline`, `notifications`, `activities`

Step	Screen / route	Component	Service function	Permission	LocalStorage written
Customer order	`/customer/cart`	`CustomerCart`	`placeCustomerOrder` → submit	`customer.order`	as above + `customer_carts` cleared
Approve / return / reject	`/orders/approval/:id`	`approvalDetail`	`approveOrder`, `returnOrderForCorrection`, `rejectOrder`	`orders.approve`, `orders.reject`	`orders`, `approvals`, `order_timeline`
Resubmit	`/orders/:id/edit`	`orderForm`	`resubmitOrder`	`orders.create`	`orders`, `order_timeline`
Consolidate	`/orders/consolidation/:id`	`consolidationDetail`	`consolidateOrder`	`orders.consolidate`	`orders`, `consolidations`, `order_timeline`
Ready for dispatch	same	same	`markReadyForDispatch`	`orders.consolidate`	`orders`, `order_timeline`
Plan dispatch	`/dispatch/create`	`dispatchForm`	`createDispatch`	`dispatch.manage`	`dispatches`, `dispatch_sequence`, `dispatch_timeline`, `orders` (dispatchId)
Assign	`/dispatch/assignments`, `/dispatch/:id`	`dispatchAssignments`, `dispatchDetail`	`assignDispatch`	`dispatch.manage`	`dispatches`, `dispatch_timeline`
Confirm	`/dispatch/:id`	`dispatchDetail`	`confirmDispatch`	`dispatch.manage`	`dispatches`, `deliveries`, `inventory`, `stock_movements`, `orders`, timelines
Start	`/driver/deliveries/:id`	`DriverDeliveryDetail`	`startDelivery`	`driver.app`	`deliveries`, `dispatches`, `orders`, `driver_tracking`
Arrive	same	same	`markArrived`	`driver.app`	`deliveries`, `delivery_otps`
Complete	`/driver/deliveries/:id/confirm`	`DriverConfirmDelivery`	`confirmDelivery`	`driver.app`	`deliveries`, `delivery_notes`, `delivery_pod`, `orders`, `dispatches`, `delivery_otps`
Fail	`/driver/deliveries/:id`	`FailDeliveryDialog`	`failDelivery`	`driver.app`	`deliveries`, `orders`, `dispatches`
Review exception	`/dispatch/deliveries/:id`	`deliveryDetail`	`reviewDelivery`	`dispatch.manage`	`deliveries`, `orders`, `inventory`, `stock_movements`
Invoice	`/invoices/create`	`invoiceCreate`	`generateInvoice`	`invoices.manage`	`invoices`, `invoice_settings`

6.5 Stock effect along the flow

Moment	Stock effect	Movement type
Submit / approve	None — a **warning** is shown if hub stock is short	—
Consolidate	None — stock review is information only	—
Confirm dispatch	Hub stock reduced for every line, exactly once (`stockDeducted`)	Stock Dispatch
Cancel a confirmed dispatch	Stock restored once (`stockReversed`)	Stock Dispatch Reversal
Partial / failed delivery reviewed with “return to hub”	Undelivered (not damaged) quantity booked back once (`returnedToHub`)	Delivery Return

6.6 Rules that stop the flow

Where	Rule	Message (from code)
Submit	Required data	“Select a customer.”, “Add at least one product.”, “Delivery date cannot be before the order date.” …
Approval	Blocking checks	“Cannot approve: …” (customer inactive, hub inactive, products inactive, total invalid, credit when blocking enabled)
Approval	Warnings	“Confirm that you have reviewed the credit and stock warnings.”
Consolidation	Checklist	“Complete the verification checklist first.”
Dispatch planning	Eligibility	“Only orders ready for dispatch can be dispatched.”, “…is already on BM-DSP-…”, “…has no complete delivery address.”
Assignment	Compliance	Driver / vehicle issues (chapter 11)
Confirmation	Review	“Review the orders, driver, vehicle, route and stock, then tick the confirmation.”
Start	Sequence / date / confirmation	“Stop N (…) comes first in the sequence.”, “Waiting for the Dispatcher to confirm the dispatch.”, “Planned for … — Start is available on that day.”, “Finish your current delivery first.”
Complete	OTP	Wrong / expired / locked OTP blocks completion
Complete	State	“Start the delivery before confirming it.”, “…is already delivered. It was not confirmed again.”
Invoice	Eligibility	“Draft orders cannot be invoiced.”, “…already has invoice … Cancel it first to issue a new one.”

07 · Approval Workflow

Bellomassi has four approval processes in which one person decides on another person's request, plus two **self-confirmation** steps (dispatch confirmation and delivery exception review) that work like approvals.

Approval	Requester	Approver	Service	Status
Order approval	Sales Executive, Customer (app), Administrator	Sales Manager (Super Admin holds all keys; Administrator views only)	`approvalService`	Implemented
Scrap approval	Administrator, Dispatcher, Warehouse Manager, Inventory Executive, QA Manager	Warehouse Manager, Administrator (never the requester)	`scrapService`	Implemented
Replenishment approval	Administrator, Dispatcher, Warehouse Manager, Inventory Executive	Administrator (`procurement.approve`)	`services/ai/replenishmentService`	Implemented
Document verification	Any uploader (incl. Driver)	Administrator, Customer Service, QA Manager	`documentService`	Implemented
Dispatch confirmation	Dispatcher	Dispatcher (review tick)	`dispatchService.confirmDispatch`	Implemented
Delivery exception review	Driver (reports failure / partial)	Dispatcher, Administrator	`deliveryService.reviewDelivery`	Implemented

7.1 Master flowchart 4 — Order approval

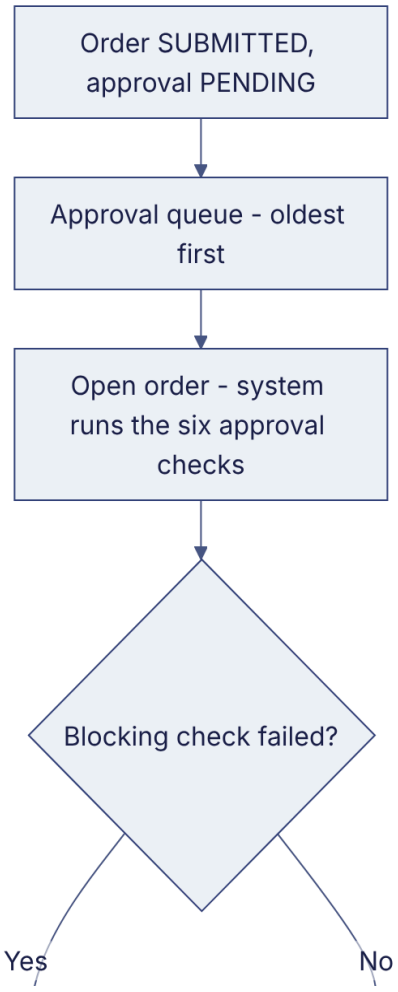


Diagram continued (1 of 3)

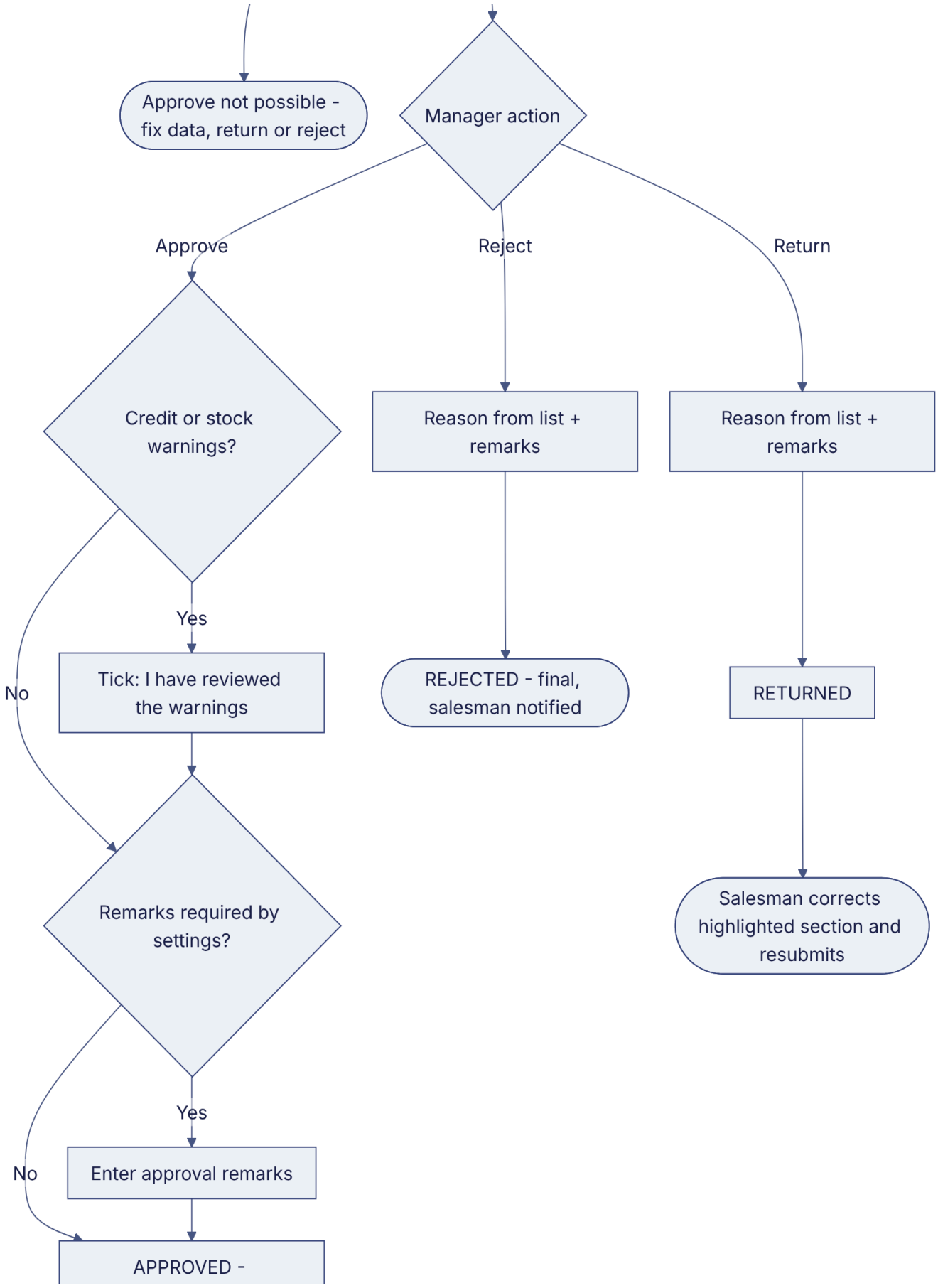


Diagram continued (2 of 3)

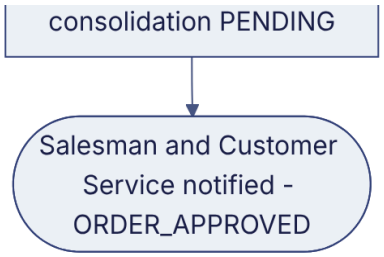


Diagram continued (3 of 3)

Approval checks (`approvalService.getApprovalChecks`)

Check	Severity	Passes when
Customer active	Block	Customer exists and status is Active
Hub active	Block	Hub exists and status is Active
Credit within limit	**Warn** (Block if Application Settings › Approval › *Block when credit exceeded* is on)	Credit enabled and the order does not take the balance over the limit
Stock available	Warn	Every inventory-tracked line is covered by hub stock
Order total valid	Block	Total > 0 and at least one line
Products active	Block	Every product is Active

Reasons

- **Rejection reasons** (`REJECTION_REASONS`): Credit limit exceeded, Duplicate order, Customer cancelled, Pricing not approved, Stock not available, Other.
- **Return reasons** (`RETURN_REASONS`): Quantity needs confirmation, Pricing or discount correction, Reduce order value (credit), Delivery details incomplete, Wrong product selected, Other. Each return reason highlights the matching section on the salesman's edit screen (`RETURN_REASON_SECTIONS`).

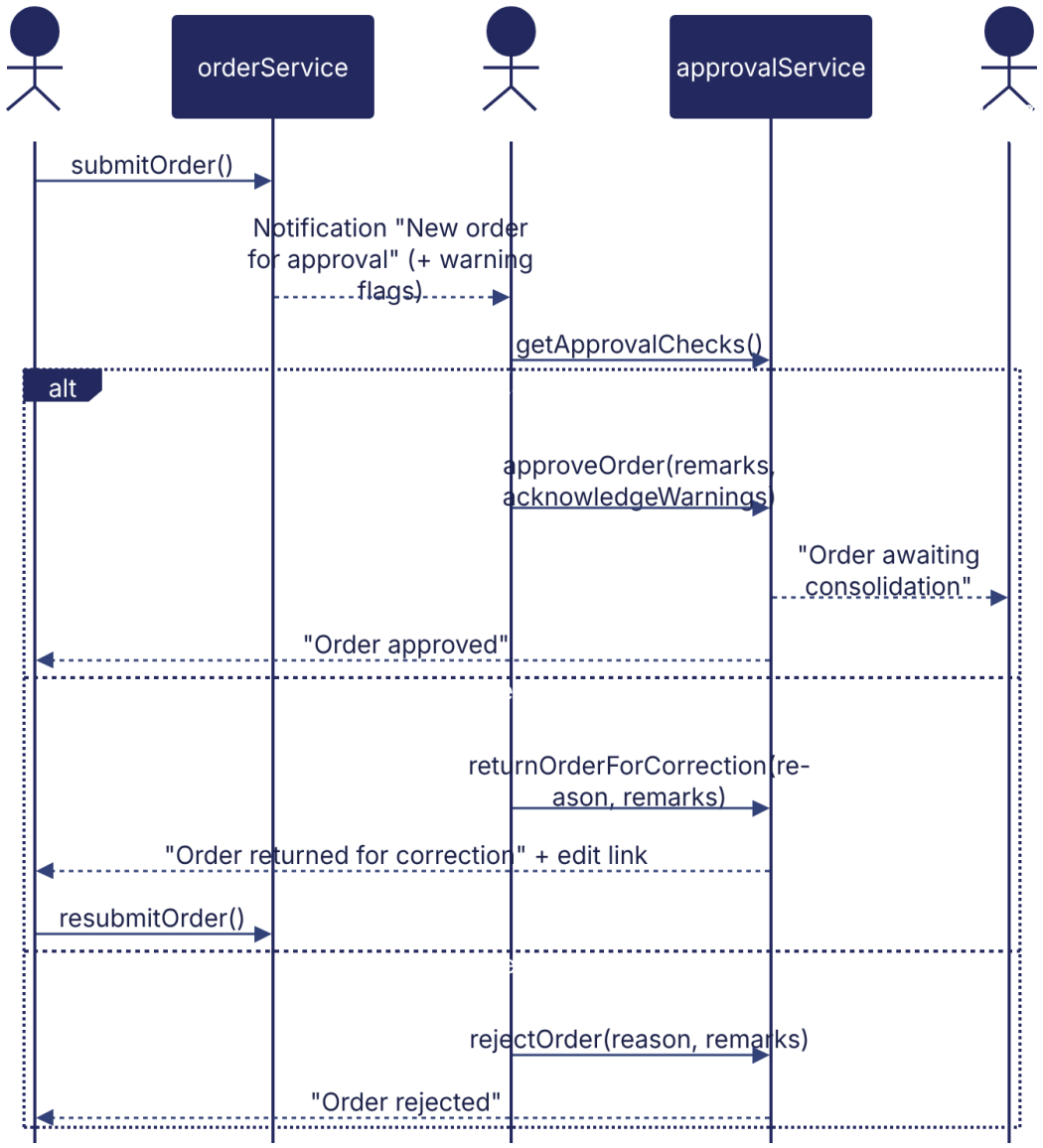
What the system records

One transaction writes the updated order, an approval record in `bellomassi_approvals` (decision, reason, remarks, decided by, order total, credit and stock warning flags) and a timeline event in `bellomassi_order_timeline`.

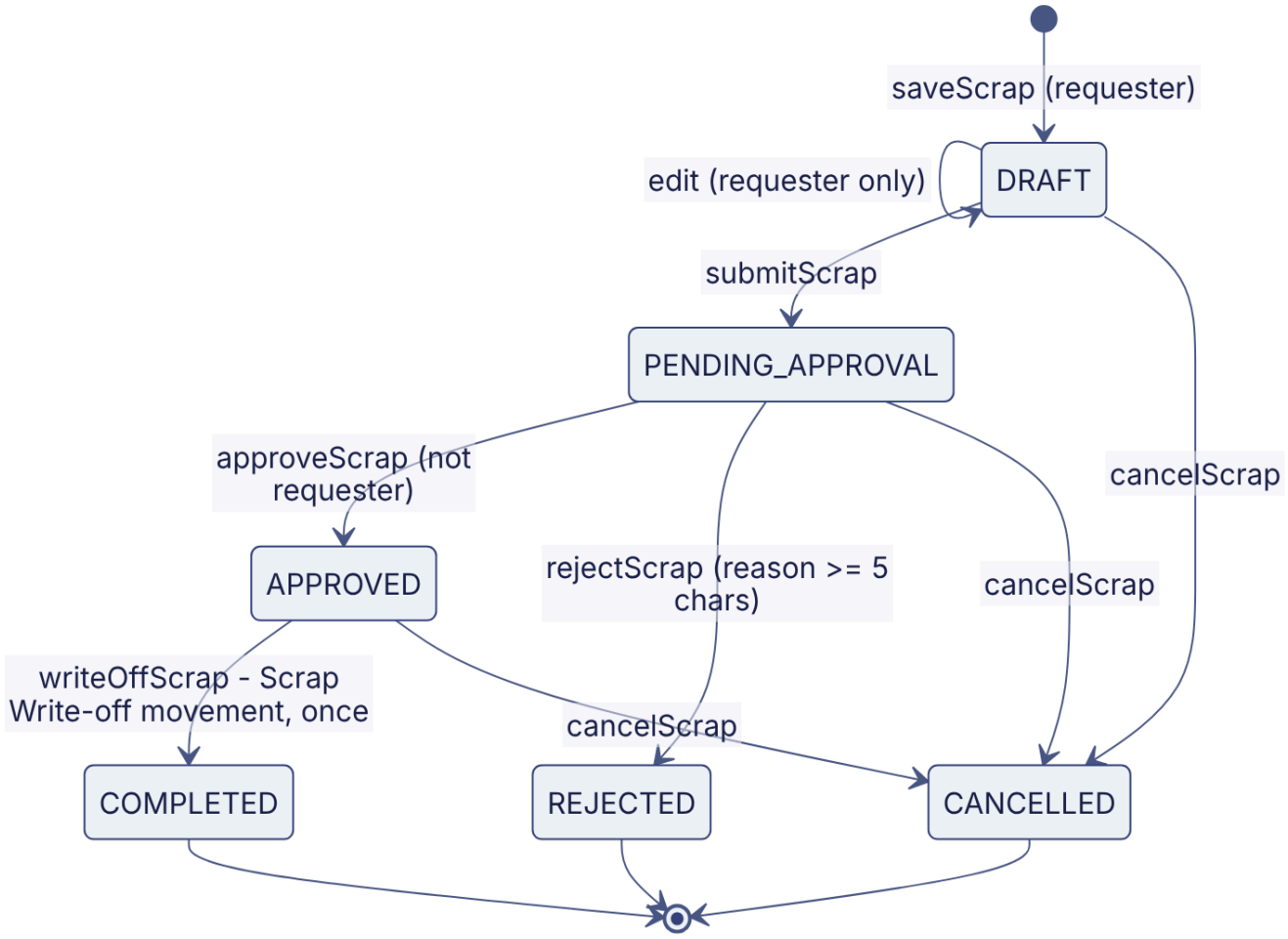
Edge cases

- A draft cannot be decided: "Draft orders cannot be decided — the salesman has not submitted it."
- A second decision is refused: "BM-ORD-… has already been approved. No change was made."
- Only `orders.approve` may approve; only `orders.reject` may return or reject.

7.2 Order approval — swimlane



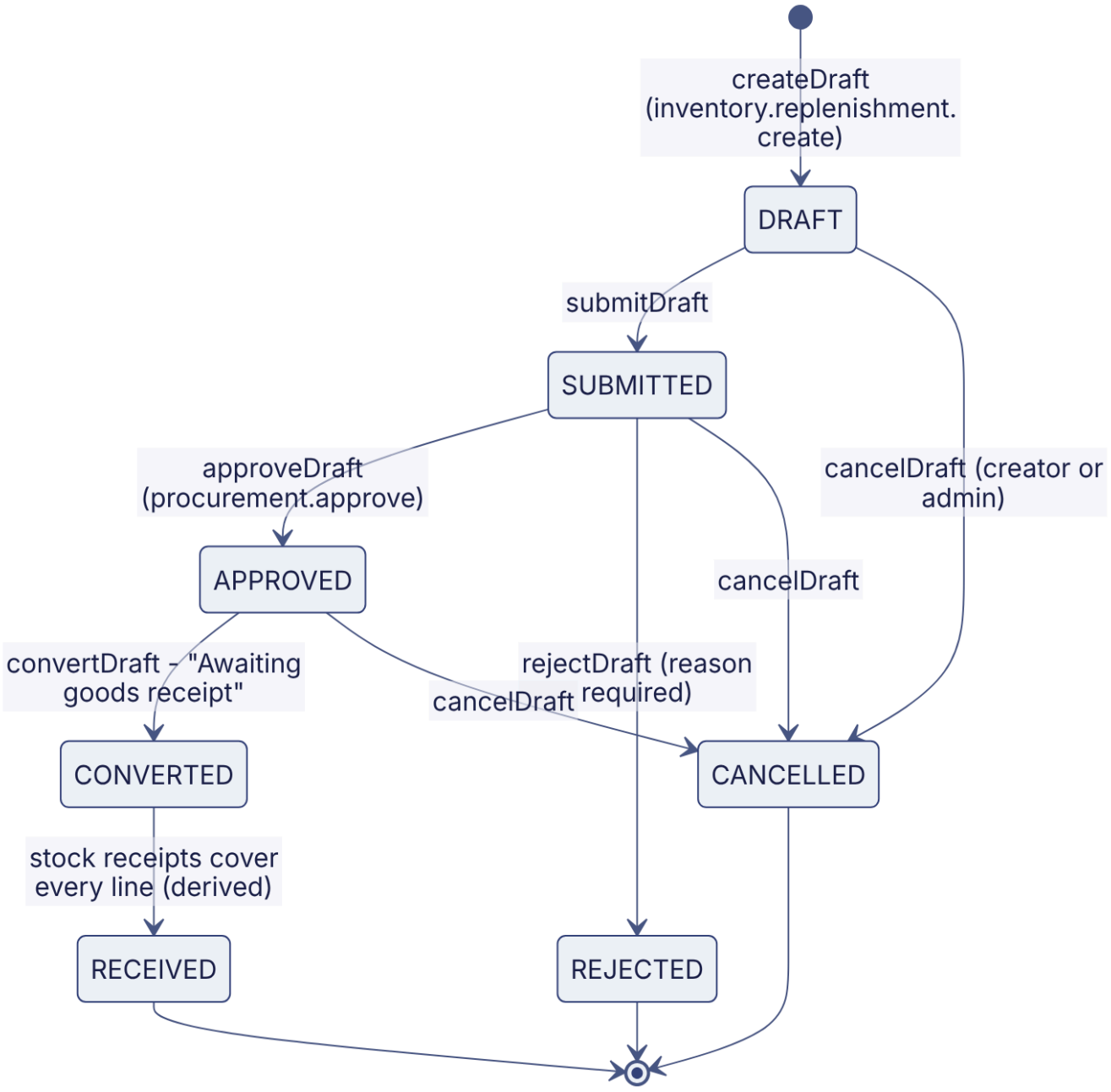
7.3 Scrap approval



Rule	Message
Requester cannot approve own request	“You cannot approve your own scrap request.”
Only drafts can be edited, only by the requester	“Only draft requests can be edited.”, “Only the requester can edit this draft.”
Rejection needs a reason	“Enter a rejection reason (at least 5 characters).”
Write-off only after approval, by an approver	“Only an approved scrap request can create a stock write-off.”, “Only an approver can post a stock write-off.”
Stock cannot go negative	“The write-off would make stock negative.”

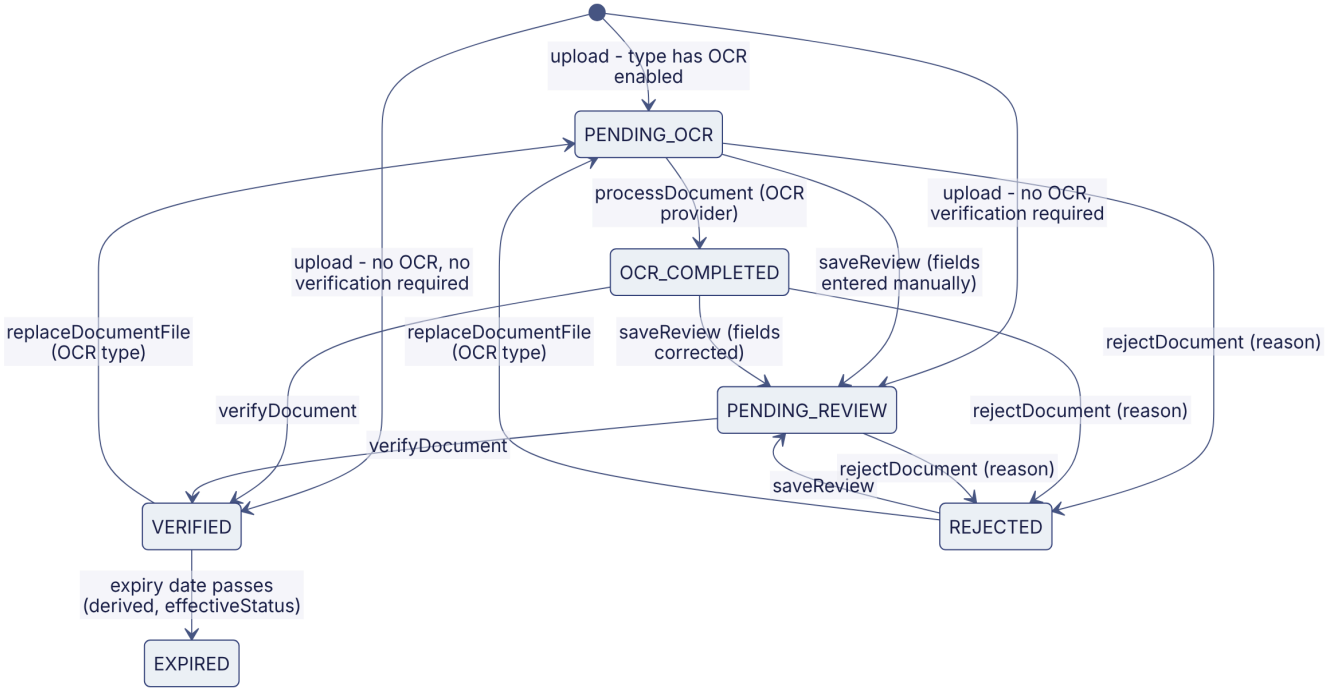
Screens: `/inventory/scrap` (dashboard), `/inventory/scrap/requests`, `/inventory/scrap/requests/:id`, print `/inventory/scrap/requests/:id/print`. Storage: `bellomassi_scrap_requests`, write-off in `bellomassi_stock_movements`.

7.4 Replenishment approval



- Only the creator or an Administrator can edit or cancel a draft.
- After conversion, each line has a link to **New Stock Receipt** pre-filled with hub, product, outstanding quantity and the request number as reference (`receiptLink`). When receipts referencing the request cover all lines, the status shows **Received**.
- The Supplier portal shows approved / converted requests for the linked supplier (read-only).

7.5 Document verification



The initial status depends on the document type (`ocrEnabled`, `verificationRequired`). Replacing the file restarts the review (PENDING_OCR for OCR types, otherwise PENDING_REVIEW). “Verified documents are locked. Replace the file to start a new review.” Verifying an OCR-type document needs OCR or manually saved fields first (“Run OCR (or enter the fields manually and save) before verifying.”). Rejection requires a reason “so the uploader knows what to fix.” EXPIRED is not stored — it is derived from the expiry date for any non-rejected document.

7.6 Dispatch confirmation and exception review

These are covered in chapters 11 and 15. Both require an explicit tick or remarks from the Dispatcher and are refused when repeated (“…is already confirmed. Stock was deducted once and was not deducted again.”, “…was already reviewed… No change was made.”).

7.7 Approval responsibilities

Approval	Approve	Return	Reject	Who is notified
Order	Sales Manager	Sales Manager	Sales Manager	Salesman; Customer Service on approval
Scrap	Warehouse Manager / Administrator	—	Warehouse Manager / Administrator	(in-app list / dashboard)
Replenishment	Administrator	—	Administrator	(in-app list); Supplier portal shows approved requests
Document	Administrator / Customer Service / QA Manager	—	same	Uploader sees status and reason

08 · Driver Workflow

Status: Implemented. GPS tracking is **Simulated** (DEMO GPS badge), the delivery OTP is **Simulated** (no SMS — the customer reads the code in the customer app). Photo and signature proof of delivery are **Implemented** and stored in the browser (`bellomassi_delivery_pod`, max 350 KB each).

8.1 The driver's day in plain English

1. Sign in on the phone at `/driver/login` (mobile number, email or username).
2. **Today** shows today's and tomorrow's stops "published by the Dispatcher", the next stop, an ETA plan and personal performance for the selected period (default **Today**).
3. Nothing can start until the Dispatcher **confirms** the dispatch (vehicle loaded, stock deducted).
4. Open the first stop → **Start delivery**. The order becomes *Out for delivery*, a DEMO GPS session starts and the customer and Dispatcher are notified.
5. At the customer → **Reach location**. Arrival time is recorded and a 6-digit OTP is issued to the customer (shown in their app, valid 10 minutes by default).
6. **Enter OTP & complete**: enter delivered and damaged quantities per line, who received the goods, remarks, optional rejected quantities and shortfall reason, optional signature and photo, the customer's remarks, then the OTP.
7. If everything was delivered → *Delivered*; otherwise → *Partially delivered*. A delivery note (BM-DN-YYYY-NNNNN) is generated and can be shown to the customer.
8. If the delivery cannot happen → **Report issue / mark failed** with a reason. The Dispatcher reviews it.
9. Move to the next stop. When every stop is closed, the dispatch closes as *Completed* or *Closed with exceptions*.
10. Sign out from **Driver Profile**.

8.2 Master flowchart 5 — Driver workflow

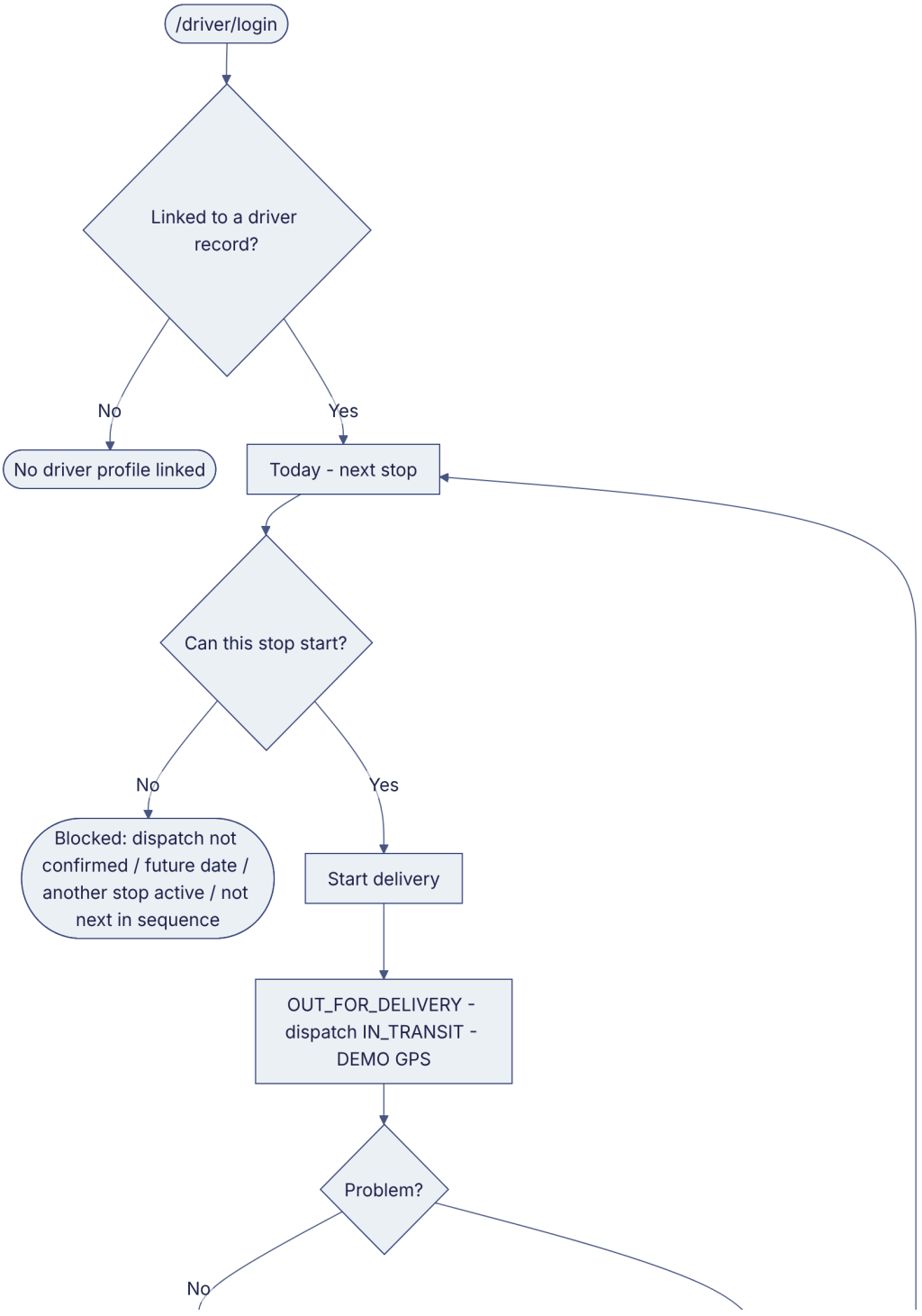


Diagram continued (1 of 2)

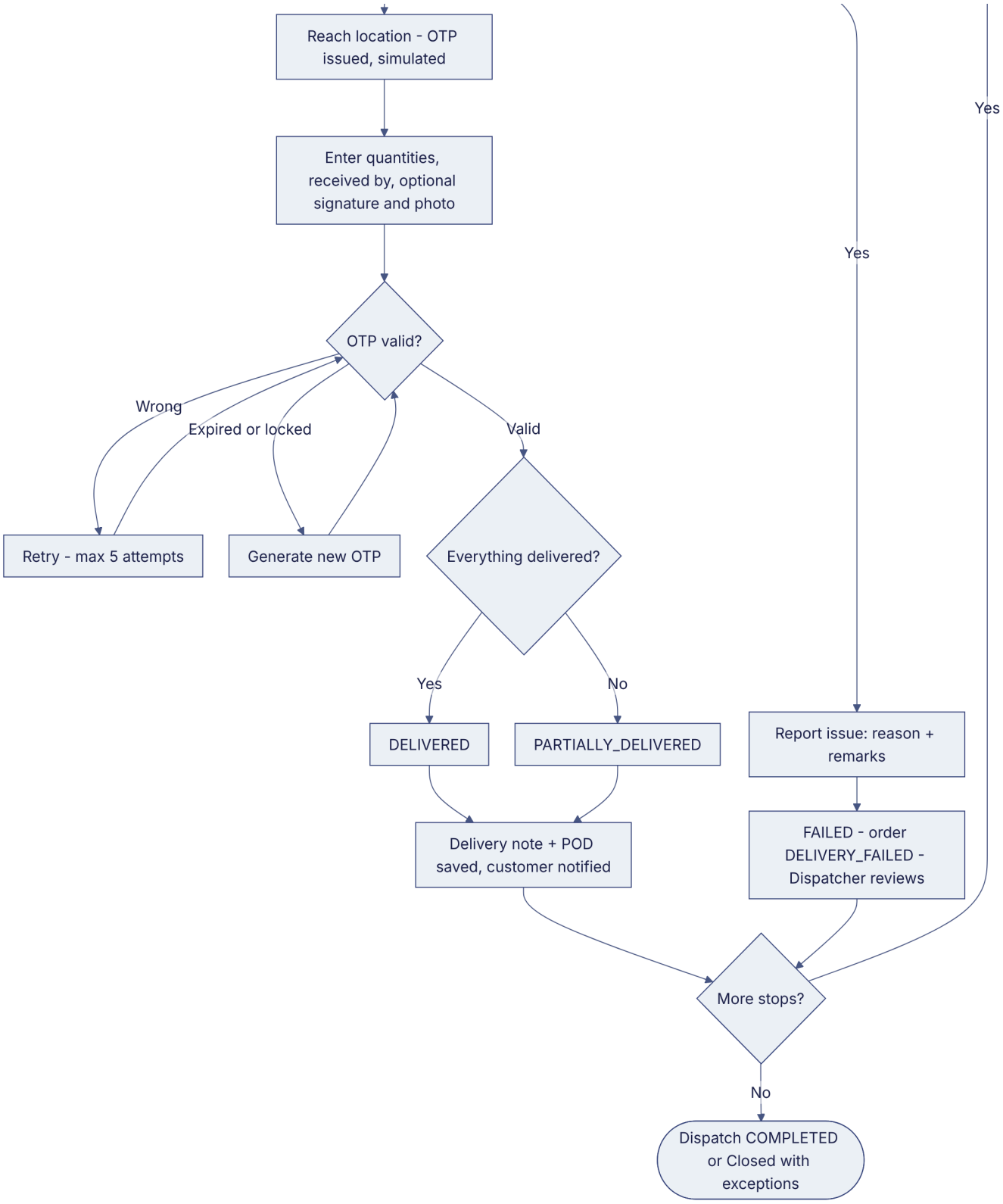


Diagram continued (2 of 2)

8.3 Driver app screens

Screen	Route	Component	What the driver does
Sign in	`/driver/login`	`AppLogin` (driver)	Mobile / email / username + password

Screen	Route	Component	What the driver does
Today	`/driver/dashboard`	`DriverHome`	See today / tomorrow plan, next stop, period performance
My deliveries	`/driver/deliveries`	`DriverDeliveries`	List of own stops
Delivery (stop)	`/driver/deliveries/:id`	`DriverDeliveryDetail`	Address, contact actions, requested time, ETA, items, Start / Reach location / Complete / Report issue
Confirm delivery	`/driver/deliveries/:id/confirm`	`DriverConfirmDelivery`	Quantities, received by, proof of delivery, OTP
Route	`/driver/route`	`DriverRoute`	DEMO route map of the dispatch's stops
Completed	`/driver/completed`	`DriverCompleted`	Closed stops
My delivery notes	`/driver/delivery-notes` (+ `/:id`)	`DriverNotes`	View / show delivery notes
Driver profile	`/driver/profile`	`DriverProfile`	Profile, vehicle, sign out
My documents	`/driver/documents` (+ `/:id`)	`DriverDocuments`	Licence, medical and other documents, upload

8.4 Rules enforced by `deliveryService`

Action	Allowed when	Refusal message
`startDelivery`	Delivery DISPATCHED; dispatch CONFIRMED or IN_TRANSIT; dispatch date not in the future; it is the next stop in sequence; no other stop of this dispatch is out	"...is already out for delivery.", "...is already delivered and cannot be restarted." and the blockers shown in 8.2
`markArrived`	Delivery OUT_FOR_DELIVERY	"Start the delivery first.", "...is already closed."
`confirmDelivery`	Delivery OUT_FOR_DELIVERY; OTP valid	"Start the delivery before confirming it.", "... is already delivered. It was not confirmed again.", "...was already marked failed."
`failDelivery`	Delivery OUT_FOR_DELIVERY	"Start the delivery before reporting a failure.", "...is already delivered and cannot be marked failed.", "...is already marked failed."

Quantity and POD validation

Field	Rule
Delivered	Required, not negative, whole units, not more than the dispatched quantity
Damaged / rejected	0 or more; rejected cannot exceed what was not delivered
Received by	"Enter the name of the person who received the goods."
Signature / photo	Optional; each ≤ 350 KB ("The photo is too large — retake it.")
Customer remarks	≤ 300 characters

OTP (`otpService` , `OTP_CONFIG`)

Setting	Value
Length	6 digits
Default expiry	10 minutes (allowed 1–60)
Maximum attempts	5 → status LOCKED (“Too many incorrect attempts. Generate a new OTP.”)
Statuses	ACTIVE (Awaiting entry), VERIFIED, EXPIRED, INVALIDATED (replaced by a new code), LOCKED
Delivery	Shown in the customer app tracking screen; the Dispatcher can inspect it (`dispatch.otp.inspect`). **No SMS.**

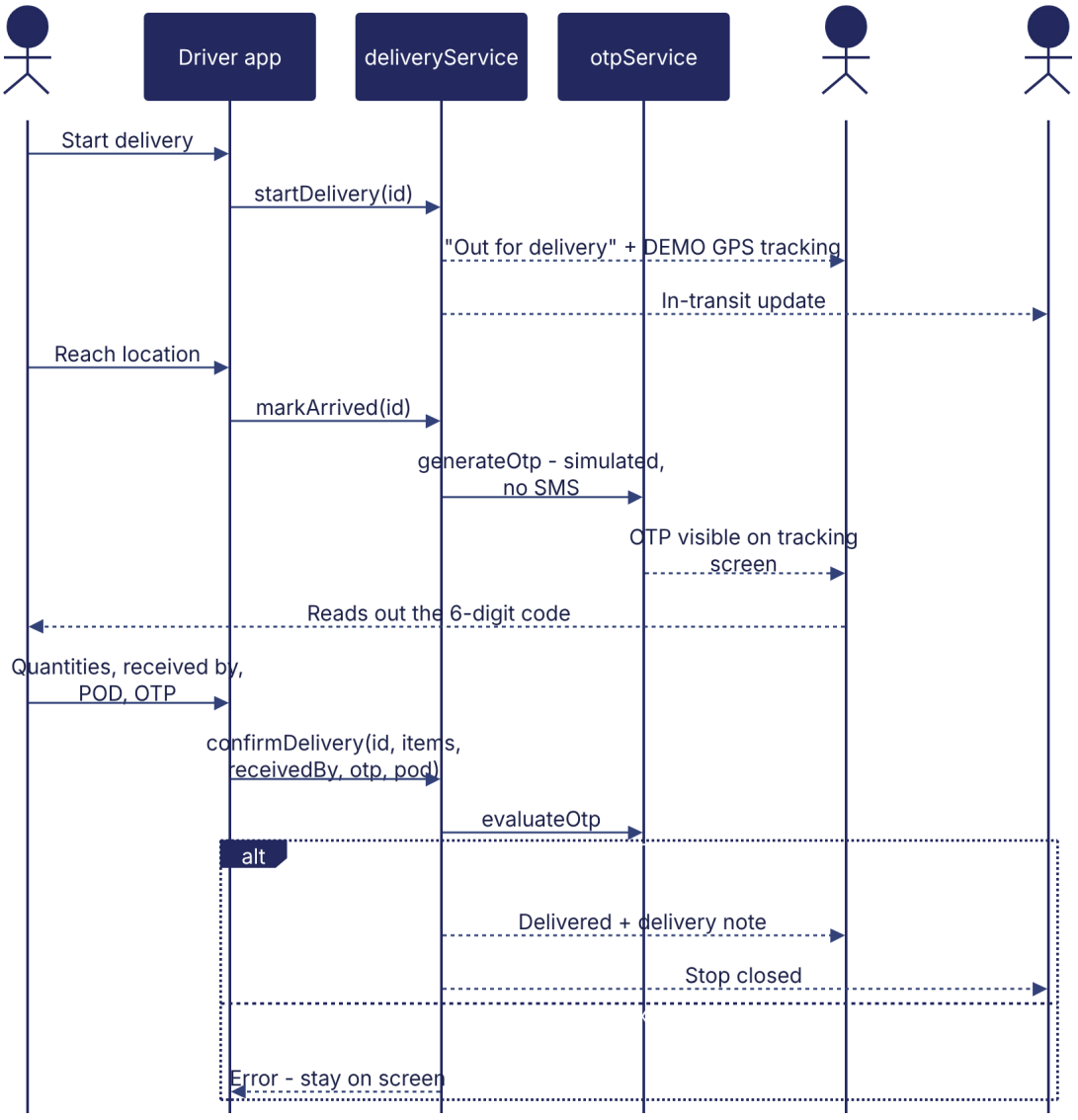
Failure reasons (`FAILURE_REASONS`)

Customer Not Available · Customer Refused · Customer Requested Reschedule · Incorrect Address · Damaged Goods · Partial Delivery · Other.

8.5 What happens in the data

Action	Delivery	Order	Dispatch	Other
Start	OUT_FOR_DELIVERY, `startedAt`	OUT_FOR_DELIVERY	IN_TRANSIT (first start)	Tracking session in `bellomassi_driver_tracking` ; event DELIVERY_STARTED
Reach location	`arrivedAt`	—	—	OTP in `bellomassi_delivery_otps`
Complete (full)	DELIVERED	DELIVERED	COMPLETED when all stops delivered	Delivery note, POD, OTP VERIFIED, tracking stopped, event DELIVERY_COMPLETED
Complete (short)	PARTIALLY_DELIVERED	PARTIALLY_DELIVERED	PARTIALLY_DELIVERED when all stops closed	Same; Dispatcher review needed
Fail	FAILED, review PENDING	DELIVERY_FAILED	Closes when all stops closed	Notifications; event DELIVERY_FAILED

8.6 Sequence — one stop



8.7 What the driver cannot do

- See or act on another driver's deliveries ("Access denied").
- Change the stop sequence (Dispatcher only), reassign the vehicle, or edit a closed stop.
- Complete without the customer's OTP.
- Use office modules (the Driver role has only `dashboard.view`, `driver.app`, `documents.upload`).

09 · Customer Workflow

Status: Implemented for ordering, tracking and documents. Delivery tracking uses **DEMO GPS** (Simulated); the OTP is shown in the app instead of being sent by SMS (Simulated). **Payments are Not Implemented** and **order cancellation by the customer is Not Implemented**.

9.1 The customer's journey in plain English

1. Sign in at `/customer/login` with mobile number, email or username (demo: `customer` / `Customer@123` or `+91 98410 22310`). The login must be linked to a customer account; otherwise the app shows "No customer account linked".
2. **Home** shows Active orders, Pending approval, Delivered (period filter) and the Outstanding balance (current), a card for any delivery on the road, recent orders, delivery notes and notifications.
3. **Catalog** lists active products with a selling price, searchable by name / code / SKU / brand and filterable by category. Availability at the customer's hub is shown; limited stock does not stop ordering.
4. **Cart**: change quantities, remove lines, see subtotal, discount, tax and total. A credit warning appears if the order takes the account over its limit — the order can still be placed and the Sales Manager reviews it.
5. Choose **delivery address** (registered or another), **requested date** (tomorrow up to 60 days ahead), optional **preferred time** slot, PO / reference and remarks → **Place order** → confirm.
6. The order is submitted exactly like a salesman's order (source `customer_app`, salesman = the customer's sales rep, priority High for VIP customers, payment terms Credit or Cash on Delivery from the customer record). Status: **Pending approval**.
7. The customer follows the order on **My orders** / **Order** with a 9-stage tracker: Order placed → Pending approval → Approved → Prepared → Dispatch planned → Out for delivery → Driver arrived → OTP verified → Delivered.
8. When the driver is on the way, **Track delivery** shows the driver, vehicle and a DEMO GPS map. When the driver arrives, the **OTP** appears in large digits with its expiry time — the customer reads it to the driver.
9. After delivery, the **delivery note** is available under Delivery notes; once Finance issues the invoice it appears under **Invoices**.
10. **Reorder** on a past order copies its still-orderable lines into the cart.

9.2 Master flowchart 6 — Customer workflow

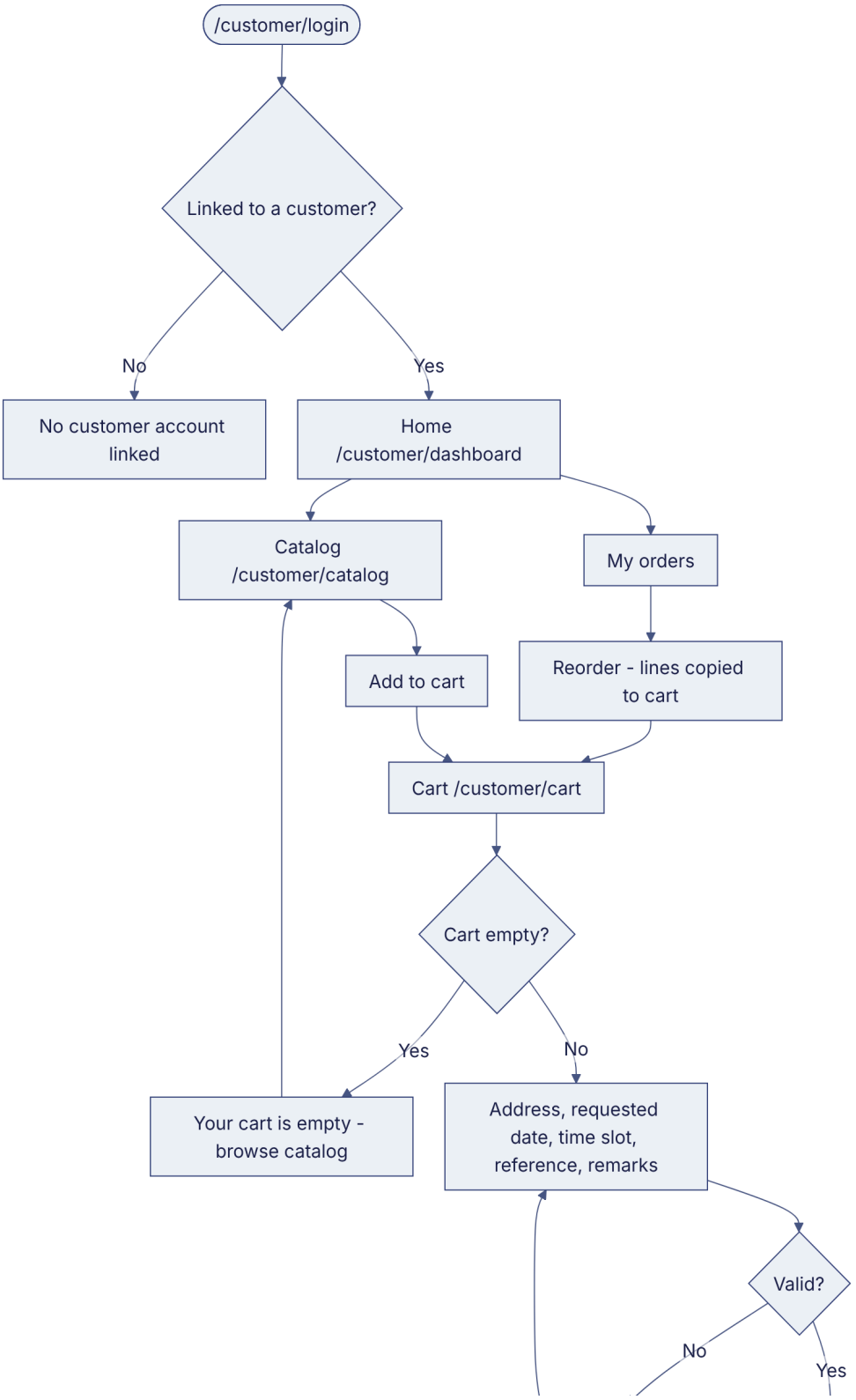


Diagram continued (1 of 3)

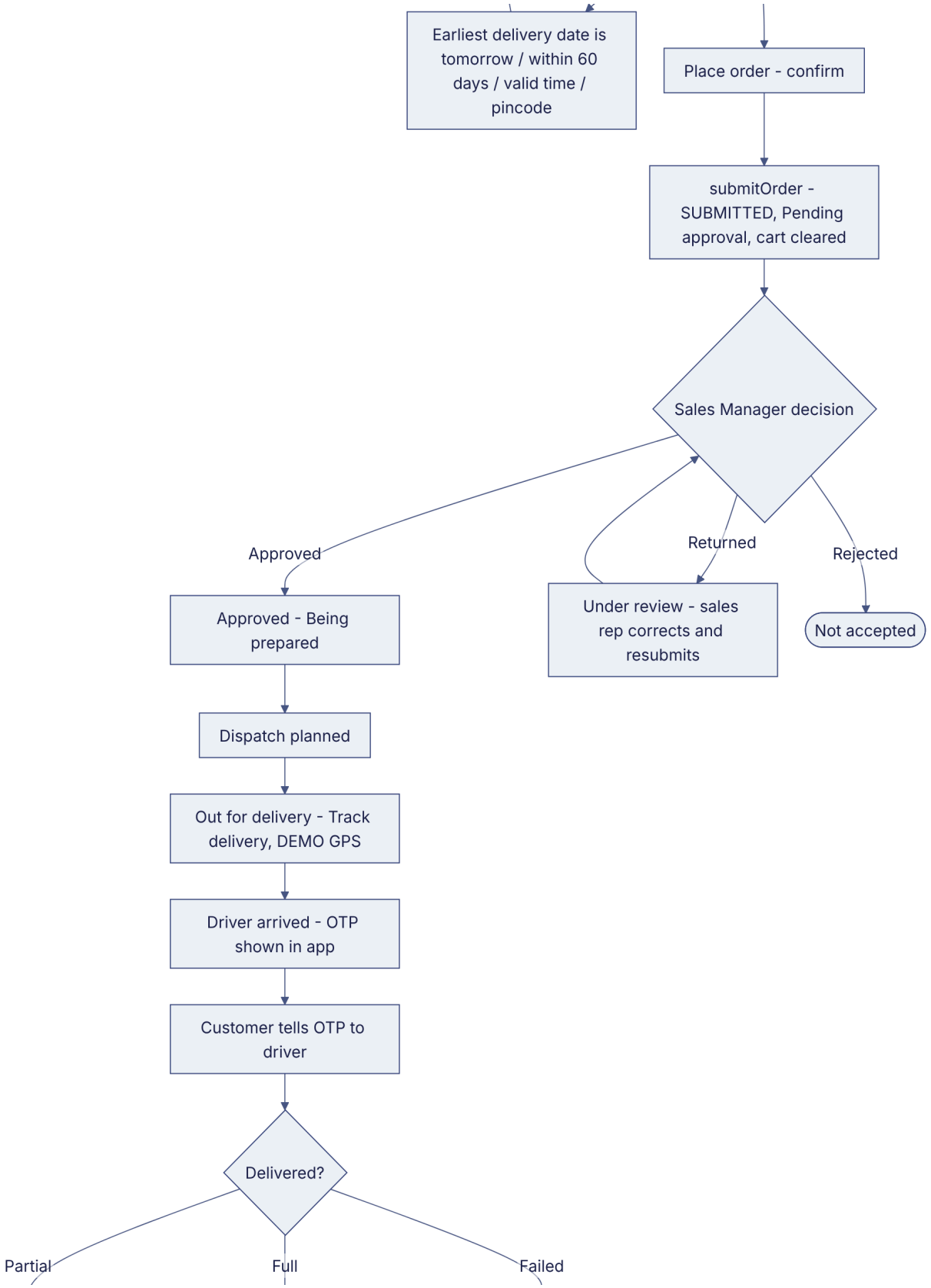


Diagram continued (2 of 3)

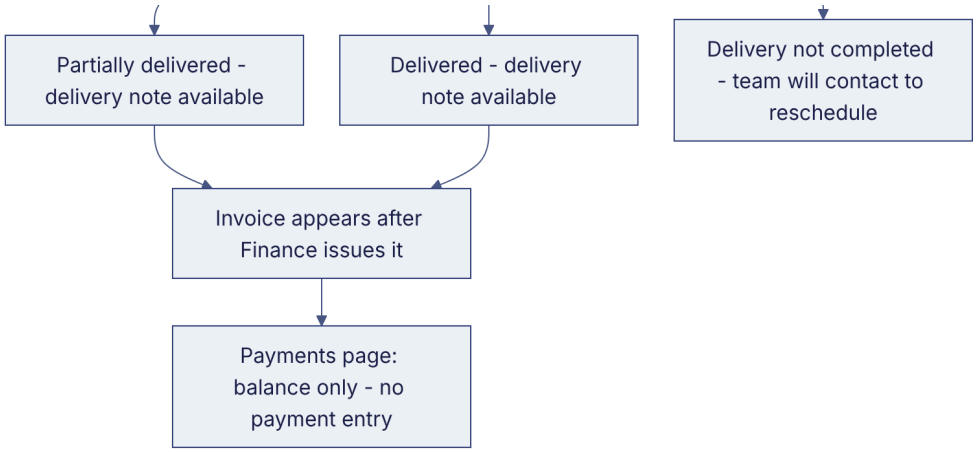


Diagram continued (3 of 3)

9.3 Screens

Screen	Route	Component	Purpose
Sign in	`/customer/login`	`AppLogin` (customer)	Mobile / email / username
Home	`/customer/dashboard`	`CustomerHome`	KPIs, on-the-road card, quick actions, recent items
Catalog	`/customer/catalog`	`CustomerCatalog`	Search, category chips, quantity stepper, add to cart
Cart	`/customer/cart`	`CustomerCart`	Lines, totals, delivery details, place order (sticky button)
My orders	`/customer/orders`	`CustomerOrders`	Filters: active, pending, delivered
Order	`/customer/orders/:id`	`CustomerOrderDetail`	Stage tracker, lines, approval / delivery labels, Reorder
Track delivery	`/customer/tracking/:id`	`CustomerTracking`	DEMO GPS map, driver, vehicle, OTP when arrived
Delivery notes	`/customer/delivery-notes` (+ `/:id`)	`CustomerDocuments`	List and view notes
Invoices	`/customer/invoices` (+ `/:id`)	`CustomerDocuments`	List and view invoices
Payments	`/customer/payments`	`CustomerPayments`	Account balance; “No payments recorded in this prototype”
Profile	`/customer/profile`	`CustomerProfile`	Account details, sign out

9.4 Customer-facing labels

Internal status	What the customer sees
SUBMITTED	Pending approval
RETURNED	Under review
REJECTED	Not accepted
APPROVED	Approved / Not scheduled

Internal status	What the customer sees
CONSOLIDATED, READY_FOR_DISPATCH	Being prepared, or Dispatch planned once on a dispatch
OUT_FOR_DELIVERY	Out for delivery, or Driver arrived
DELIVERED / PARTIALLY_DELIVERED	Delivered / Partially delivered
DELIVERY_FAILED	Delivery failed

9.5 Validation (`customerAppService.placeCustomerOrder`)

Rule	Message
Login linked to a customer	“Your login is not linked to a customer account.”
Cart has items	“Your cart is empty.”
Requested date	“Choose a delivery date.”, “The earliest delivery date is tomorrow.”, “Choose a date within the next 60 days.”
Time	“Enter a valid time.”
Remarks	“Use 300 characters or fewer.”
Order rules	All order validation from `orderService.validateOrder` (address, pincode, products, total)
Reorder	“This order belongs to another account.”, “None of the products on this order can be ordered now.”

9.6 Notifications the customer receives

When	Title
Driver starts the stop	“Out for delivery: BM-ORD-…” (track in the app, DEMO GPS)
Driver reaches the address	“Driver arrived: …” — share the delivery OTP shown in the app (demo OTP — no SMS)
Delivery completed	“Delivered: …” or “Partially delivered: …” — OTP verified, delivery note available
Delivery failed	“Delivery not completed: …” — “Our team will contact you to reschedule.”
Automation (if rules enabled)	Email / WhatsApp per rule for ORDER_CREATED, ORDER_APPROVED, ORDER_REJECTED, ORDER_READY, DISPATCH_ASSIGNED, DELIVERY_STARTED, DELIVERY_COMPLETED, DELIVERY_FAILED, INVOICE_GENERATED (demo channels)

9.7 What the customer cannot do

- Cancel or edit an order after placing it (**Not Implemented**).
- Pay online or see a payment history (**Not Implemented**).
- See other customers’ orders, notes or invoices.
- Approve anything.

9.8 Customer portal on the office web app

The CUSTOMER role's office dashboard (`CustomerPortalDashboard`: My orders, Open orders, Deliveries, Deliveries on the way, Invoices, Credit available, Account) exists, but a customer who opens `/dashboard` is redirected to `/customer/dashboard`, so in practice the customer always uses the customer app.

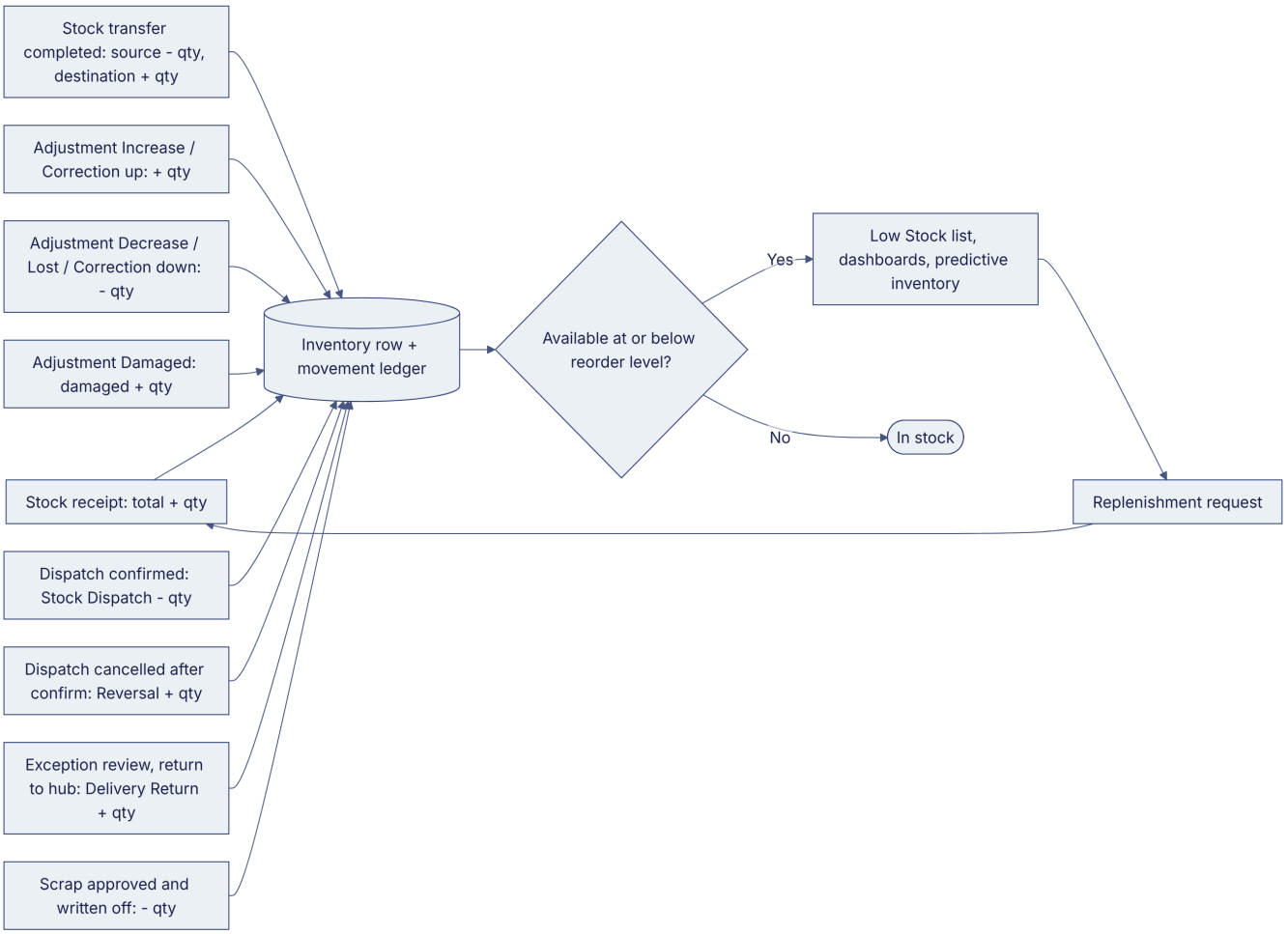
10 · Inventory Workflow

Status: Implemented. All stock lives in `bellomassi_inventory` (one row per product per hub) and every change is written to the movement ledger `bellomassi_stock_movements`. Batch and expiry tracking on stock rows is **Not Implemented** (only scrap requests carry a batch reference); cycle counting is **Not Implemented** (use adjustments).

10.1 Stock figures

Figure	Formula / meaning
Total quantity	Physical quantity at the hub
Damaged quantity	Part of total held as damaged (quarantine)
Reserved quantity	Part of total reserved
Available	Total − Reserved − Damaged (`utils/stock.getAvailable`)
Minimum / Reorder level / Maximum	Set per row on Stock details (`updateStockLevels` ; button shown to holders of `inventory.receipts.manage`)
Stock status	Out of Stock (available ≤ 0) → Low Stock (available ≤ reorder level) → Overstock (total > maximum) → In Stock

10.2 Master flowchart 7 — Inventory workflow



10.3 Stock receipt

Item	Details
Who	Administrator, Warehouse Manager, Inventory Executive (`inventory.receipts.manage`)
Screens	`/inventory/stock-receipts` (list), `/inventory/stock-receipts/create`
Steps	Choose hub, product, quantity, receipt date, reference (e.g. supplier invoice or replenishment number) → save
Validation	“Select a hub.”; inactive hub “…is inactive and cannot receive stock.”; product must track inventory; valid quantity (whole units for whole-unit products); valid date; reference ≤ 40 characters
Effect	Total quantity increases; movement **Stock Receipt**; receipt numbered GRN-… in `bellomassi_stock_receipts`
Approval	None — posted immediately

10.4 Stock transfer

Item	Details
Who	Administrator, Warehouse Manager, Inventory Executive (`inventory.transfers.manage`)
Screens	`/inventory/stock-transfers` , `/inventory/stock-transfers/create`

Item	Details
Steps	Source hub, destination hub, product, quantity, date, notes → **Save draft** or **Complete**
Validation	Both hubs required and different; destination active; at completion the quantity must be available at the source ("Only N … available at …")
Effect	On completion: source total − qty (**Stock Transfer Out**), destination total + qty (**Stock Transfer In**)
Statuses	Draft → Completed / Cancelled (`TRANSFER_STATUS`)
Guards	"…is already completed — stock was not changed again."; "Only draft transfers can be cancelled."

10.5 Stock adjustment

Item	Details
Who	Administrator, Warehouse Manager, QA Manager (`inventory.adjustments.manage`)
Screens	`/inventory/stock-adjustments` , `/inventory/stock-adjustments/create`
Types	Increase, Decrease, Damaged, Lost, Correction (with direction increase / decrease)
Reasons	Physical Stock Correction, Damaged Stock, Lost Stock, Expired Stock, Data Correction, Other (notes required)
Validation	Product must track inventory; cannot remove more than available unless the product allows negative stock; cannot mark more than available as damaged; stock can only be reduced at an inactive hub
Effect	Movement Stock Adjustment Increase / Decrease, Lost Stock or Damage
Approval	None — posted immediately (**no adjustment approval exists**)

10.6 Movements, stock overview and low stock

Screen	Route	Purpose
Inventory dashboard	`/inventory`	Stock value, status mix, alerts (period filter on dated documents)
Stock overview	`/inventory/stock`	All rows by product and hub with status; filters
Stock details	`/inventory/stock/:id`	Row history and min / reorder / max levels
Low stock	`/inventory/low-stock`	Rows that are Low or Out
Stock movements	`/inventory/movements`	Full ledger with type, sign, reference

Movement types (`MOVEMENT_TYPES`): Stock Receipt, Stock Transfer In, Stock Transfer Out, Stock Adjustment Increase, Stock Adjustment Decrease, Damage, Lost Stock, Stock Dispatch, Stock Dispatch Reversal, Delivery Return, Scrap Write-off.

10.7 Scrap

Scrap takes stock from the **available** or **damaged** source. Eligible quantity = on-hand in that source minus the quantity already in other open requests. Full lifecycle and approval rules are in chapter 07 §7.3. Write-off posts a single **Scrap Write-off** movement (`inventoryService.postScrapWriteOff`).

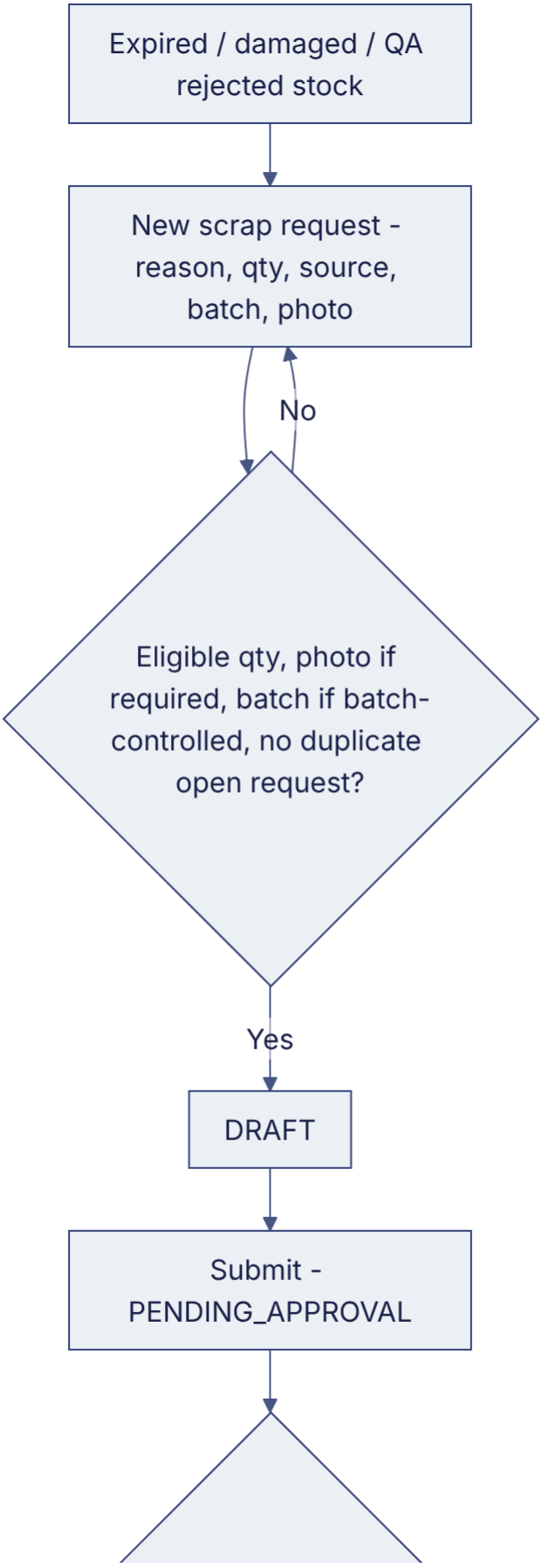


Diagram continued (1 of 2)

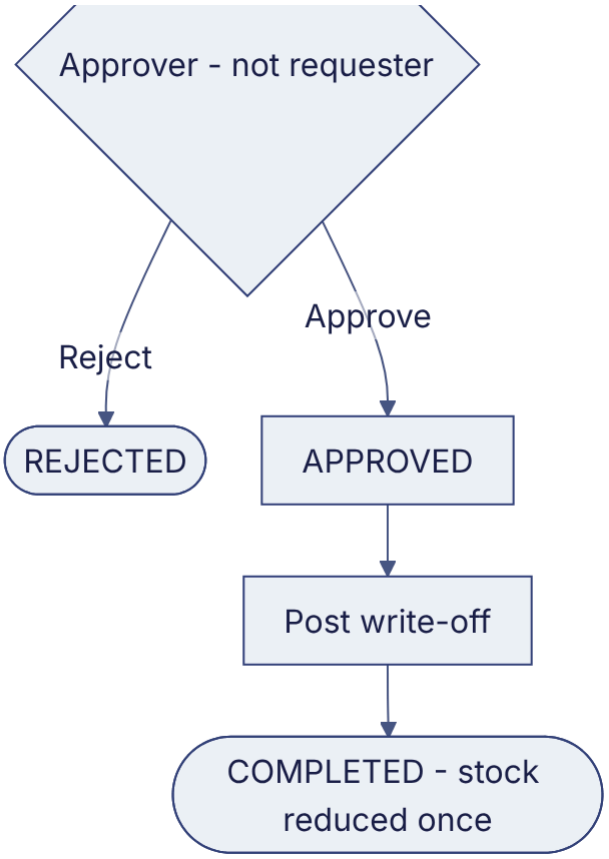


Diagram continued (2 of 2)

10.8 Predictive inventory and replenishment

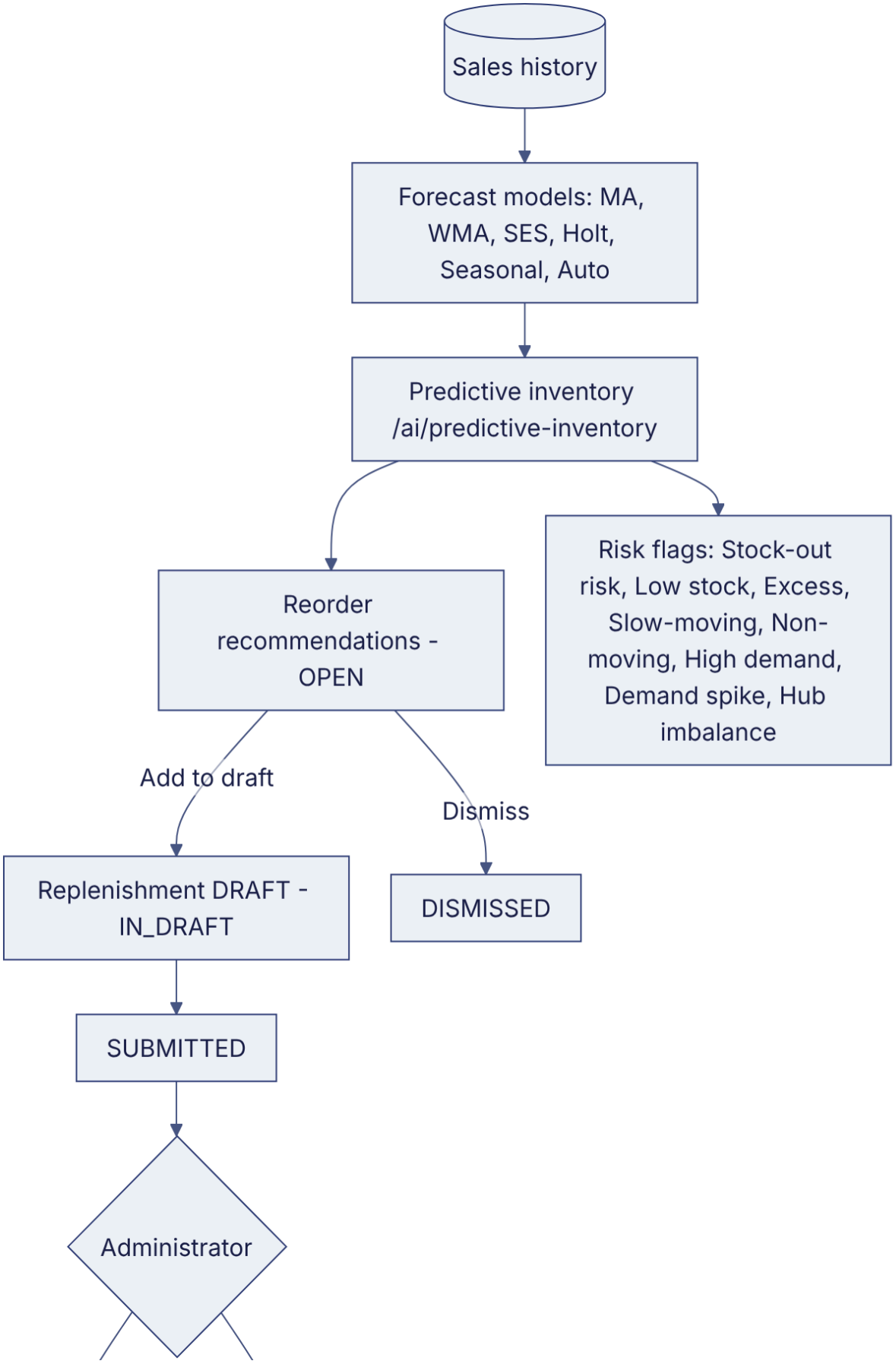


Diagram continued (1 of 2)

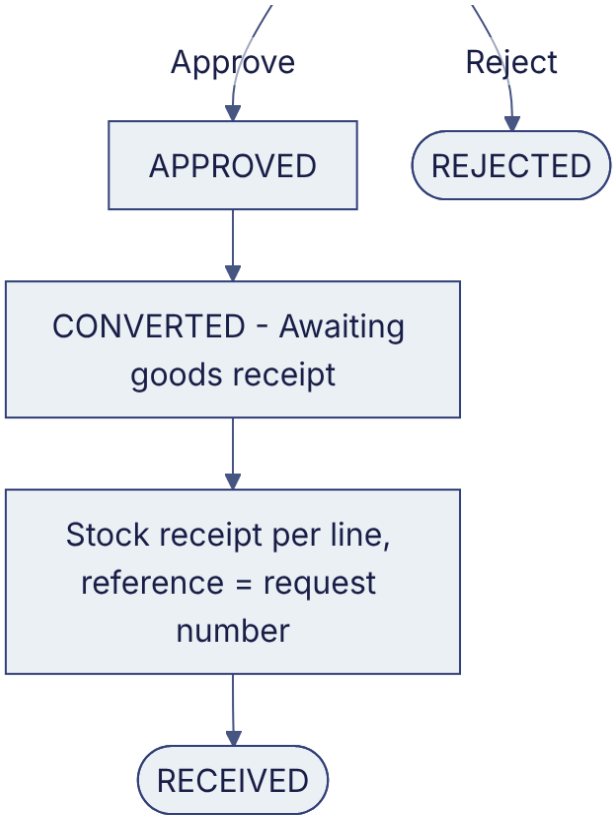


Diagram continued (2 of 2)

Step	Who	Route
Review predictions and risks	Roles with `inventory.predictions.view`	`/ai/predictive-inventory`, `/risks`, `/reorder`, `/demand`
Change planning parameters	Administrator (`inventory.predictions.manage`)	`/ai/predictive-inventory/settings`
Create replenishment draft	Administrator, Dispatcher, Warehouse Manager, Inventory Executive	`/ai/replenishment/create`
Approve / reject / convert	Administrator	`/ai/replenishment/:id`
Receive goods	Administrator, Warehouse Manager, Inventory Executive	Stock receipt link from the request
See approved requests	Supplier	Supplier portal dashboard

10.9 Inventory responsibilities

Activity	Inventory Executive	Warehouse Manager	QA Manager	Dispatcher	Administrator
Stock receipt	✓	✓	view	view	✓
Stock transfer	✓	✓	view	view	✓
Stock adjustment	view	✓	✓	view	✓
Stock levels (min / reorder / max)	✓	✓	—	—	✓
Scrap request	✓	✓	✓	✓	✓
Scrap approval / write-off	—	✓	—	—	✓

Activity	Inventory Executive	Warehouse Manager	QA Manager	Dispatcher	Administrator
Replenishment draft	✓	✓	—	✓	✓
Replenishment approval	—	—	—	—	✓
Dispatch stock deduction	—	—	—	✓ (by confirming)	✓

11 · Dispatch & VMS Workflow

Status: Dispatch planning, assignment, confirmation, cancellation, sequencing, delivery review and VMS compliance are **Implemented**. Live tracking is **Simulated** (DEMO GPS). Route optimisation is a suggested sequence, not a road routing engine.

11.1 Business summary

The Dispatcher works from the **Ready for Dispatch** queue. A dispatch is one vehicle run from one hub on one date. It holds a set of orders in a delivery sequence, one driver and one vehicle. Before anything leaves the hub, the system checks that every order is still ready, that the driver and vehicle are free, from the same hub, legally compliant and that the load fits the vehicle. **Confirming** the dispatch is the loading sign-off: hub stock is deducted and the driver can start. The Dispatcher then follows the run on Delivery Tracking / Live Tracking and resolves failed and partial stops.

11.2 Master flowchart 8 — Dispatch / VMS workflow

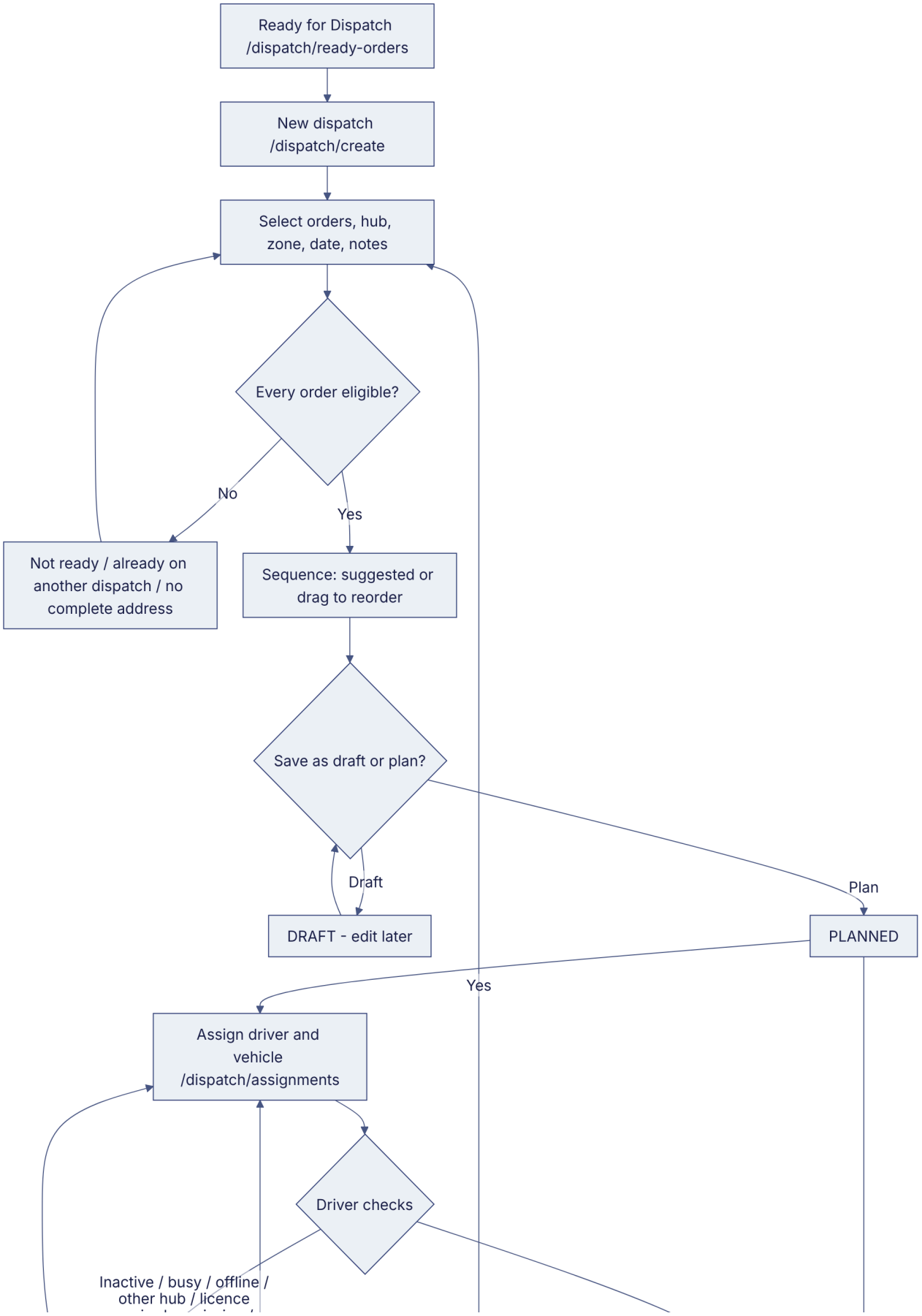


Diagram continued (1 of 3)

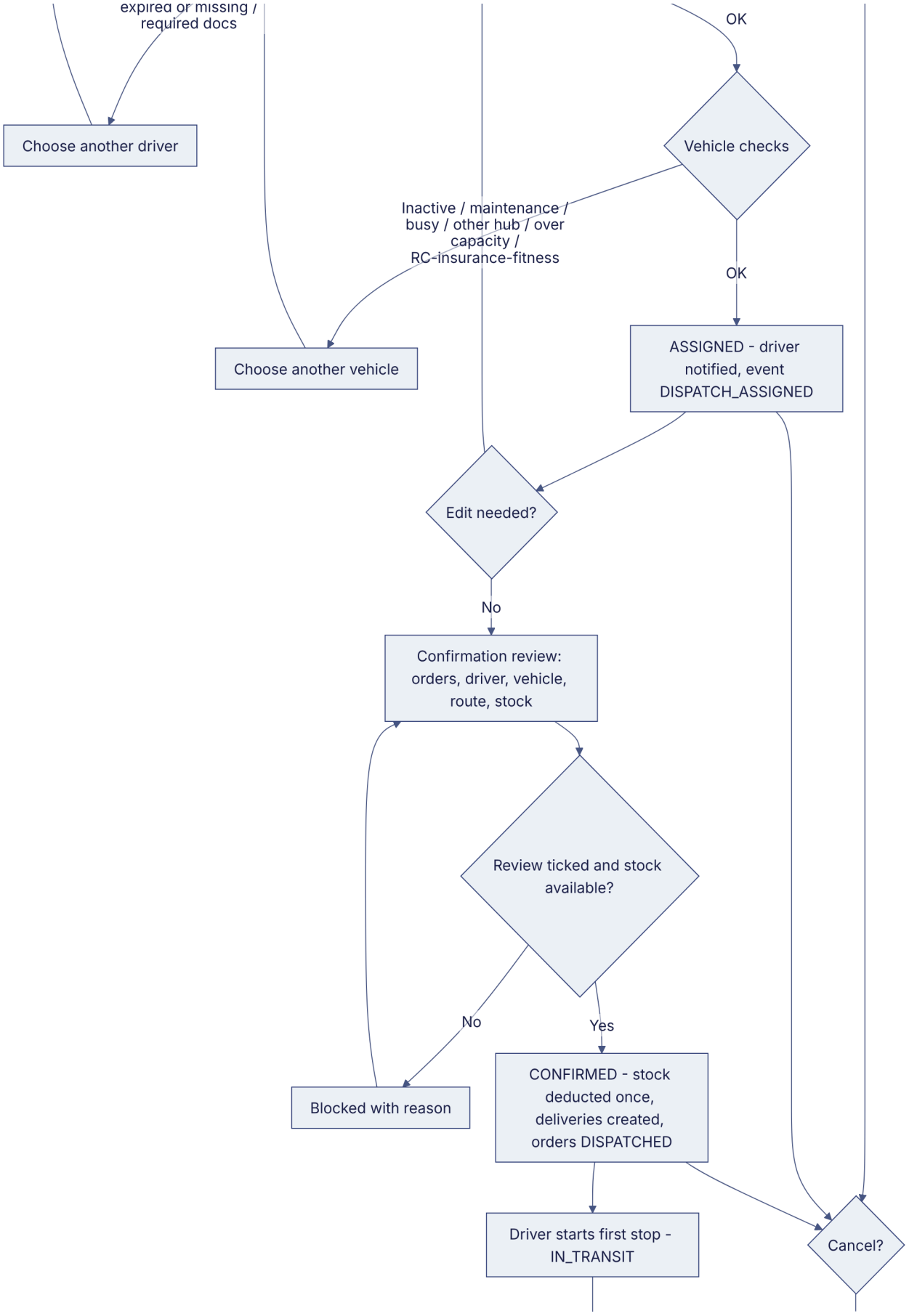


Diagram continued (2 of 3)

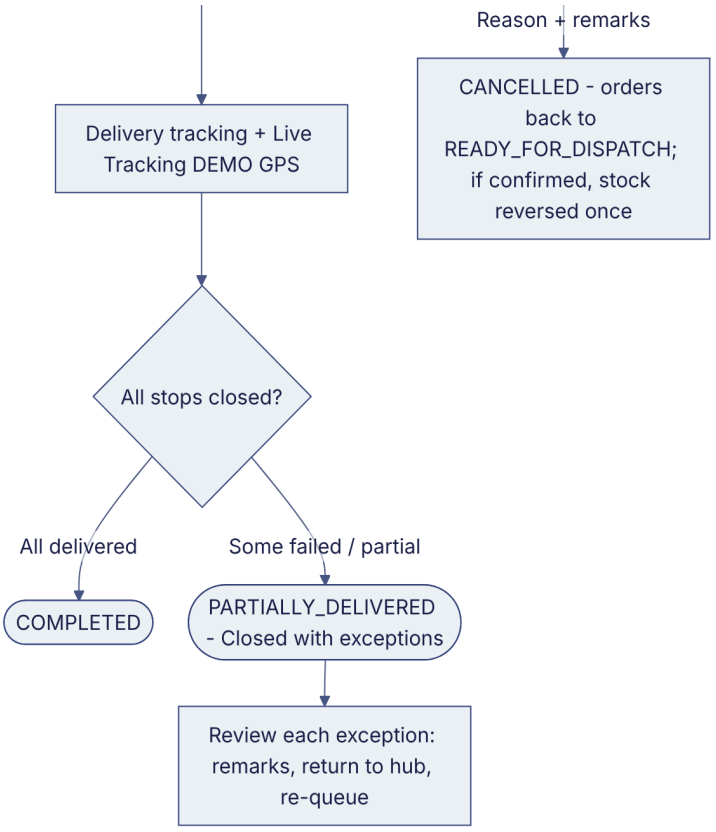


Diagram continued (3 of 3)

11.3 Dispatch statuses (`DISPATCH_STATUS_META`)

Status	Label	Meaning	Editable?
DRAFT	Draft	Saved, incomplete plan	Yes
PLANNED	Planned	Orders and sequence set	Yes
ASSIGNED	Assigned	Driver and vehicle assigned	Yes
CONFIRMED	Confirmed	Loaded; stock deducted; deliveries created	No (cancel only while no stop has started)
IN_TRANSIT	In transit	First stop started	No
PARTIALLY_ DELIVERED	Closed with exceptions	All stops closed; at least one failed / partial	No
COMPLETED	Completed	All stops delivered	No
CANCELLED	Cancelled	Cancelled with reason	No

Resource-holding statuses (driver / vehicle busy): ASSIGNED, CONFIRMED, IN_TRANSIT.

11.4 Step reference

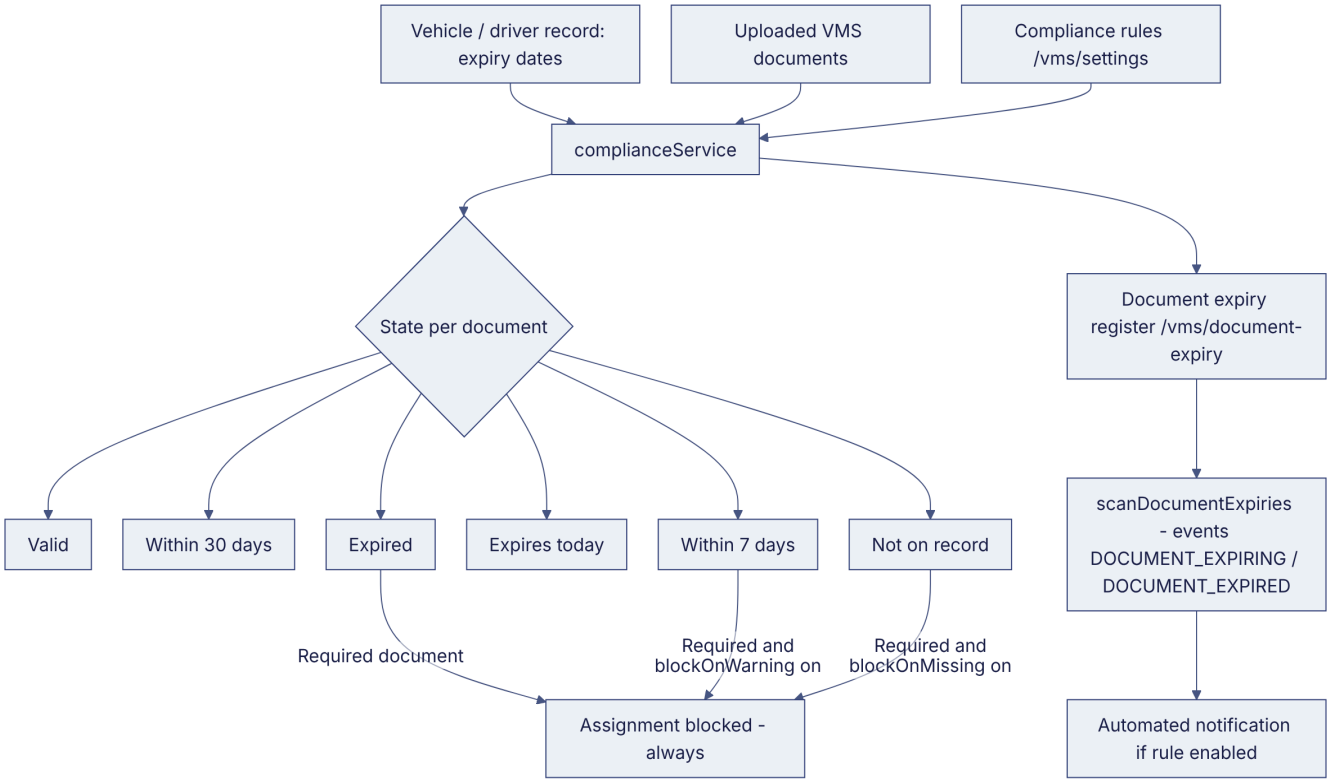
Step	Route	Service	Rules and messages
Create	`/dispatch/create`	`createDispatch`	"Only the Dispatcher or Admin can create dispatches."; orders must be READY_FOR_DISPATCH, not on another active dispatch, with a complete address; neighbouring-zone **warning**
Edit	`/dispatch/:id/edit`	`updateDispatch`	Only DRAFT / PLANNED / ASSIGNED: "...is confirmed and can no longer be edited."
Assign / reassign	`/dispatch/assignments`, `/dispatch/:id`	`assignDispatch`	Only PLANNED / ASSIGNED: "Complete the draft plan before assigning a driver."; driver and vehicle issue checks; previous driver notified "Removed from ..."
Change sequence	`/dispatch/:id` (sequence editor)	`updateSequence`	"Invalid delivery sequence: every stop must appear exactly once.", "The sequence has not changed."; driver notified
Confirm	`/dispatch/:id`	`confirmDispatch`	Must be ASSIGNED; review tick required; stock checked per line ("...needs N, only M available at the hub."); stock deducted once
Cancel	`/dispatch/:id`	`cancelDispatch`	Reason from list + remarks; not after a stop has started; confirmed dispatch → stock reversed once; orders back to READY_FOR_DISPATCH
Review exception	`/dispatch/deliveries/:id`	`reviewDelivery`	Chapter 15

Cancel reasons (`CANCEL_REASONS`): Vehicle breakdown, Driver unavailable, Customer requested reschedule, Stock issue at hub, Planning error, Other.

11.5 Driver and vehicle assignment checks

Resource	Check	Message
Driver	Selected	"Select a driver."
Driver	Active	"... is inactive."
Driver	Not on another active dispatch	"... already has an active dispatch (BM-DSP-...)."
Driver	Not offline	"... is offline."
Driver	Same hub	"... works from another hub."
Driver	Licence	"...'s driving licence expired on ...", "... has no valid licence expiry on record."
Driver	VMS required documents	From `driverComplianceIssues` (default: licence)
Vehicle	Selected	"Select a vehicle."
Vehicle	Not inactive / maintenance	"... is inactive.", "... is under maintenance."
Vehicle	Not on another active dispatch	"... is already on BM-DSP-...."
Vehicle	Same hub	"... belongs to another hub."
Vehicle	Capacity	"Load of N exceeds ...'s capacity of C ..."
Vehicle	VMS required documents	From `vehicleComplianceIssues` (default: RC, insurance, fitness)

11.6 VMS compliance



Setting (`DEFAULT_VMS_SETTINGS`)	Default
Required vehicle documents	Registration (RC), Insurance, Fitness certificate
Required driver documents	Driving licence
Block on expired required document	Always on
Block when expiring within warning days	Off (warning days 7)
Block when not on record	Off

Tracked expiry fields — vehicle: Registration (RC), Insurance, Pollution (PUC), Fitness certificate, Permit; driver: Driving licence, Medical certificate.

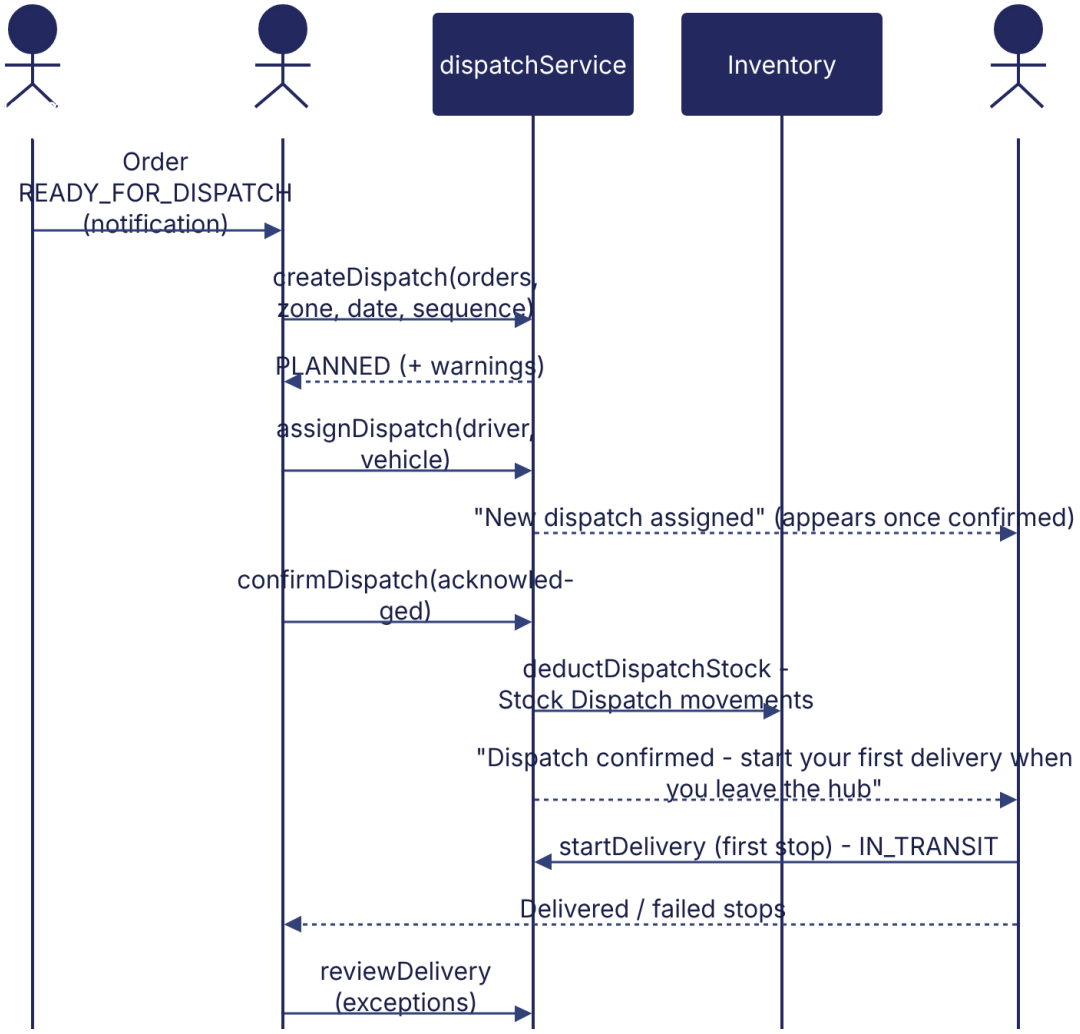
VMS screens: VMS dashboard `/vms`, Vehicles `/vms/vehicles` (details and documents), Driver documents `/vms/drivers`, Document expiry `/vms/document-expiry`, VMS reports `/vms/reports`, Compliance rules `/vms/settings` (Administrator).

11.7 Tracking

Screen	Route	Who	What it shows
Dispatch dashboard	`/dispatch`	Dispatch viewers	KPIs and charts (chapter 04 §4.6)
Delivery tracking	`/dispatch/delivery-tracking`	Administrator, Sales Manager, Customer Service, Dispatcher	Every stop with status, driver, time
Live Tracking (Demo GPS)	`/dispatch/live-tracking`	Administrator, Dispatcher	Map with simulated vehicle positions from `bellomassi_driver_tracking`

Screen	Route	Who	What it shows
Delivery history	`/dispatch/delivery-history`	Dispatch viewers, Finance, Warehouse Manager	Closed stops, exceptions to review
Delivery details	`/dispatch/deliveries/:id`	`deliveries.view`	Items, POD, OTP inspector (Dispatcher / Admin), review panel
Delivery reports	`/dispatch/reports`	Administrator, Dispatcher, Finance	Delivery performance

11.8 Dispatch — swimlane



12 · Finance & Invoice Workflow

Status: Invoice templates, numbering, generation, printing and cancellation are **Implemented**. Tax is a configurable **demo** calculation (not GST-compliant e-invoicing). Emailing an invoice uses the email module (**Simulated** in demo mode). **Payments, receipts, credit notes, refunds, collections posting and expenses are Not Implemented.** The customer's credit balance is a field on the customer record and is **not** updated by invoices.

12.1 Business summary

Finance (or an Administrator) picks an approved-or-later order and issues one invoice for it. The invoice is a frozen snapshot of the order, customer, hub, template and — when available — the delivery note, so later changes to master data never change an issued invoice. Finance chooses whether to bill **ordered** quantities or **delivered** quantities (from the delivery note), can add an order discount, shipping and additional charges, and sets invoice and due dates. An invoice that is wrong is **cancelled** with a reason; a new one can then be issued.

12.2 Master flowchart 9 — Finance / invoice workflow

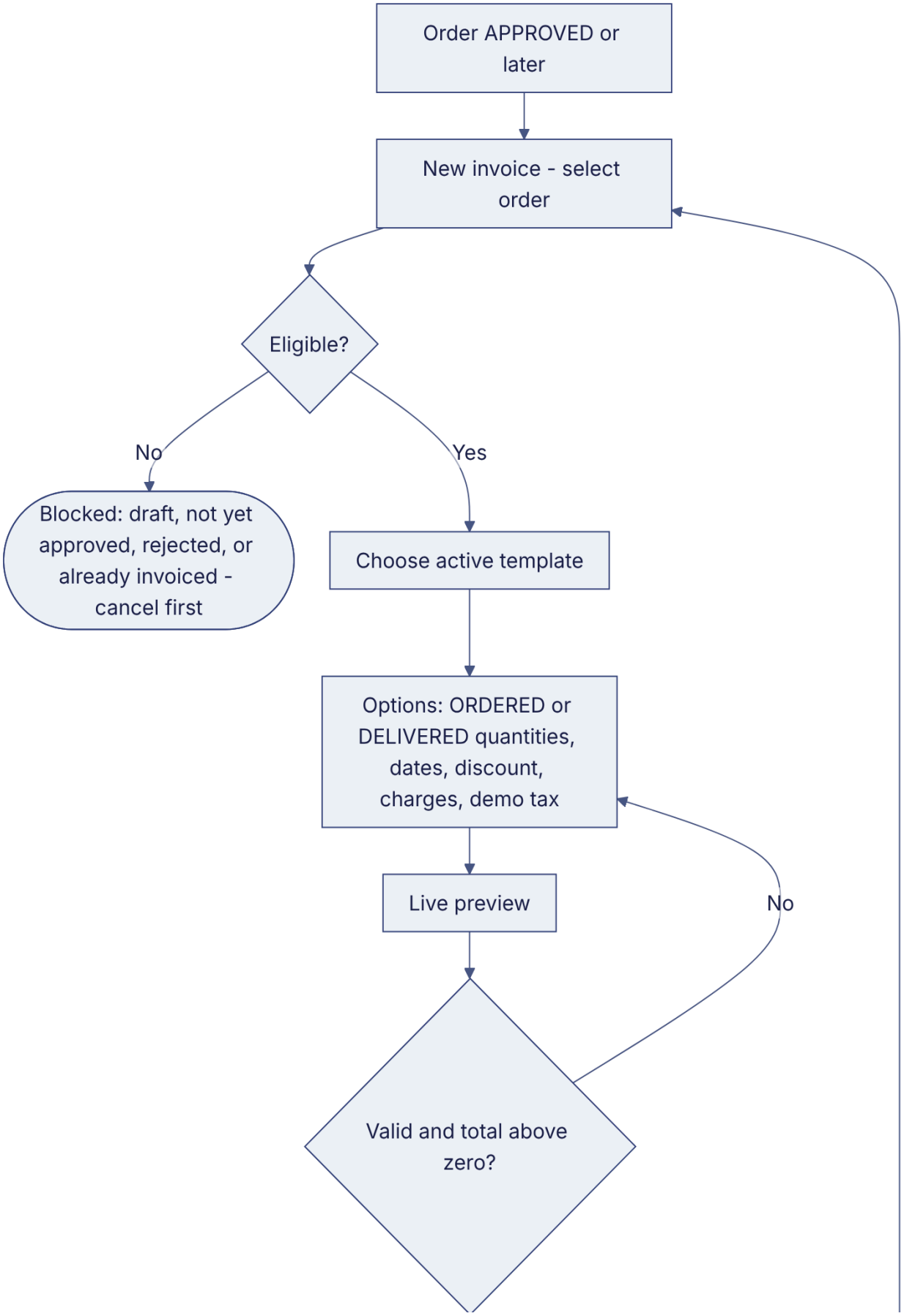


Diagram continued (1 of 2)

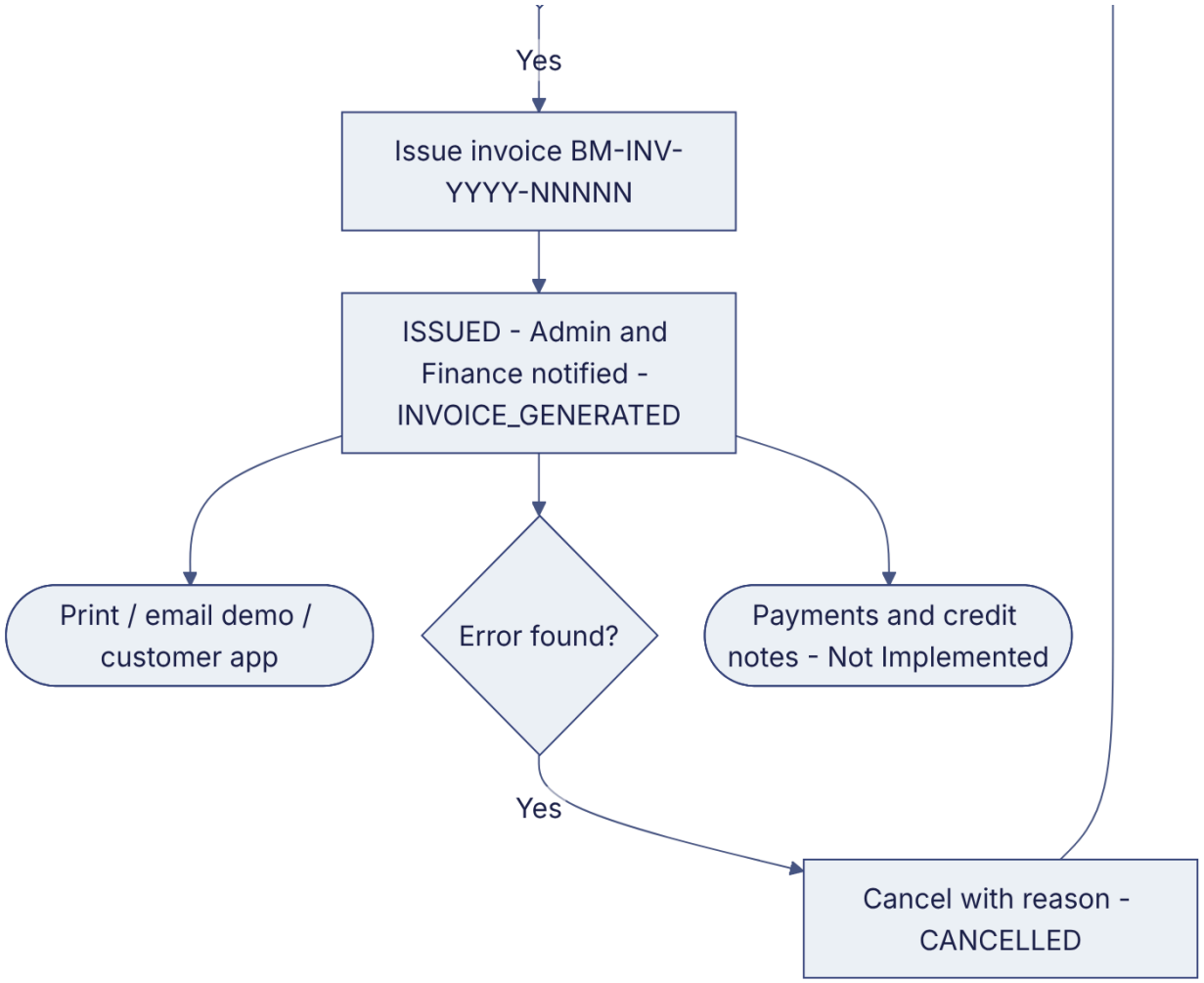


Diagram continued (2 of 2)

12.3 Templates and numbering

Item	Details
Templates	`/invoices/templates` — code, company details (GSTIN, PAN, email), page format (A4 portrait, A4 landscape, custom 100–500 mm), font (Inter, Georgia, Helvetica, Courier), font size 8–14 pt, primary / accent colours, header and footer layouts, sections (e.g. signature, QR code placeholder), active flag, default flag. Preview at `/invoices/templates/:id/preview`.
Who	Administrator, Finance (`invoices.templates`)
Numbering	`bellomassi_invoice_settings` : prefix **BM-INV**, include year, starting number, 5 digits → e.g. BM-INV-2026-00001. Only Administrator or Finance can change numbering.

12.4 Calculation (`services/invoiceCalculationService.js`)

Line	Formula
Line subtotal	quantity × unit price
Item discount	line subtotal × item discount %
Order discount	NONE, flat AMOUNT or PERCENT of (Σ subtotal − Σ item discount), allocated to lines by value

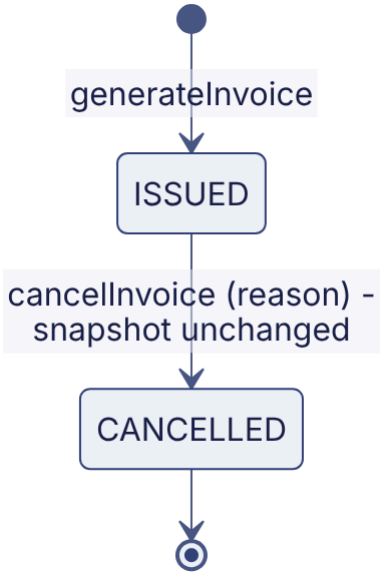
Line	Formula
Taxable amount	line subtotal − item discount − allocated order discount
Tax	taxable × tax % (product rate from the order, or a flat demo rate ≤ 40%)
Charges	shipping + additional charges, taxed at charges tax % (default 0, ≤ 40%)
Grand total	Σ taxable + Σ tax + charges + charges tax

Quantity basis: **DELIVERED** bills what the delivery note recorded as delivered; it falls back to **ORDERED** when there is no delivery note (`utils/invoiceSnapshot.js`).

12.5 Validation and messages

Rule	Message
Permission	“Only Finance or Admin can issue invoices.” / “...cancel invoices.”
Template	“Create an invoice template first.”, “... is inactive.”
Customer	“The order’s customer record no longer exists.”
Dates	“Enter a valid date.”, “Due date cannot be before the invoice date.”
Charges / tax	“Enter 0 or more.”, “Demo tax rate cannot exceed 40%.”
Discount	“Discount must be 0–100%.”, “Discount cannot be negative.”
Total	“The invoice total must be greater than zero (check delivered quantities and discounts).”
Cancel	“Give a reason for cancelling.”, “... is already cancelled.”

12.6 Statuses

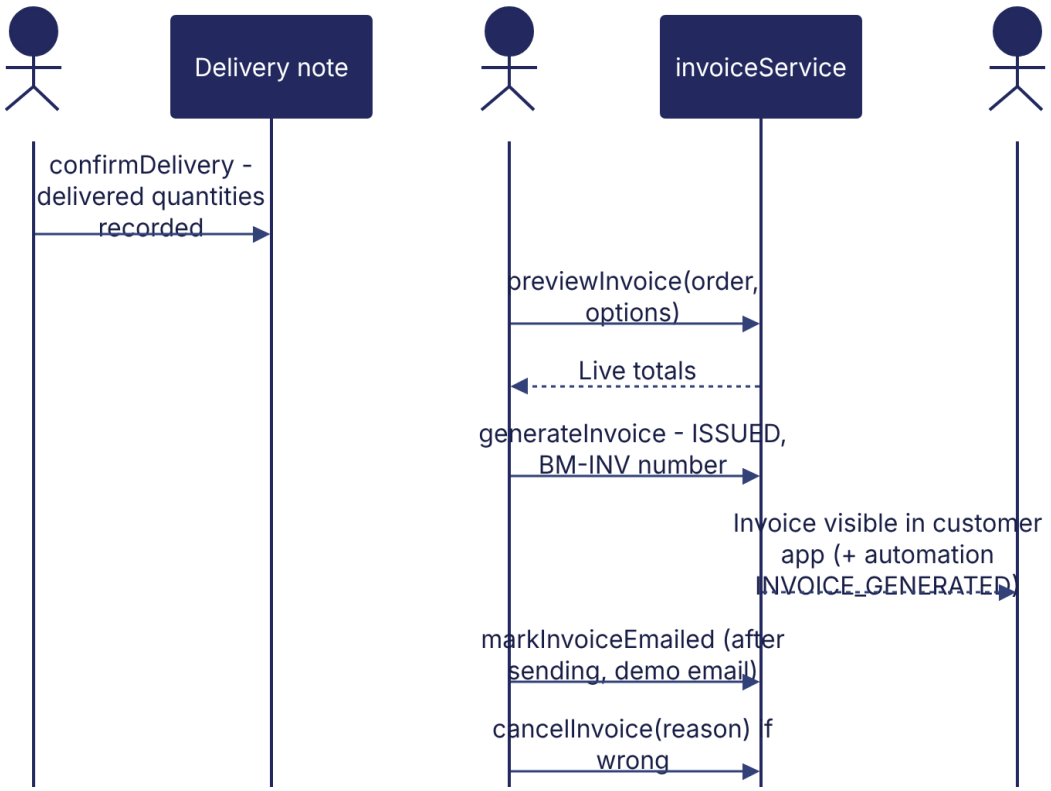


Invoiceable order statuses (`INVOICEABLE_ORDER_STATUSES`): APPROVED, CONSOLIDATED, READY_FOR_DISPATCH, DISPATCHED, OUT_FOR_DELIVERY, PARTIALLY_DELIVERED, DELIVERED, DELIVERY_FAILED, RETURNED_TO_HUB.

12.7 Credit and outstanding

Topic	How it works today
Credit limit, days, enabled	Customer master (Administrator)
Credit balance	Customer master field; seeded demo values; **not** changed by invoices or deliveries
Order credit check	`getOrderCreditPosition`: projected balance = balance + order total; warning (or block, if configured) at approval
Customer Credit page	`/customers/credit` — limits, balances, utilisation (Administrator, Sales Manager, Finance)
Finance dashboard	Outstanding (customer balances), Credit utilisation, Over credit limit, Collections
Customer app	Outstanding balance on Home and Payments; “No payments recorded in this prototype”
Payments, receipts, allocation, credit notes, refunds	**Not Implemented** — the related Finance reports show “no source data”

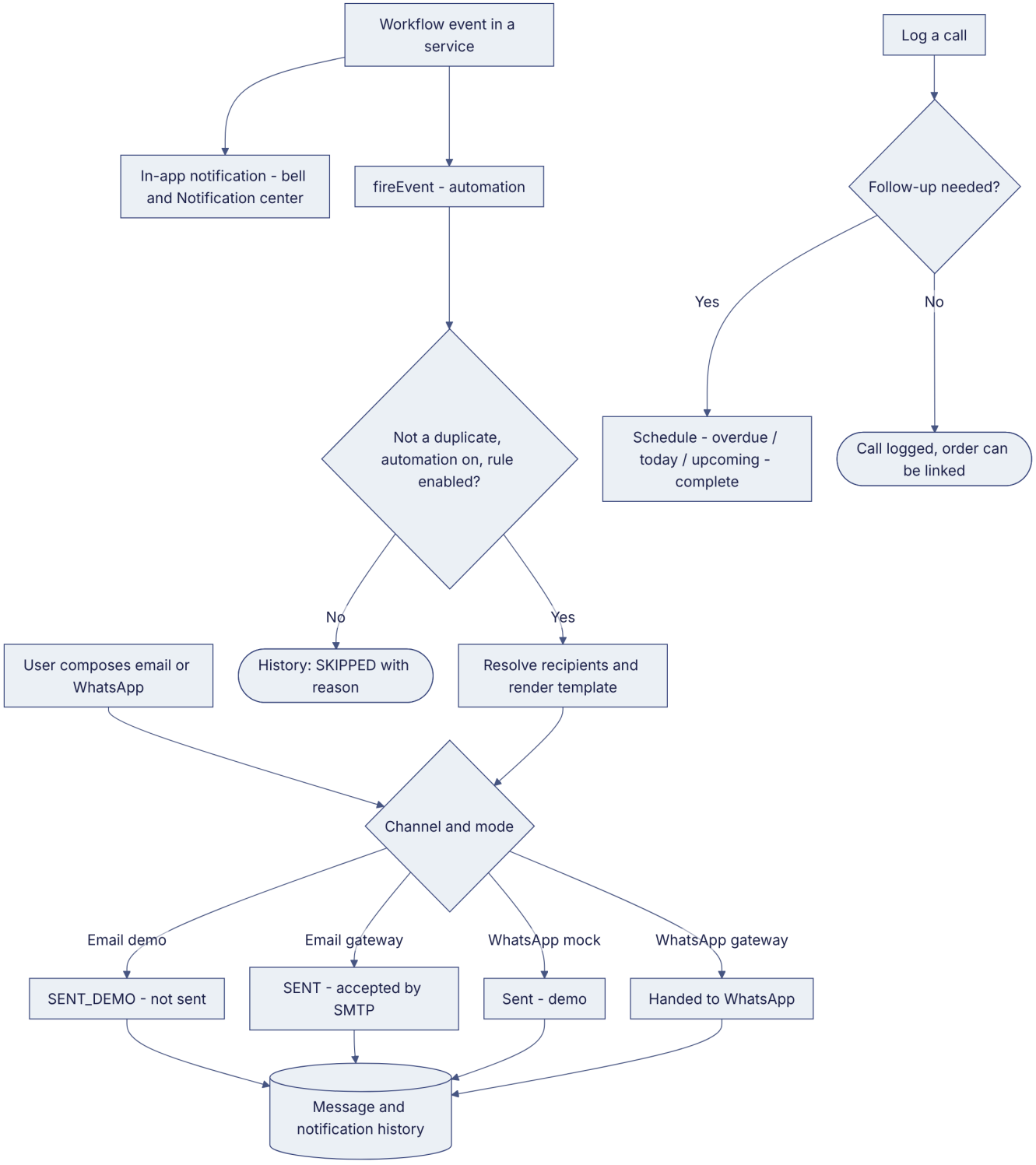
12.8 Finance — swimlane



13 · Communication & Documents Workflow

Feature	Status
In-app notifications	Implemented
Automated notifications (rules per event)	Implemented — delivered through the simulated channels in demo mode
Email (compose, drafts, sent, history, templates, inbox)	**Simulated** in demo mode (“Sent · demo (not sent)”); gateway SMTP / IMAP mode **Not Verified**
WhatsApp via OpenWA	**Simulated** (mock session); gateway mode **Not Verified**
Call history and follow-ups	Implemented as a manual log (no telephony)
Documents repository, upload, review, verify, reject	Implemented
OCR	**Simulated** (MOCK provider); browser OCR unavailable; backend OCR **Not Verified**

13.1 Master flowchart 10 — Communication workflow



13.2 In-app notifications

Item	Details
Created by	Workflow services (`notifyOrder` , `notifyDispatch` , `notifyCustomer` , `pushNotification`)
Addressed to	Roles, a specific user (e.g. the order's salesman), a driver, or a customer

Item	Details
Storage	`bellomassi_notifications`; preferences `bellomassi_notification_preferences` (muted categories, toasts, digest)
Screens	Header bell dropdown; `/notifications` (Notification Center)
Actions	Mark read, mark all read, filter by category, open linked record

Key notifications by event:

Event	Recipients
Order submitted / resubmitted	Sales Manager (“New order for approval”), salesman
Order approved	Salesman, Customer Service (“Order awaiting consolidation”)
Order returned / rejected	Salesman
Order consolidated / ready for dispatch	Dispatcher; salesman on ready
Dispatch assigned / changed / sequence updated / cancelled	Driver; Administrator / Dispatcher
Dispatch confirmed	Driver; Administrator
Out for delivery / driver arrived / delivered / not completed	Customer; Dispatcher
Delivery failed	Administrator, Dispatcher (“Review required”), customer
Invoice issued	Administrator, Finance

13.3 Email

Step	Route	Notes
Home	`/communication/email`	Counts, mode (Demo / Live)
Compose	`/communication/email/compose`	To, CC, subject, body, template, attachments from documents
Drafts	`/communication/email/drafts`	`bellomassi_email_drafts`
Sent / History	`/communication/email/sent`, `/history`	`bellomassi_email_history` with status
Inbox	`/communication/email/inbox`	Seeded demo mailbox; “sync” may add a simulated message (`bellomassi_email_inbox`)
Templates	`/communication/email/templates`	Variables such as customer, order, invoice fields
Settings	`/communication/email/settings`	SMTP / IMAP settings, encryption; credentials are only used by the gateway

13.4 WhatsApp (OpenWA)

Step	Route	Notes
Home	`/communication/whatsapp`	Session state; “Connected” only when OpenWA reports `ready`

Step	Route	Notes
Compose	`/communication/whatsapp/compose`	Recipient (country code default 91), message, template
History	`/communication/whatsapp/history`	`bellomassi_whatsapp_messages`
Templates	`/communication/whatsapp/templates`	`bellomassi_whatsapp_templates`
OpenWA settings	`/communication/whatsapp/settings`	Session start / stop / QR through the gateway; mock in demo

13.5 Call history

Step	Details
Log a call	Customer, direction (Incoming / Outgoing), status (Connected, Missed, Busy, No Answer, Voicemail), outcome, duration, notes
Outcomes	Order Received, Interested, Follow-up Required, No Response, Busy, Not Interested, Complaint, Payment Follow-up, Other
Follow-up	Suggested for Follow-up Required, Interested, Complaint, Payment Follow-up, No Response; date must not be before the call
Notes	Add notes (≤ 500 characters)
Order link	Link an order created from the call
Visibility	Salesmen see only their own calls; Sales Manager, Customer Service and Administrator see all
Storage	`bellomassi_call_history`

13.6 Documents and OCR

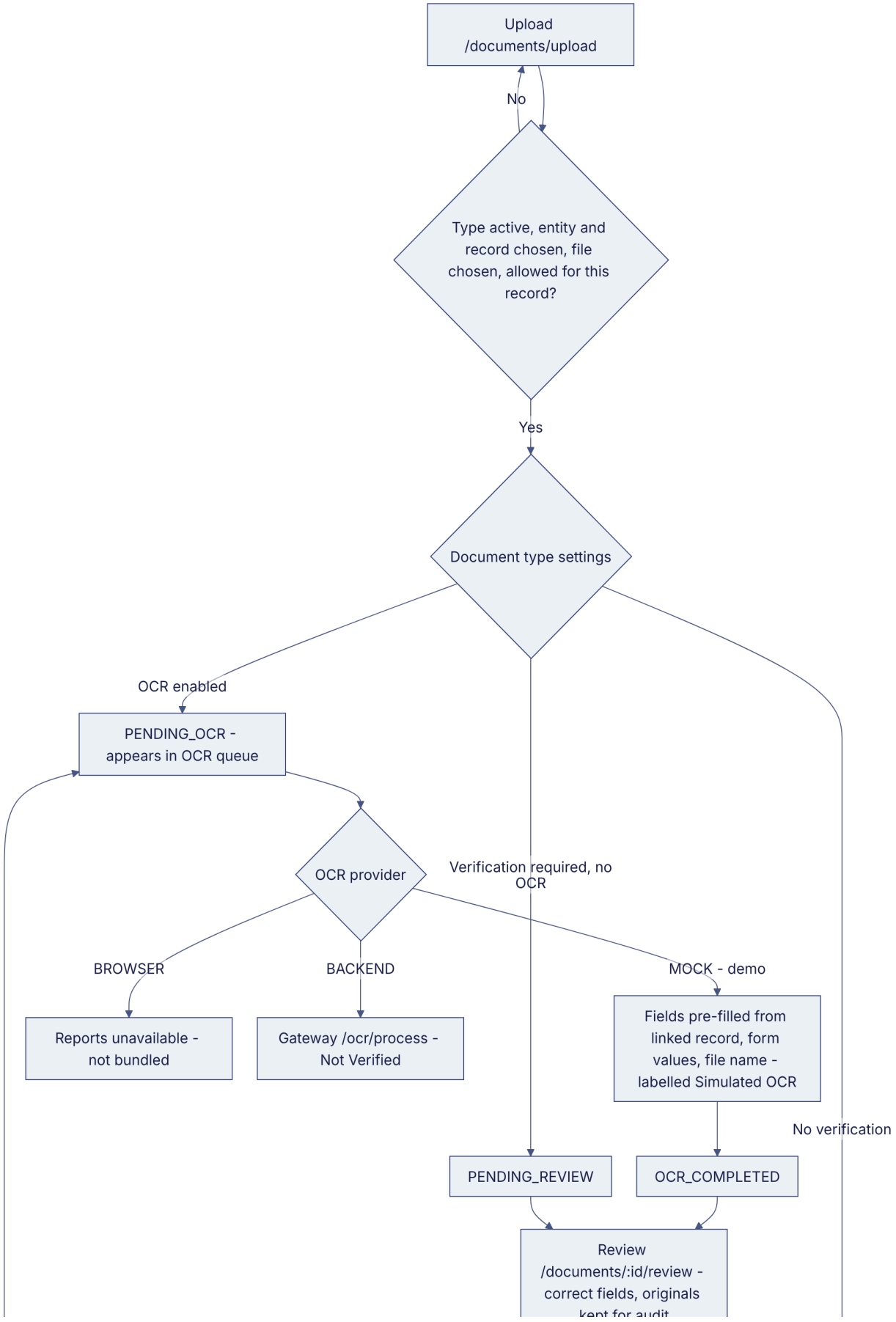


Diagram continued (1 of 2)

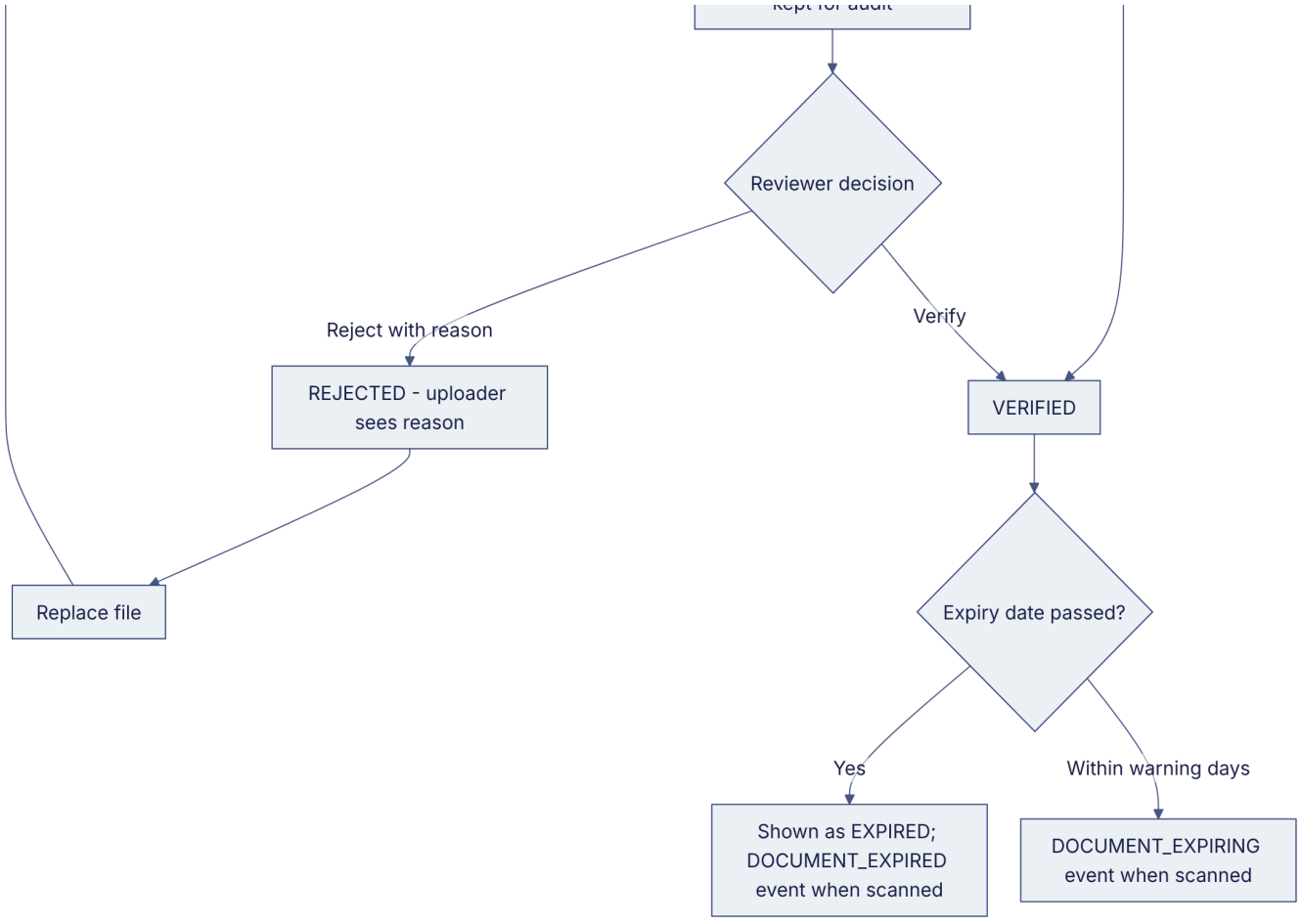
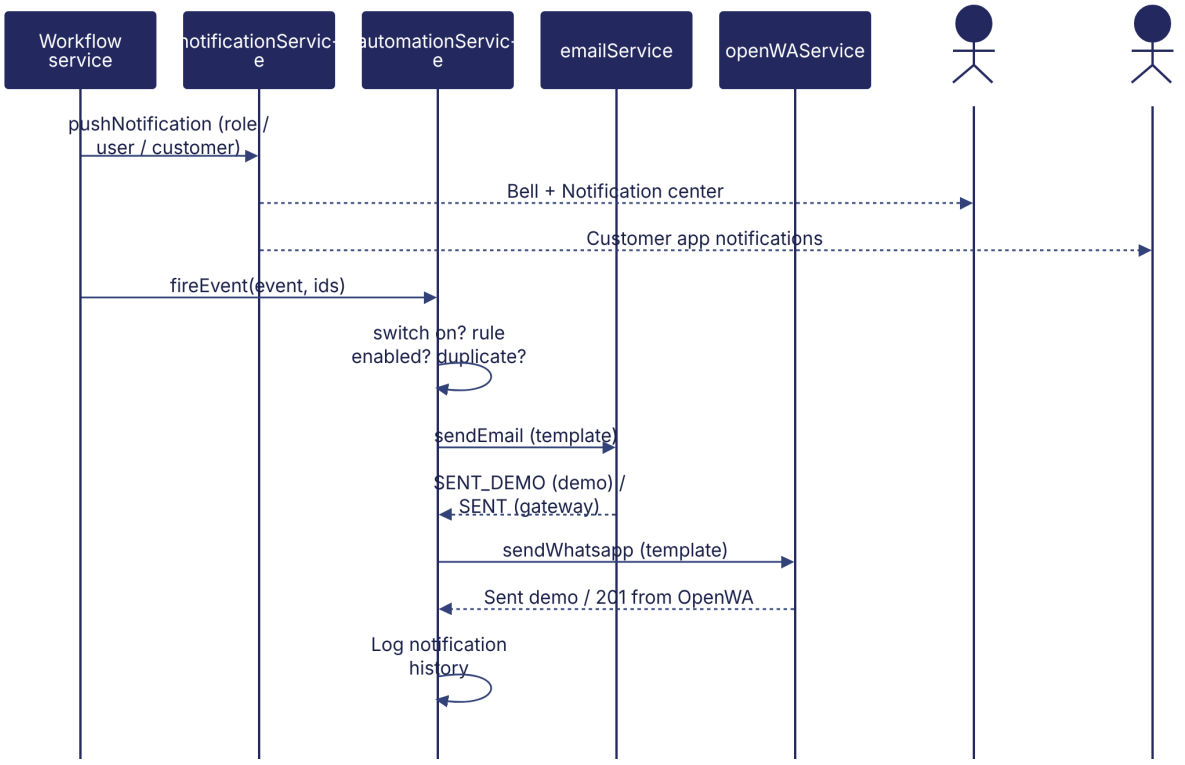


Diagram continued (2 of 2)

Item	Details
Who uploads	Most office roles and the Driver (own records only for scoped roles)
Who reviews	Administrator, Customer Service, QA Manager (`documents.review` , `documents.reject`)
Document types	`/documents/types` (Administrator): code, name, entity, fields to capture, expiry required, warning days, OCR enabled, verification required. Built-in codes (e.g. VEH_RC, VEH_INSURANCE, DRV_LICENSE) cannot change because VMS rules refer to them.
Storage	`bellomassi_documents` , files via `documentStorageService` , OCR results `bellomassi_ocr_results` , types `bellomassi_document_types`
Lists	Repository, OCR queue (PENDING_OCR, OCR_COMPLETED, PENDING_REVIEW), Verified, Rejected, Document reports

13.7 Communication — swimlane



14 · Reports & Dashboard Workflow

Feature	Status
Role dashboards (12 dashboards + generic launcher)	Implemented
Universal dashboard period filter with comparison	Implemented (selection kept in memory, not stored)
Reports Center — 118 reports in 11 categories	Implemented; **24 reports have no source data** in the prototype and say so
Module report pages (orders, delivery, invoices, documents, VMS)	Implemented
CSV export, print	Implemented
Forecasting and predictive inventory	Implemented (in-browser statistical models)
AI Assistant / Jarvis	**Simulated** (demo intent engine; external providers Not Verified)
AI insights	Implemented (rule-based)

14.1 Dashboards

Role	Dashboard component	Main KPIs / queues
Super Admin, Administrator	`AdminDashboard` + `OpsOverview`	Total orders, Pending approval, Approved, Rejected, Consolidated, Ready for dispatch, Order pipeline; Sales (delivered), Delivery, Inventory value, Finance — invoiced, Purchase (replenishment), AI insights
Sales Executive	`SalesmanDashboard`	My orders, Draft, Pending approval, Approved, Returned, Rejected, Needs your action, My order trend
Sales Manager	`SalesManagerDashboard`	Pending approvals, Returned orders, Recent pending orders, Active accounts and credit position
Customer Service	`CustomerServiceDashboard`	Approved orders, Awaiting consolidation, Consolidated, Ready for dispatch, Queue by salesman, Consolidation queue
Dispatcher	`DispatcherDashboard` / `DispatchOverview`	Ready for dispatch, Planned, Assigned, Confirmed, In transit, Completed, Failed, Partial deliveries, Active drivers, Available vehicles; trends; driver and zone performance
Finance	`FinanceDashboard`	Order value (submitted), Approved order value, Delivered to invoice, Invoices, Collections, Outstanding, Credit exposure / utilisation, Over credit limit
Warehouse Manager / Inventory Executive	`WarehouseDashboard` (inventory variant)	Stock value, Low / out of stock, Damaged / held, Expiry, Inventory alerts, Picking & loading
QA Manager	`QaDashboard`	Inspections pending, Quarantine, Expired documents, Expiry, Cold chain card (“No sensor integration”)
Supplier	`SupplierPortalDashboard`	Open purchase requests, Approved replenishment, Lines requested, Lead time, Goods receipts
Customer	Customer app Home	Active orders, Pending approval, Delivered, Outstanding
Driver	Driver app Today	Today / tomorrow stops, next stop, performance

Role	Dashboard component	Main KPIs / queues
Custom role	`GenericDashboard`	Workspace launcher (permission-driven)

Period filter

Options: Today, Yesterday, This Week, Last Week, Last 7 Days, Last 30 Days, This Month, Last Month, Month to Date, This Quarter, Last Quarter, Quarter to Date, This Half-Year, Last Half-Year, Half-Year to Date, This Year, Last Year, Year to Date, Custom Date Range, All Time, plus **Specific period** (day, week, month, quarter, half-year, calendar year, financial year; financial year starts in April by default). **Compare** shows % change, difference and previous value against the previous equivalent period. Defaults: office and customer dashboards **All Time**; driver **Today**.

Each dated collection is filtered on its own business date: orders (order date), approvals (decision date), dispatches (dispatch date), deliveries (completion / failure date), invoices (invoice date), receipts / transfers / adjustments (their dates), replenishment (created date), documents (upload date), delivery notes (delivery date), AI insights (generated date). Stock, inventory value, credit balances, customers, products, vehicles and drivers are **Current** positions. Source: `utils/dashboardFilter.js`, `context/DashboardPeriodContext.jsx`.

14.2 Master flowchart 11 — Reports workflow

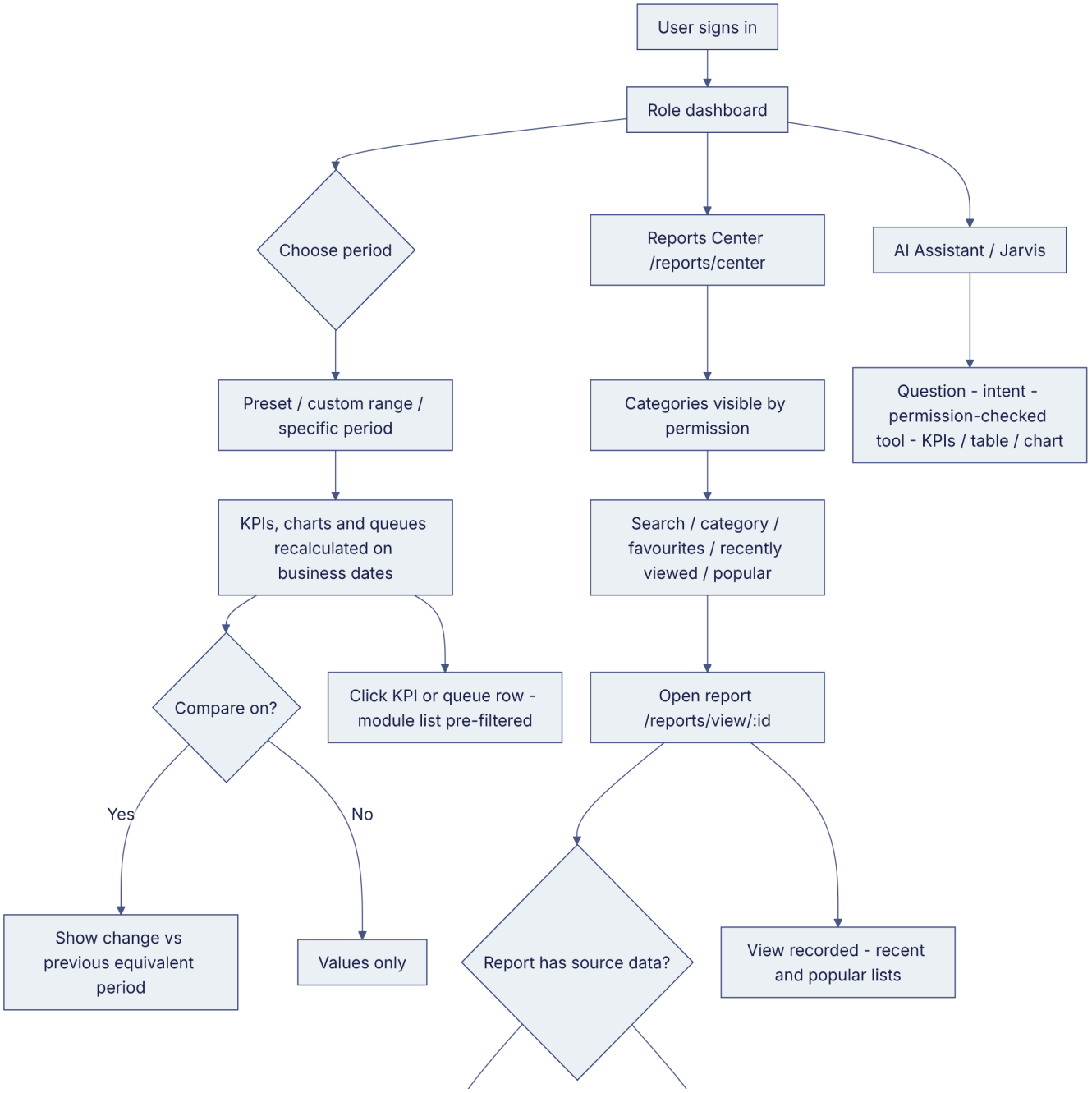


Diagram continued (1 of 2)

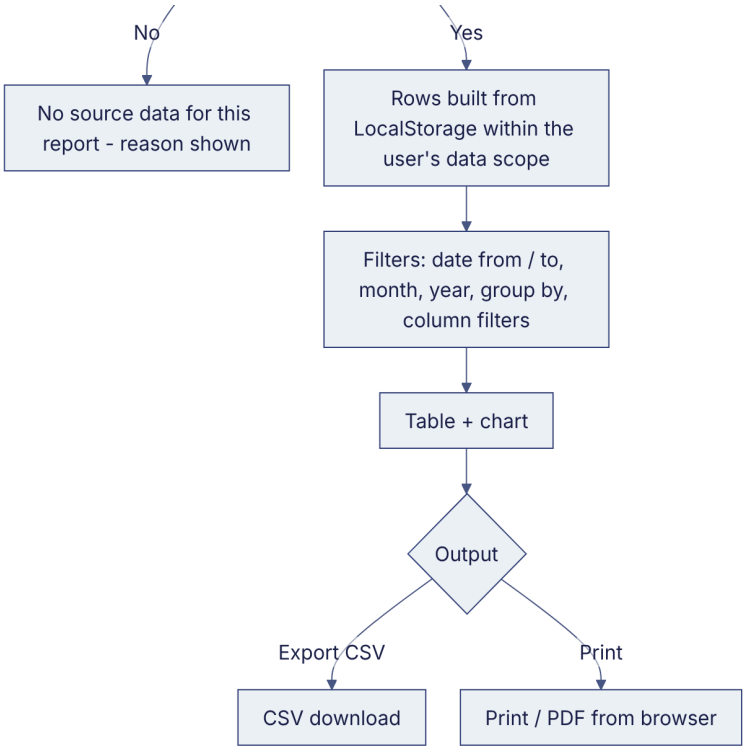


Diagram continued (2 of 2)

14.3 Reports Center catalogue

Entries marked * have **no source data** in this prototype; the report opens and states why (payments, credit notes, refunds, expenses, supplier invoices, purchase / customer returns, batch / expiry on stock, cycle counts, fuel and maintenance are not captured).

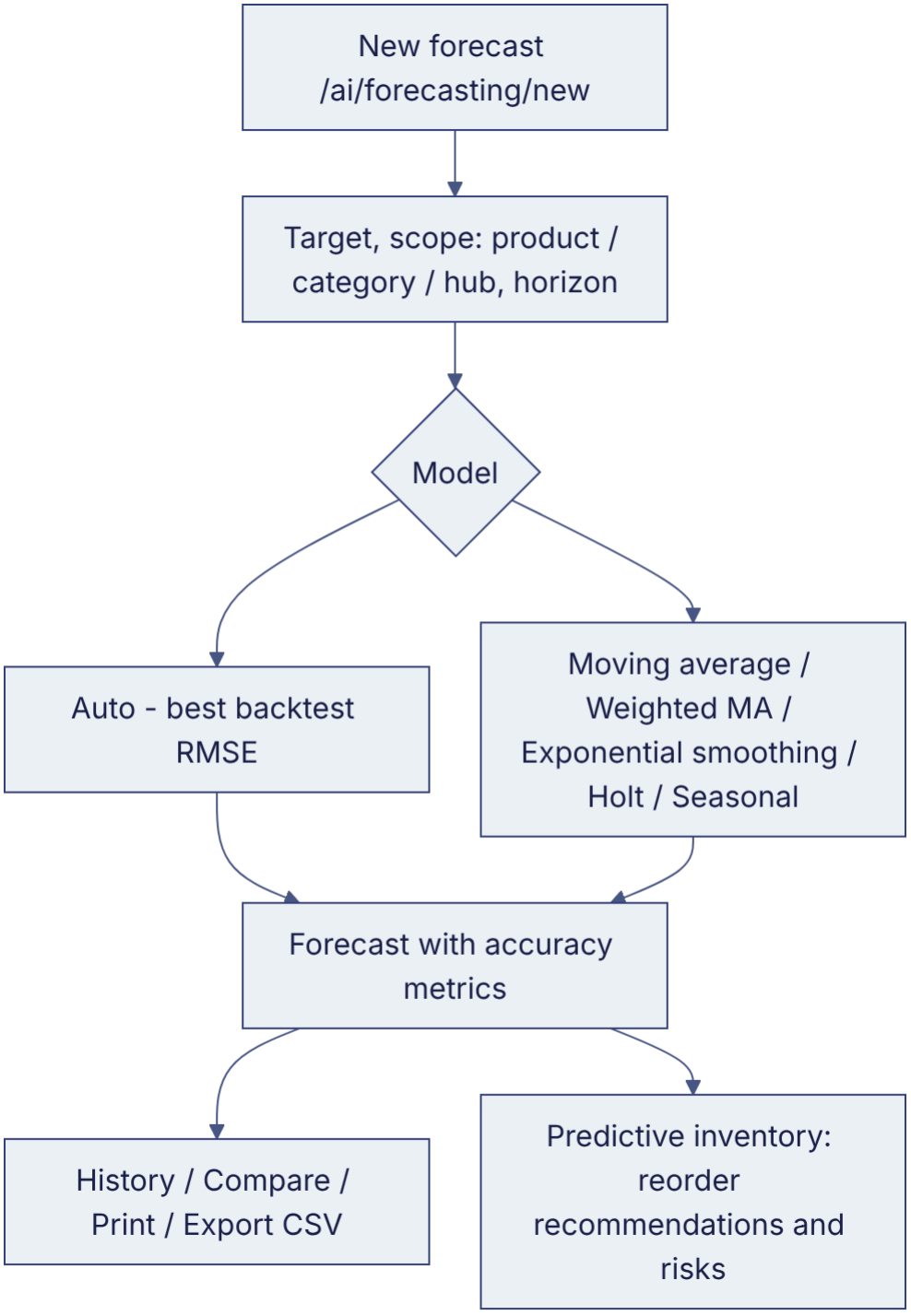
Category	Reports	Visible with	Report titles
Sales	14	`orders.reports`	Sales Order Register, Sales Order Summary, Sales by Customer, Sales by Salesman, Sales by Product, Sales by Category, Sales by Month, Sales Order Status, Pending Approval, Rejected Orders, Credit Limit Exceptions, Order Conversion, Customer Sales Analysis, VIP Orders
Inventory	18	`reports.inventory`	Stock Overview, Stock Valuation, Stock Movement, Stock Ageing*, Batch Inventory*, Expiry Report*, Near Expiry*, Expired Stock, Slow Moving Stock, Non Moving Stock, Stock-Out Report, Stock Transfer, Stock Adjustment, Scrap Register, Scrap Valuation, Warehouse Utilization, Cycle Count Variance*, Inventory Transaction History
Procurement	10	`procurement.view`	Purchase Order Register, Purchase Order Status, Purchase by Supplier, Purchase by Product, Goods Receipt Register, Pending Goods Receipts, Purchase Return*, Supplier Invoice Register*, Supplier Outstanding*, Procurement Performance
Dispatch & Delivery	19	`dispatch.reports` or `dispatch.history`	Dispatch Register, Dispatch Summary, Dispatch by Zone, Dispatch by Driver, Dispatch by Vehicle, Delivery Register, Delivered Orders, Pending Deliveries, Failed Deliveries, Partial Deliveries, Delivery Performance, On-Time Delivery, VIP Delivery Performance, Delivery Proof Report, OTP Verification Report, Delivery Note Register, Driver Performance, Route Performance, Customer Delivery History

Category	Reports	Visible with	Report titles
Finance	14	`finance.view or invoices.reports`	Customer Outstanding, Customer Ageing, Receivable Summary, Invoice Register, Payment Register*, Payment Allocation*, Credit Note Register*, Refund Register*, Supplier Outstanding*, Supplier Ageing*, Supplier Payment Register*, Expense Register*, Expense Category Report*, Finance Summary
Customer	8	`customers.view`	Customer Master, Customer Sales, Customer Outstanding, Customer Order History, Customer Delivery History, Customer Payment History*, Customer Return History*, Customer Performance
Supplier	5	`suppliers.view`	Supplier Master, Supplier Purchase, Supplier Outstanding*, Supplier Payment*, Supplier Performance
Fleet	9	`fleet.view or vms.view`	Vehicle Register, Driver Register, Vehicle Utilization, Vehicle Maintenance*, Vehicle Expenses*, Fuel Register*, Vehicle Document Expiry, Driver Performance, Delivery per Vehicle
Communication	7	`communication.calls`	Call History, Calls by Salesman, Calls by Customer, Call Outcome, Follow-up Pending, Overdue Follow-ups, Call-to-Order Conversion
Scrap	7	`inventory.scrap.view`	Scrap Register, Scrap by Product, Scrap by Batch, Scrap by Hub, Scrap by Reason, Scrap Valuation, Scrap Approval History
AI	7	`ai.insights.view`	Sales Insights, Inventory Insights, Demand Forecast, Expiry Risk, Slow-Moving Products, Delivery Insights, Business Summary

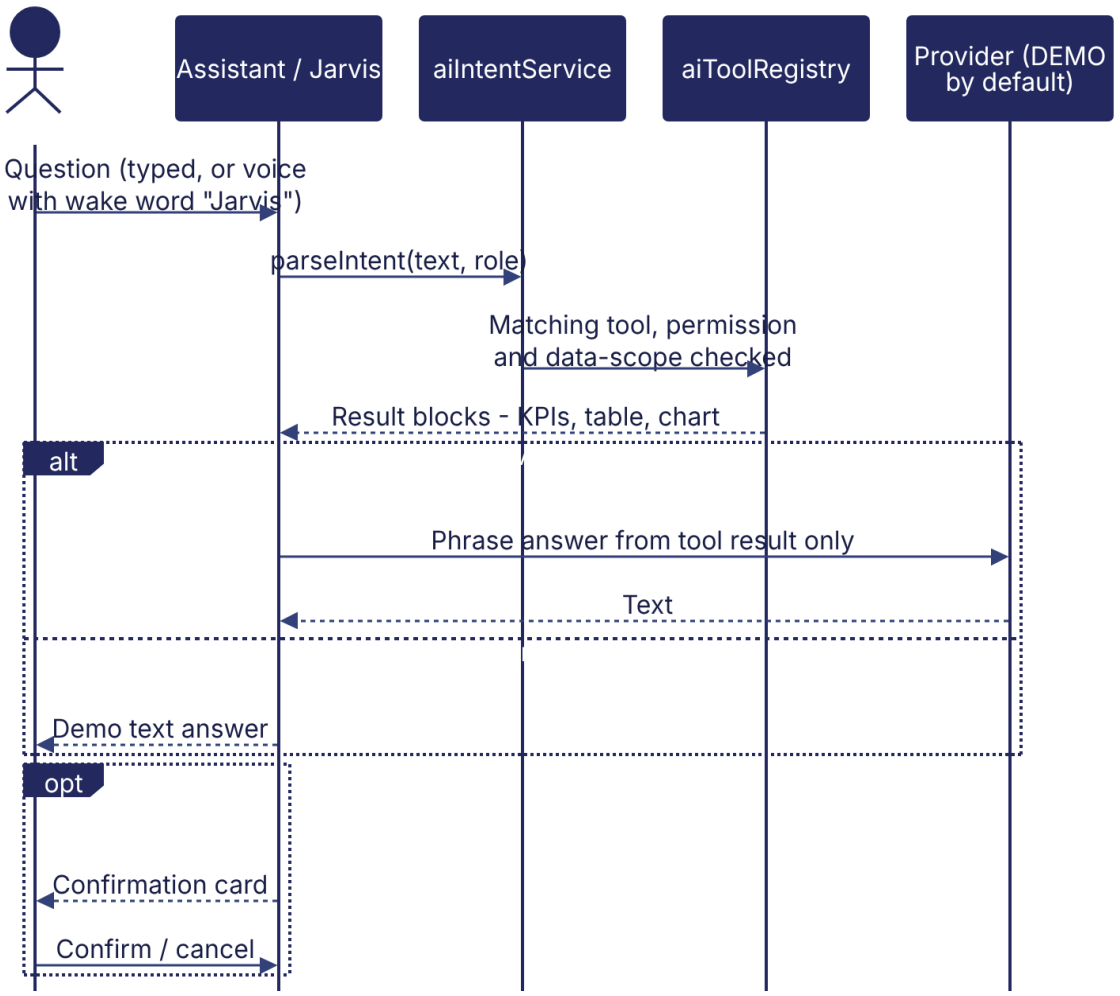
Total: 118 reports; 94 with data, 24 marked *.

Module report pages also exist: Order Reports `/orders/reports`, Delivery Reports `/dispatch/reports`, Invoice Reports `/invoices/reports`, Document Reports `/documents/reports`, VMS Reports `/vms/reports`. Legacy routes `/reports`, `/reports/deliveries`, `/reports/inventory`, `/reports/invoices` redirect to the current pages.

14.4 Forecasting



14.5 AI Assistant and Jarvis



Conversations are private to each user. Every assistant action is logged in AI Activity (`/ai/activity`). AI Settings (`/ai/settings`, Administrator) choose the provider: Demo, OpenJarvis, Ollama, Gemini, OpenAI-compatible or Custom — only Demo was verified.

15 · Exception Workflow

An exception is anything that stops the normal path. In Bellomassi every exception is handled by the service that owns the record: the action is refused with a message and nothing is saved, or the record moves to an exception status that a named role must resolve. Repeating an action that already happened never changes data twice (“No change was made.”).

15.1 Master flowchart 12 — Exception workflow

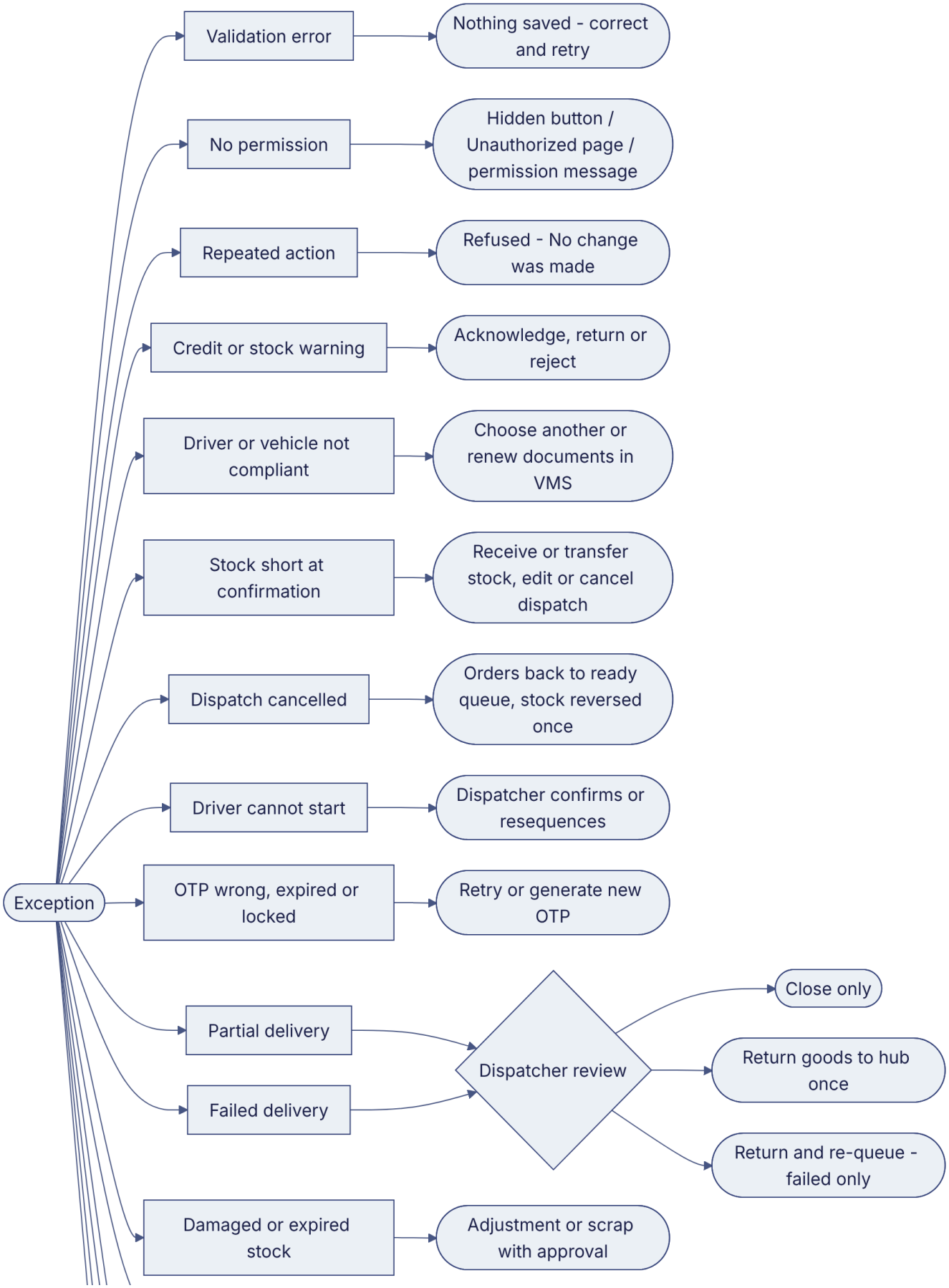


Diagram continued (1 of 2)

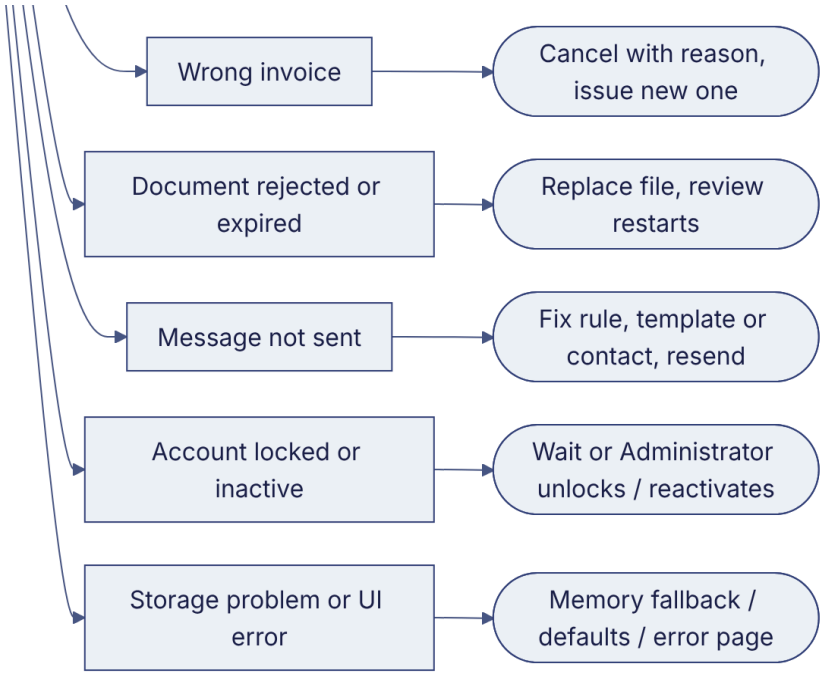


Diagram continued (2 of 2)

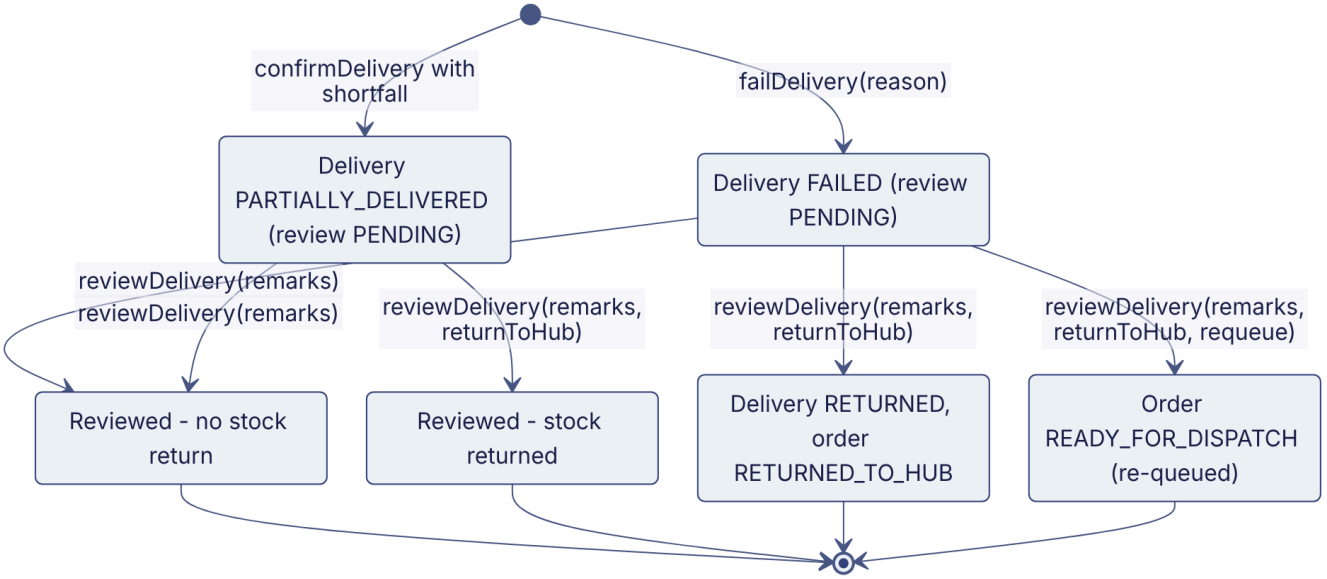
15.2 Exception register

#	Exception	Detected in	What the user sees	Who resolves	Resolution	Status
1	Wrong password	`authService.login`	“The email/username or password is incorrect…”	User	Retry; reset password	Implemented
2	Account locked (5 failures)	`authService`	“Too many failed attempts. This account is locked for N minutes.”	User / Administrator	Wait, or **Unlock** in User Management	Implemented
3	Inactive account	`authService`	“This account is inactive…”	Administrator	Reactivate user	Implemented
4	Idle session	`SessionWatcher`	Signed out	User	Sign in again	Implemented
5	No permission	`ProtectedRoute`, services	`/unauthorized` or “You do not have permission to…”	Administrator	Grant permission / role	Implemented
6	Order data invalid	`orderService.validateOrder`	Field messages	Sales Executive	Correct and submit	Implemented
7	Credit limit exceeded	`getApprovalChecks`	Warning (or block if configured)	Sales Manager	Acknowledge, return (“Reduce order value (credit)”) or reject	Implemented
8	Stock short at approval	`getApprovalChecks`	Warning	Sales Manager	Acknowledge or return / reject (“Stock not available”)	Implemented
9	Inactive customer / hub / product	`getApprovalChecks`	“Cannot approve: …”	Administrator / Sales Manager	Fix master data, or reject	Implemented

#	Exception	Detected in	What the user sees	Who resolves	Resolution	Status
10	Order returned	`returnOrderForCorrection`	Salesman notification with edit link, highlighted section	Sales Executive	Correct and resubmit	Implemented
11	Order rejected	`rejectOrder`	Rejected Orders list, notification	—	Final; a new order can be created (or duplicated)	Implemented
12	Customer wants to cancel a submitted order	—	—	—	**Not Implemented** (Sales Manager may reject with "Customer cancelled")	Not Implemented
13	Consolidation checklist incomplete	`consolidateOrder`	"Complete the verification checklist first."	Customer Service	Tick all checks	Implemented
14	Order not eligible for dispatch	`orderIneligibility`	"Only orders ready for dispatch can be dispatched." / already on another dispatch / no complete address	Dispatcher / Customer Service	Wait for release, fix address	Implemented
15	Driver / vehicle not compliant or busy	`driverAssignmentIssues`, `vehicleAssignmentIssues`	Specific issue messages	Dispatcher / Administrator	Choose another, renew documents, maintenance	Implemented
16	Load exceeds capacity	`vehicleAssignmentIssues`	"Load of N exceeds … capacity"	Dispatcher	Bigger vehicle or split dispatch	Implemented
17	Stock short at confirmation	`deductDispatchStock`	"…needs N, only M available at the hub."	Warehouse / Dispatcher	Receive or transfer stock; edit dispatch	Implemented
18	Dispatch cancelled	`cancelDispatch`	Driver and Admin notified; orders back in ready queue	Dispatcher	Re-plan; stock reversed if it was deducted	Implemented
19	Driver starts out of order / too early	`startDelivery`	Blocker text under the Start button	Driver / Dispatcher	Follow sequence; Dispatcher resequences or confirms	Implemented
20	Wrong OTP	`otpService.evaluateOtp`	Error on confirm screen; attempt counted	Driver / customer	Re-enter	Simulated OTP
21	OTP expired / locked	`otpService`	"This OTP has expired. Generate a new one." / "Too many incorrect attempts. Generate a new OTP."	Driver	Generate a new OTP	Simulated OTP
22	Partial delivery	`confirmDelivery`	PARTIALLY_ DELIVERED; remaining stays with Dispatcher	Dispatcher	Review; return undelivered stock to hub	Implemented
23	Failed delivery	`failDelivery`	FAILED; reasons list; customer told "Our team will contact you to reschedule."	Dispatcher	Review; return to hub; re-queue	Implemented

#	Exception	Detected in	What the user sees	Who resolves	Resolution	Status
24	Damaged goods at delivery	`confirmDelivery` (damaged qty)	Recorded on the delivery and note	Dispatcher / Warehouse	Damaged quantity is not returned to available stock by review; use adjustment / scrap	Implemented
25	Transfer more than available	`validateTransfer`	“Only N … available at …”	Inventory staff	Reduce quantity	Implemented
26	Adjustment would go negative	`validateAdjustment`	“Cannot remove more than the N available…”	Inventory staff	Correct quantity	Implemented
27	Scrap exceeds eligible / duplicate / missing photo or batch	`validateScrap`	“Only N eligible…”, photo / batch messages	Requester	Correct request	Implemented
28	Scrap rejected	`rejectScrap`	Rejection reason	Requester	New request if needed	Implemented
29	Invoice wrong	`cancelInvoice`	CANCELLED with reason	Finance	Issue a new invoice	Implemented
30	Invoice for ineligible / already invoiced order	`invoiceEligibility`	Eligibility message	Finance	Wait or cancel existing invoice	Implemented
31	Document rejected	`rejectDocument`	Reason shown to uploader	Uploader	Replace file	Implemented
32	Document expired	`effectiveStatus`, VMS	EXPIRED badge; assignment blocked for required VMS documents	Administrator / Dispatcher / QA	Upload renewed document, update expiry	Implemented
33	OCR not possible	`ocrService`	“The file is not stored in this browser…”, provider unavailable	Reviewer	Enter fields manually	Implemented
34	Automated message skipped / failed	`automationService`	History entry with note	Administrator	Enable automation / rule, add template or contact	Implemented
35	Cold-chain temperature excursion	—	“No sensor integration”	—	Record a scrap request with reason “Temperature Excursion”	Not Implemented (monitoring)
36	Browser storage blocked or corrupted	`storageService`	App keeps working in memory; corrupted key falls back to defaults	User / Administrator	Allow storage; reset demo data if needed	Implemented
37	Unknown page	Router	Not Found page	User	Use the menu	Implemented

15.3 Delivery exception review — detail



Rules (`deliveryService.reviewDelivery`): only Dispatcher / Administrator; only FAILED or PARTIALLY_DELIVERED; remarks required; goods can be returned once; re-queue only for a failed delivery whose goods are returned; a second review is refused.

16 · Role Responsibility Matrix

Role abbreviations: **SA** Super Admin · **AD** Administrator · **SE** Sales Executive · **SM** Sales Manager · **CS** Customer Service · **DP** Dispatcher · **DR** Driver · **FI** Finance · **WM** Warehouse Manager · **IE** Inventory Executive · **QA** QA Manager · **CU** Customer · **SU** Supplier.

Legend: **R** responsible (does the work) · **A** approves / decides · **C** can act or contribute · **V** view only · blank = no access. Derived from the default permissions in `src/config/permissions.js` and the service rules.

16.1 Process responsibility matrix

Process step	SA	AD	SE	SM	CS	DP	DR	FI	WM	IE	QA	CU	SU
Maintain customers	R	R	V	V	V								
Maintain products	R	R	V	V	V	V		V	V	V	V		
Maintain hubs	R	R		V	V	V			V	V	V		
Maintain suppliers	R	R		V		V		V	V				
Maintain zones, drivers, vehicles	R	R				V							
Create / submit order	C	C	R									R	
Approve / return / reject order	A	V		A									
Correct returned order	C	C	R										
Consolidate, mark ready for dispatch	C	V			R								
Plan dispatch, assign driver and vehicle	R	R		V	V	R		V	V				
Confirm dispatch (stock deducted)	R	R				R							
Cancel dispatch	R	R				R							
Deliver (start, arrive, OTP, POD, complete / fail)	C*						R					C (shares OTP)	
Review failed / partial delivery	A	A				A							
Issue / cancel invoice	R	R		V				R				V	
Invoice templates and numbering	R	R						R					
Stock receipt	R	R				V			R	R	V		
Stock transfer	R	R				V			R	R	V		
Stock adjustment	R	R				V			R	V	R		
Scrap request	R	R				R			R	R	R		
Scrap approval / write-off	A	A							A				
Replenishment draft	R	R				R			R	R			
Replenishment approval / convert	A	A											
View approved replenishment													V
Vehicle / driver compliance rules	R	R											
Upload VMS documents	R	R			R	R							
Upload documents	R	R	R	R	R	R	R	R	R	R	R		
Verify / reject documents	A	A			A						A		
Document types	R	R											
Send email / WhatsApp	R	R	R	R	R	R		R					
Automation rules	R	R											
Call log and follow-ups	R	R	R (own)	R	R								
Reports Center	V	V	V	V	V	V		V	V	V	V		
Generate forecasts	R	R		R									

Process step	SA	AD	SE	SM	CS	DP	DR	FI	WM	IE	QA	CU	SU
Users, roles, permissions	R	R											
Application settings, organization	R	R											
Audit log	V	V											

* Super Admin holds `driver.app` and `customer.app` but can only act in those apps if the user is linked to a driver or customer record.

16.2 Hand-off matrix (who passes work to whom)

From	Hand-off	To	Trigger in code
Sales Executive / Customer	Submitted order	Sales Manager	`submitOrder` notification
Sales Manager	Returned order	Sales Executive	`returnOrderForCorrection` notification with edit link
Sales Manager	Approved order	Customer Service	`approveOrder` notification
Customer Service	Ready order	Dispatcher	`markReadyForDispatch` notification
Dispatcher	Assigned / confirmed dispatch	Driver	`assignDispatch`, `confirmDispatch` notifications
Driver	Delivery started / arrived / completed	Customer	`startDelivery`, `markArrived`, `confirmDelivery` notifications
Driver	Failed / partial delivery	Dispatcher	`failDelivery` notification (“Review required”)
Driver	Delivered order	Finance	Delivery note; “Delivered, to invoice” on Finance dashboard
Finance	Invoice	Customer	Invoice visible in customer app; automation INVOICE_GENERATED
Inventory Executive / QA	Scrap request	Warehouse Manager	Scrap queue (PENDING_APPROVAL)
Warehouse / Inventory / Dispatcher	Replenishment draft	Administrator	Replenishment list (SUBMITTED)
Administrator	Approved replenishment	Supplier, warehouse	Supplier portal; receipt links
Any uploader	Document	Customer Service / QA / Administrator	OCR queue

16.3 Permission matrix (default grants)

All 110 permission keys from `src/config/rbacCatalog.js` with the default grant per system role (`src/config/permissions.js`). Super Admin always holds every key. Administrators can change grants for non-locked roles in **Role Management**.

Permission key	Meaning	SA	AD	SE	SM	CS	DP	DR	FI	WM	IE	QA	CU	SU
`dashboard.view`	Role dashboard	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
`customers.view`	Customer list & details	✓	✓	✓	✓	✓								
`customers.manage`	Create, edit and delete customers	✓	✓											

Permission key	Meaning	SA	AD	SE	SM	CS	DP	DR	FI	WM	IE	QA	CU	SU
customers.credit.view	Customer credit	✓	✓		✓				✓					
products.view	Product catalogue	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓		
products.manage	Create, edit and delete products	✓	✓											
suppliers.view	Supplier directory	✓	✓		✓		✓		✓	✓				
suppliers.manage	Edit supplier records	✓	✓											
hubs.view	Hubs & warehouses	✓	✓		✓	✓	✓			✓	✓	✓		
hubs.manage	Create, edit and delete hubs	✓	✓											
inventory.dashboard	Inventory dashboard	✓	✓				✓		✓	✓	✓			
inventory.stock.view	Stock overview	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓		
inventory.lowstock.view	Low stock	✓	✓		✓	✓	✓			✓	✓	✓		
inventory.movements.view	Stock movement ledger	✓	✓				✓			✓	✓	✓		
inventory.documents.view	Receipts, transfers & adjustments lists	✓	✓				✓			✓	✓	✓		
inventory.receipts.manage	Receive stock (GRN)	✓	✓							✓	✓			
inventory.transfers.manage	Stock transfers between hubs	✓	✓							✓	✓			
inventory.adjustments.manage	Post stock adjustments	✓	✓							✓		✓		
inventory.scrap.view	Scrap management	✓	✓				✓			✓	✓	✓		
scrap.create	Create, edit & submit scrap requests	✓	✓				✓			✓	✓	✓		
scrap.approve	Approve scrap & post the stock write-off	✓	✓							✓				
scrap.reject	Reject scrap requests	✓	✓							✓				
warehouse.dashboard	Warehouse dashboard	✓	✓							✓	✓	✓		
warehouse.qa.view	QA & cold-chain overview	✓	✓							✓		✓		
orders.view	Order list & details	✓	✓	✓	✓	✓	✓		✓	✓				
orders.create	Create & edit sales orders	✓	✓	✓										
orders.mine	My orders	✓	✓	✓										
orders.corrections.view	Returned & rejected orders	✓	✓	✓	✓									
orders.approval.view	Approval queue	✓	✓		✓									
orders.approve	Approve orders	✓			✓									
orders.reject	Reject or return orders	✓			✓									
orders.consolidation.view	Consolidation queue	✓	✓			✓								
orders.consolidate	Consolidate & release orders	✓				✓								
orders.reports	Order reports	✓	✓		✓				✓					
sales.targets.manage	Set sales targets	✓	✓		✓									
customer.app	Customer app — catalogue, own orders, tracking & documents	✓											✓	
customer.order	Customer app — place orders	✓											✓	
tracking.fleet	Live fleet map (demo GPS)	✓	✓				✓							
dispatch.view	Dispatch dashboard & planning	✓	✓		✓	✓	✓		✓	✓				
dispatch.ready.view	Ready-for-dispatch queue	✓	✓			✓	✓			✓				
dispatch.manage	Create dispatches & assign drivers / vehicles	✓	✓				✓							
dispatch.tracking	Delivery tracking	✓	✓		✓	✓	✓							
dispatch.history	Delivery history	✓	✓		✓	✓	✓		✓	✓				
dispatch.reports	Delivery reports	✓	✓				✓		✓					
dispatch.otp.inspect	Inspect simulated delivery OTPs	✓	✓				✓							
deliveries.view	Delivery details	✓	✓	✓	✓	✓	✓		✓	✓				
fleet.view	Zones, drivers & vehicles	✓	✓				✓							
fleet.manage	Manage zones, drivers & vehicles	✓	✓											

Permission key	Meaning	SA	AD	SE	SM	CS	DP	DR	FI	WM	IE	QA	CU	SU
deliveryNotes.view	Delivery notes	✓	✓	✓	✓	✓	✓		✓	✓				
driver.app	Driver app — run deliveries & confirm with OTP	✓						✓						
vms.view	VMS dashboard, register & expiry	✓	✓				✓					✓		
vms.manage	Manage vehicle register	✓	✓											
vms.documents.upload	Upload vehicle & driver documents	✓	✓			✓	✓							
vms.settings	Compliance rules	✓	✓											
documents.view	Document repository	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓		
documents.upload	Upload documents	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
documents.review	Run OCR review & verify documents	✓	✓			✓						✓		
documents.reject	Reject documents	✓	✓			✓						✓		
documents.types	Document types & OCR fields	✓	✓											
documents.reports	Document reports	✓	✓			✓						✓		
invoices.view	Invoices	✓	✓		✓				✓					
invoices.manage	Generate & cancel invoices	✓	✓						✓					
invoices.templates	Invoice templates & numbering	✓	✓						✓					
invoices.reports	Invoice reports	✓	✓						✓					
communication.view	Communication center, sent & history	✓	✓	✓	✓	✓	✓		✓					
communication.send	Compose & send email	✓	✓	✓	✓	✓	✓		✓					
communication.inbox	Email inbox	✓	✓		✓	✓								
communication.templates	Email & WhatsApp templates	✓	✓											
communication.settings	SMTP / IMAP & OpenWA settings	✓	✓											
communication.automation	Automated notification rules	✓	✓											
communication.calls	Call history	✓	✓	✓	✓	✓								
calls.manage	Log, edit and delete call records & follow-ups	✓	✓	✓	✓	✓								
whatsapp.view	WhatsApp home & history	✓	✓	✓	✓	✓	✓		✓					
whatsapp.send	Compose & send WhatsApp	✓	✓	✓	✓	✓	✓		✓					
ai.assistant.view	AI overview & assistant	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓		
ai.assistant.use	Ask the AI assistant	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓		
ai.insights.view	AI insights	✓	✓		✓	✓	✓		✓	✓		✓		
ai.insights.manage	Acknowledge & resolve insights	✓	✓		✓		✓							
jarvis.view	Jarvis	✓	✓	✓	✓	✓	✓		✓	✓				
jarvis.use	Use Jarvis commands	✓	✓	✓	✓	✓	✓		✓	✓				
forecasting.view	Forecasting	✓	✓		✓		✓		✓	✓				
forecasting.generate	Generate forecasts	✓	✓		✓									
forecasting.export	Export & print forecasts	✓	✓		✓				✓					
inventory.predictions.view	Predictive inventory	✓	✓		✓		✓		✓	✓				
inventory.predictions.manage	Planning parameters & recommendation status	✓	✓											
inventory.replenishment.create	Create & submit replenishment drafts	✓	✓				✓			✓	✓			
procurement.view	Replenishment drafts	✓	✓		✓		✓		✓	✓	✓			
procurement.approve	Approve & convert replenishment drafts	✓	✓											
procurement.reject	Reject replenishment drafts	✓	✓											
ai.settings.manage	AI provider settings	✓	✓											
ai.models.manage	Model management	✓	✓											
reports.inventory	Inventory reports	✓	✓		✓				✓	✓				

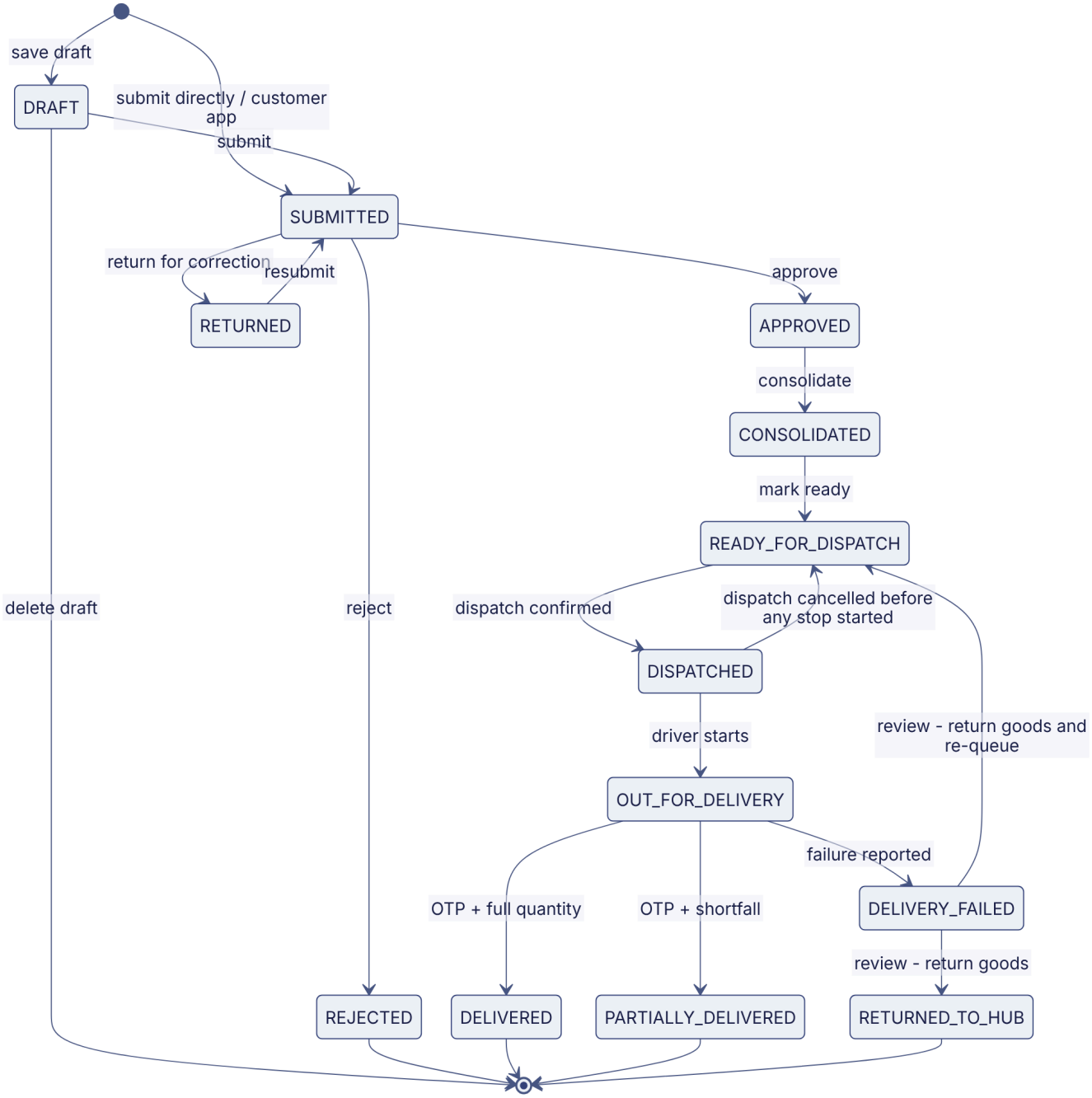
Permission key	Meaning	SA	AD	SE	SM	CS	DP	DR	FI	WM	IE	QA	CU	SU
`reports.center`	Reports center (each report also needs its module permission)	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓		
`data.export`	Export CSV files	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓		
`data.import`	Import a backup	✓	✓											
`finance.view`	Finance overview (outstanding & collections)	✓	✓						✓					
`settings.view`	Application settings (read-only)	✓	✓											
`settings.manage`	Change application settings	✓	✓											
`organization.view`	Organization master	✓	✓											
`organization.manage`	Edit organization master	✓	✓											
`users.view`	Users	✓	✓											
`users.create`	Create users	✓	✓											
`users.edit`	Edit users	✓	✓											
`users.delete`	Delete users	✓	✓											
`users.manage`	Activate / deactivate users, reset demo passwords, assign roles & hubs	✓	✓											
`roles.view`	Roles & permission matrix	✓	✓											
`roles.create`	Create & duplicate roles	✓	✓											
`roles.edit`	Edit roles & permissions	✓	✓											
`roles.delete`	Delete custom roles	✓	✓											
`audit.view`	Audit log	✓	✓											

17 · Status Lifecycle

All values below are the exact codes stored in LocalStorage, with the label the UI shows. Sources are given for each set.

17.1 Order status (`orderStatus`, `ORDER_STATUS_META` in `src/config/orderConfig.js`)

Code	Label	Step	Set by
DRAFT	Draft	1	`createOrder`, `updateOrder`
SUBMITTED	Submitted	2	`submitOrder`, `resubmitOrder`, `placeCustomerOrder`
RETURNED	Returned for correction	2	`returnOrderForCorrection`
REJECTED	Rejected	0	`rejectOrder`
APPROVED	Approved	3	`approveOrder`
CONSOLIDATED	Consolidated	4	`consolidateOrder`
READY_FOR_DISPATCH	Ready for dispatch	5	`markReadyForDispatch`; `cancelDispatch`; `reviewDelivery` (re-queue)
DISPATCHED	Dispatched	6	`confirmDispatch`
OUT_FOR_DELIVERY	Out for delivery	7	`startDelivery`
PARTIALLY_DELIVERED	Partially delivered	8	`confirmDelivery` (shortfall)
DELIVERED	Delivered	8	`confirmDelivery` (full)
DELIVERY_FAILED	Delivery failed	7	`failDelivery`
RETURNED_TO_HUB	Returned to hub	7	`reviewDelivery` (failed + return to hub, no re-queue)



Notes: while an order sits on a DRAFT / PLANNED / ASSIGNED dispatch it stays READY_FOR_DISPATCH (with `dispatchId` set). Cancelling a dispatch that was never confirmed also leaves orders READY_FOR_DISPATCH.

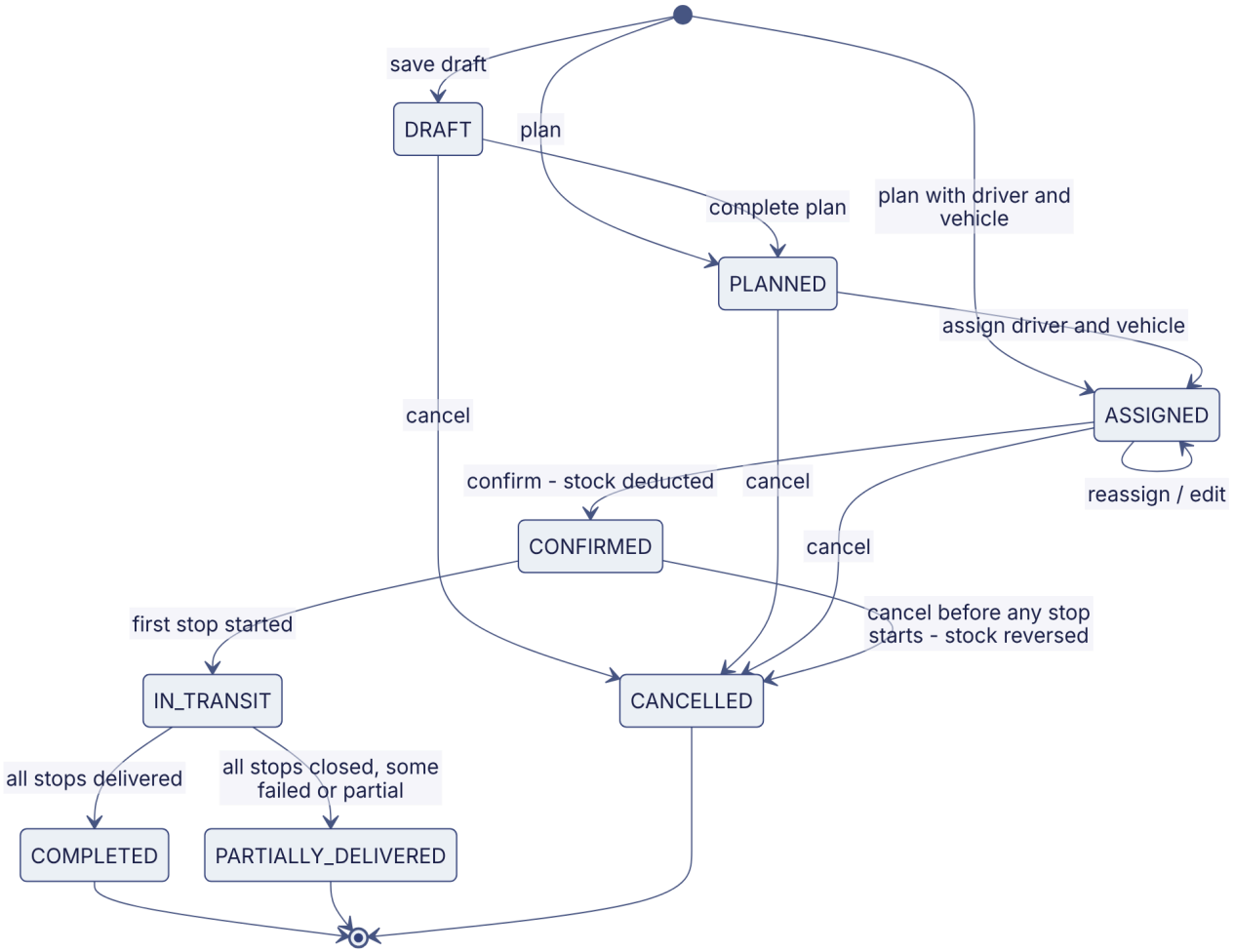
Companion fields

Field	Values	Meaning
`approvalStatus`	NOT_SUBMITTED, PENDING, APPROVED, REJECTED, RETURNED	Sales Manager decision
`consolidationStatus`	NOT_STARTED, PENDING, COMPLETED	PENDING after approval; COMPLETED after consolidation
`orderSource`	e.g. `customer_app` for customer orders	Where the order came from

Field	Values	Meaning
`creditWarning`, `stockWarning`	true / false	Flags shown to the approver

17.2 Dispatch status (`DISPATCH_STATUS_META`, `src/config/dispatchConfig.js`)

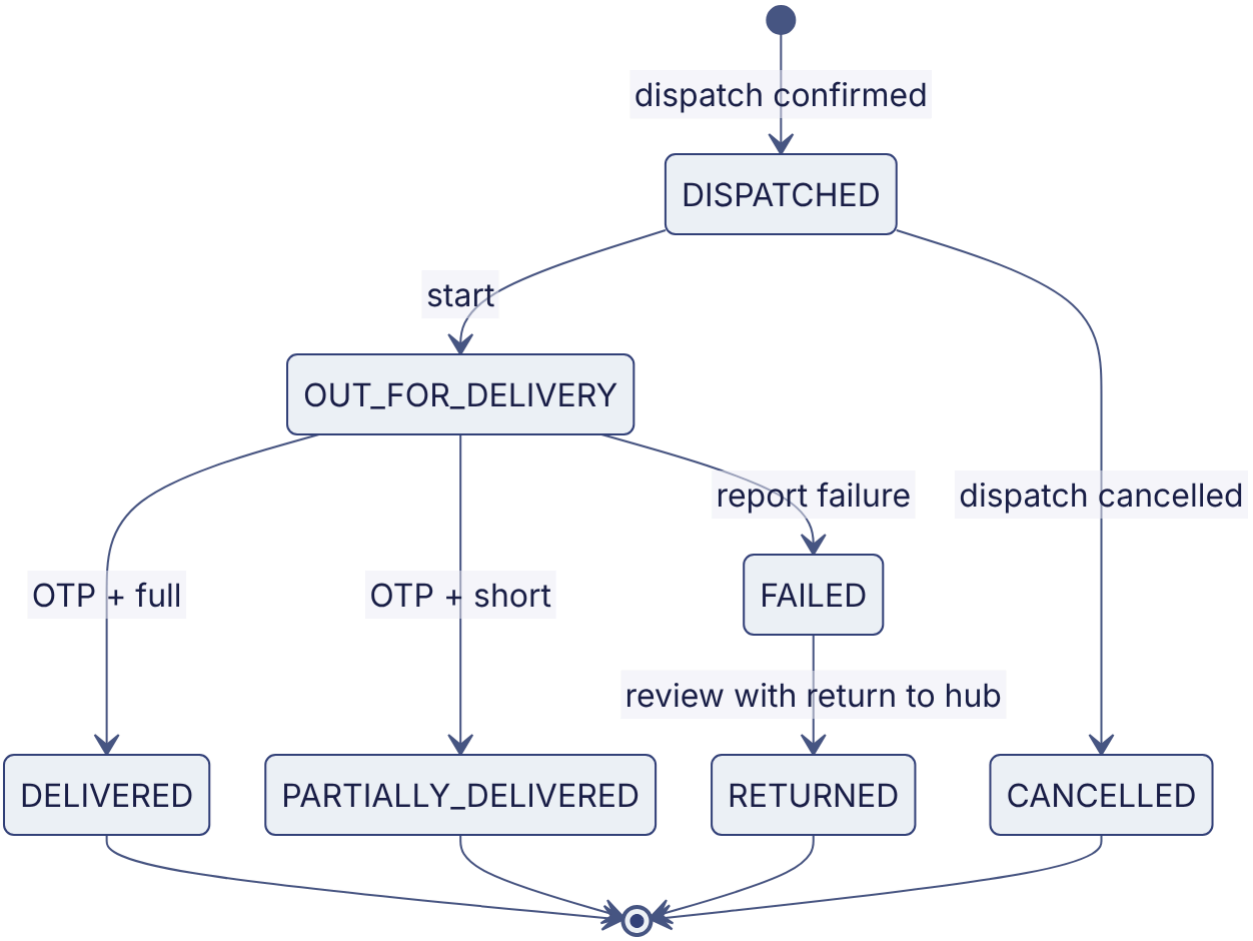
Code	Label	Step
DRAFT	Draft	0
PLANNED	Planned	1
ASSIGNED	Assigned	2
CONFIRMED	Confirmed	3
IN_TRANSIT	In transit	4
PARTIALLY_DELIVERED	Closed with exceptions	5
COMPLETED	Completed	5
CANCELLED	Cancelled	−1



17.3 Delivery status (`DELIVERY_STATUS`)

Code	Label	Set by
DISPATCHED	Dispatched	`confirmDispatch` (one delivery per order)
OUT_FOR_DELIVERY	Out for delivery	`startDelivery`
DELIVERED	Delivered	`confirmDelivery`
PARTIALLY_DELIVERED	Partially delivered	`confirmDelivery`
FAILED	Failed	`failDelivery`
RETURNED	Returned	`reviewDelivery` (failed + return to hub)
CANCELLED	Cancelled	`cancelDispatch` (confirmed dispatch)

Review status on failed / partial deliveries: `PENDING` → `REVIEWED`. Terminal delivery statuses (`TERMINAL_DELIVERY_STATUSES`) cannot be restarted.



17.4 Delivery OTP (`OTP_STATUS`)

Code	Label
ACTIVE	Awaiting entry
VERIFIED	Verified

Code	Label
EXPIRED	Expired (derived when `expiresAt` passes)
INVALIDATED	Replaced by a newer OTP
LOCKED	Too many attempts (5)

17.5 Scrap request (`SCRAP_STATUS`, `scrapService`)

DRAFT → PENDING_APPROVAL → APPROVED → COMPLETED; PENDING_APPROVAL → REJECTED; DRAFT / PENDING_APPROVAL / APPROVED → CANCELLED. Diagram: chapter 07 §7.3.

17.6 Replenishment (`REPLENISHMENT_META`, `src/config/aiConfig.js`)

Code	Label
DRAFT	Draft
SUBMITTED	Submitted for approval
APPROVED	Approved
REJECTED	Rejected
CONVERTED	Awaiting goods receipt
RECEIVED	Received (derived from stock receipts)
CANCELLED	Cancelled

Reorder recommendation (`RECOMMENDATION_STATUS`): OPEN, IN_DRAFT, DISMISSED. Diagram: chapter 07 §7.4.

17.7 Document (`DOCUMENT_STATUS_META`, `src/config/documentConfig.js`)

Code	Label
PENDING_OCR	Pending OCR
OCR_COMPLETED	OCR completed
PENDING_REVIEW	Pending review
VERIFIED	Verified
REJECTED	Rejected
EXPIRED	Expired (derived from expiry date)

Diagram: chapter 07 §7.5.

17.8 Invoice (`INVOICE_STATUS_META`, `src/config/invoiceConfig.js`)

Code	Label
ISSUED	Issued

Code	Label
CANCELLED	Cancelled

17.9 Stock transfer (`TRANSFER_STATUS` , `src/config/masterData.js`)

Draft → Completed; Draft → Cancelled.

17.10 Stock status (`STOCK_STATUS` , `utils/stock.getStockStatus`)

In Stock · Low Stock · Out of Stock · Overstock (first match: Out → Low → Over → In).

17.11 Messages — email and WhatsApp (`MESSAGE_STATUS_META` , `src/config/communicationConfig.js`)

Code	Label
DRAFT	Draft
QUEUED	Queued
SENDING	Sending
SENT	Sent (accepted by provider)
DELIVERED	Delivered (provider confirmed)
FAILED	Failed
QUEUED_DEMO	Queued · demo
SENT_DEMO	Sent · demo (not sent)
FAILED_DEMO	Failed · demo

Automation history status: SKIPPED, SENT_DEMO / SENT / DELIVERED, PARTIAL, FAILED.

17.12 VMS expiry state (`EXPIRY_META` , `src/config/vmsConfig.js`)

Code	Label
EXPIRED	Expired
TODAY	Expires today
WEEK	Within 7 days
MONTH	Within 30 days
VALID	Valid
MISSING	Not on record

17.13 Fleet

Set	Values
Vehicle status	Available, Assigned, In Transit (derived from dispatches), Maintenance, Inactive
Driver status	Active, Inactive
Driver availability	Available, Assigned, On Delivery, Offline

17.14 Master data and users

Set	Values
Customer status	Active, Inactive, Blocked
Hub status	Active, Inactive
Product status	Active, Inactive, Discontinued (only Active products can be ordered and approved)
Zone status	Active, Inactive
User status	active / inactive; temporary lock after failed sign-ins
Role status	active / inactive; system, protected and locked flags

17.15 AI and calls

Set	Values
AI insight	NEW (New), REVIEWED (Reviewed), RESOLVED (Resolved)
Call status	Connected, Missed, Busy, No Answer, Voicemail
Call outcome	Order Received, Interested, Follow-up Required, No Response, Busy, Not Interested, Complaint, Payment Follow-up, Other
Call follow-up	Open (overdue / today / upcoming) → Completed

18 · Screen-Wise User Guide

Every screen in the application, in simple English. “Who” uses the role abbreviations from chapter 16 (SA Super Admin can open every screen and is not repeated). Screens reached from a list (details, edit, print) are included after their list.

18.1 Before you sign in

Screen	Route	What you do here
Sign in	`/login`	Type your email or username and password. Tick “Keep me signed in” on your own computer. Pick a demo account from the list if you are trying the demo.
Driver sign in	`/driver/login`	Drivers sign in with mobile number, email or username.
Customer sign in	`/customer/login`	Customers sign in with mobile number, email or username.
Forgot password	`/forgot-password`	Get a 6-digit reset code (shown on screen in the demo) or restore the demo password.
Reset password	`/reset-password`	Enter the code and your new password.
Unauthorized	`/unauthorized`	You opened a page your role cannot use. Go back to your dashboard.
Not found	any unknown address	The page does not exist. Use the menu.

18.2 Common controls

- **Sidebar** (office app): menu groups for your role; collapse it with the arrow. **Bottom bar** (driver and customer apps).
- **Header**: global search, notifications bell, theme, your name menu (profile, password, sign out).
- **Lists**: search box, filters, column sorting, pagination, export where allowed.
- **Forms**: required fields are marked; errors appear under the field; the save buttons stay at the bottom.
- **Status badges**: colours follow the status (green done, amber waiting, red problem, grey closed).
- **DEMO badges**: “DEMO GPS”, “Simulated OCR”, “Sent · demo” mark simulated behaviour.

18.3 Dashboard

Screen	Route	Who	What you do here
Dashboard	`/dashboard`	All	Your home page. Shows the numbers and work queues for your role. Pick a period at the top; click any card to open the list behind it.

18.4 Order Management

Screen	Route	Who	What you do here
Create Order	`/orders/create`	AD, SE	Fill in customer, hub, delivery date and address, add products and quantities. Check the totals and warnings, then Save draft or Submit for approval.
My Orders	`/orders/my-orders`	AD, SE	All orders you created, with their status. Open one to see details, edit a draft or a returned order.
Order List	`/orders`	AD, SE, SM, CS, DP, FI, WM	Every order you are allowed to see. Search and filter by status, customer, salesman, hub or date.
Order Approval	`/orders/approval`	AD, SM	Orders waiting for your decision, oldest first. Open an order to approve, return or reject it.
Order Consolidation	`/orders/consolidation`	AD, CS	Approved orders waiting to be checked against the warehouse. Open one to consolidate it.
Returned Orders	`/orders/returned`	AD, SE, SM	Orders the Sales Manager sent back. Open one to see the reason, correct it and resubmit.
Rejected Orders	`/orders/rejected`	AD, SE, SM	Orders that were rejected, with the reason. They cannot be changed; duplicate one to start again.
Order Reports	`/orders/reports`	AD, SM, FI	Order totals and trends by status, salesman, customer and period.
Edit Order	`/orders/:id/edit`	AD, SE	Change a draft or a returned order. For a returned order the section to fix is highlighted. Resubmit when done.
Review Order	`/orders/approval/:id`	AD, SM	Review one order: approval checks (customer, hub, credit, stock, total, products), lines and history. Approve with optional remarks, or return / reject with a reason.
Consolidate Order	`/orders/consolidation/:id`	AD, CS	Tick the five verification checks, review hub stock, add notes, Consolidate, then Mark ready for dispatch.
Order Details	`/orders/:id`	AD, SE, SM, CS, DP, FI, WM	Everything about one order: lines, totals, approval, dispatch, delivery, invoice and the full timeline.

18.5 Customers

Screen	Route	Who	What you do here
Customer List	`/customers`	AD, SE, SM, CS	List of customers. Search, filter and open a customer.
Customer Credit	`/customers/credit`	AD, SM, FI	Credit limit, balance and utilisation for every customer. Spot who is over the limit.
Add Customer	`/customers/create`	AD	Add a new customer: details, contacts, address, GST / PAN, hub, zone, sales rep and credit settings.
Customer Details	`/customers/:id`	AD, SE, SM, CS	One customer: profile, credit position, orders, deliveries and documents.
Edit Customer	`/customers/:id/edit`	AD	Change a customer's details or credit settings.

18.6 Master data

Screen	Route	Who	What you do here
Products	`/products`	AD, SE, SM, CS, DP, FI, WM, IE, QA	Product catalogue with price, tax, unit and status. Open or add a product.
Suppliers	`/suppliers`	AD, SM, DP, FI, WM	Suppliers used for replenishment. Add or edit a supplier and its lead time.
Hubs	`/hubs`	AD, SM, CS, DP, WM, IE, QA	Hubs (warehouses / distribution points). Open or add a hub.
Add Product	`/products/create`	AD	Add a product: code, SKU, barcode, category, unit, price, tax, discount limit, storage and stock settings.
Product Details	`/products/:id`	AD, SE, SM, CS, DP, FI, WM, IE, QA	One product: details and stock at each hub.
Edit Product	`/products/:id/edit`	AD	Change a product.
Add Hub	`/hubs/create`	AD	Add a hub: code, name, type, address and contacts.
Hub Details	`/hubs/:id`	AD, SM, CS, DP, WM, IE, QA	One hub: details, stock, zones and fleet.
Edit Hub	`/hubs/:id/edit`	AD	Change a hub.

18.7 Inventory

Screen	Route	Who	What you do here
Inventory Dashboard	`/inventory`	AD, DP, FI, WM, IE	Inventory overview: stock value, status mix, alerts and recent stock documents.
Warehouse Dashboard	`/warehouse`	AD, WM, IE, QA	Warehouse view: picking and loading for today, stock alerts, damaged stock and document expiry.
QA & Cold Chain	`/quality`	AD, WM, QA	QA view: documents to inspect, quarantine (damaged stock), expiries and the cold-chain card (no live sensors).
Stock Overview	`/inventory/stock`	AD, SE, SM, CS, DP, FI, WM, IE, QA	Stock for every product at every hub, with available quantity and status.
Low Stock	`/inventory/low-stock`	AD, SM, CS, DP, WM, IE, QA	Lines at or below the reorder level, or out of stock. Start a receipt, transfer or replenishment from here.
Stock Receipts	`/inventory/stock-receipts`	AD, DP, WM, IE, QA	List of goods received (GRN). Open one or create a new receipt.
Stock Transfers	`/inventory/stock-transfers`	AD, DP, WM, IE, QA	Transfers between hubs. Complete or cancel drafts.
Stock Adjustments	`/inventory/stock-adjustments`	AD, DP, WM, IE, QA	Stock corrections (increase, decrease, damaged, lost, correction) with reasons.
Stock Movements	`/inventory/movements`	AD, DP, WM, IE, QA	The stock ledger: every movement in and out with reference.

Screen	Route	Who	What you do here
Scrap Management	`/inventory/scrap`	AD, DP, WM, IE, QA	Scrap overview: requests by status, value and reason.
Scrap Requests	`/inventory/scrap/requests`	AD, DP, WM, IE, QA	All scrap requests. Filter by status and open one.
New Scrap Request	`/inventory/scrap/requests/create`	AD, DP, WM, IE, QA	Request scrap: product, hub, stock source, quantity, reason, batch and photo. Save draft or submit.
Edit Scrap Request	`/inventory/scrap/requests/:id/edit`	AD, DP, WM, IE, QA	Change your draft scrap request.
Scrap Note	`/inventory/scrap/requests/:id/print`	AD, DP, WM, IE, QA	Printable scrap note.
Scrap Request	`/inventory/scrap/requests/:id`	AD, DP, WM, IE, QA	One scrap request: details, photo, history; approve, reject, write off or cancel depending on your role.
Stock Details	`/inventory/stock/:id`	AD, SE, SM, CS, DP, FI, WM, IE, QA	One stock line: quantities, recent movements, and minimum / reorder / maximum levels.
New Receipt	`/inventory/stock-receipts/create`	AD, WM, IE	Record goods received: hub, product, quantity, date and reference. Stock goes up immediately.
New Transfer	`/inventory/stock-transfers/create`	AD, WM, IE	Move stock from one hub to another. Save as draft or complete now.
New Adjustment	`/inventory/stock-adjustments/create`	AD, WM, QA	Correct stock with a type and reason. Posted immediately.

18.8 Dispatch

Screen	Route	Who	What you do here
Dispatch Dashboard	`/dispatch`	AD, SM, CS, DP, FI, WM	Dispatch overview: ready orders, dispatches by status, deliveries, drivers, vehicles and performance charts.
Ready for Dispatch	`/dispatch/ready-orders`	AD, CS, DP, WM	Orders released by Customer Service and waiting to be planned. Select orders to start a dispatch.
Dispatch Planning	`/dispatch/planning`	AD, SM, CS, DP, FI, WM	All dispatches by date and status. Open one or create a new dispatch.
Driver Assignment	`/dispatch/assignments`	AD, DP	Dispatches that need a driver and vehicle. Assign only compliant, free resources from the same hub.
Delivery Tracking	`/dispatch/delivery-tracking`	AD, SM, CS, DP	Every stop on active dispatches with live status.
Delivery History	`/dispatch/delivery-history`	AD, SM, CS, DP, FI, WM	Closed stops. Failed and partial deliveries waiting for review are highlighted.
Dispatch Zones	`/dispatch/zones`	AD, DP	Delivery zones (pincode groups) for each hub.
Drivers	`/dispatch/drivers`	AD, DP	Driver list with availability, hub and licence status.
Vehicles	`/dispatch/vehicles`	AD, DP	Vehicle list with status, capacity and hub.
Delivery Reports	`/dispatch/reports`	AD, DP, FI	Delivery performance: on-time, failed, partial, by driver and zone.

Screen	Route	Who	What you do here
New Dispatch	`/dispatch/create`	AD, DP	Build a dispatch: pick ready orders, hub, zone, date, put the stops in order, optionally assign driver and vehicle. Save draft or plan.
New Zone	`/dispatch/zones/create`	AD	Add a zone with its pincodes.
Edit Zone	`/dispatch/zones/:id/edit`	AD	Change a zone.
Live Tracking (Demo GPS)	`/dispatch/live-tracking`	AD, DP	Map with simulated (DEMO GPS) positions of vehicles on the road.
New Driver	`/dispatch/drivers/create`	AD	Add a driver: code, contact, hub, zone, licence, vehicle and app login.
Edit Driver	`/dispatch/drivers/:id/edit`	AD	Change a driver.
Driver Details	`/dispatch/drivers/:id`	AD, DP	One driver: profile, compliance and performance.
New Vehicle	`/dispatch/vehicles/create`	AD	Add a vehicle: number, type, capacity, hub, category, fuel, ownership and document dates.
Edit Vehicle	`/dispatch/vehicles/:id/edit`	AD	Change a vehicle.
Delivery Details	`/dispatch/deliveries/:id`	AD, SE, SM, CS, DP, FI, WM	One stop: items, delivered / damaged quantities, proof of delivery, OTP (for the Dispatcher) and the review panel for exceptions.
Delivery Note	`/dispatch/delivery-notes/:id`	AD, SE, SM, CS, DP, FI, WM	Printable delivery note (BM-DN number) with what was delivered and who received it.
Edit Dispatch	`/dispatch/:id/edit`	AD, DP	Change a dispatch that is still Draft, Planned or Assigned.
Dispatch Details	`/dispatch/:id`	AD, SM, CS, DP, FI, WM	One dispatch: stops, sequence, driver, vehicle, stock check, timeline. Confirm (loading sign-off) or cancel here.

18.9 My Deliveries

Screen	Route	Who	What you do here
Today	`/driver/dashboard`	DR	Driver home: today’s and tomorrow’s stops, your next stop and your performance.
My Deliveries	`/driver/deliveries`	DR	All your stops. Tap one to open it.
Route	`/driver/route`	DR	Map of your stops in order (DEMO GPS).
Completed	`/driver/completed`	DR	Stops you have finished.
My Delivery Notes	`/driver/delivery-notes`	DR	Delivery notes for your completed stops.
Driver Profile	`/driver/profile`	DR	Your details and vehicle. Sign out from here.
My Documents	`/driver/documents`	DR	Your licence and other documents. Upload new ones.

Screen	Route	Who	What you do here
Document	`/driver/documents/:id`	DR	One of your documents.
Confirm Delivery	`/driver/deliveries/:id/confirm`	DR	Enter what the customer accepted, who received it, optional signature and photo, then the customer's OTP to complete.
Delivery	`/driver/deliveries/:id`	DR	One stop: address, contact, items. Start delivery, Reach location, Complete with OTP, or Report issue.
Delivery Note	`/driver/delivery-notes/:id`	DR	A delivery note to show the customer.

18.10 My Account (Customer)

Screen	Route	Who	What you do here
Home	`/customer/dashboard`	CU	Customer home: your orders, deliveries on the way, notes and balance.
Catalog	`/customer/catalog`	CU	Browse and search products and add them to your cart.
Cart	`/customer/cart`	CU	Check your cart, choose the delivery address and date, and place the order.
My Orders	`/customer/orders`	CU	Your orders with filters (active, pending, delivered).
Order	`/customer/orders/:id`	CU	One order: progress tracker, items, status. Reorder from here.
Track Delivery	`/customer/tracking/:id`	CU	Follow the driver on the map (DEMO GPS). When the driver arrives, your OTP is shown here.
Delivery Notes	`/customer/delivery-notes`	CU	Delivery notes for your deliveries.
Delivery Note	`/customer/delivery-notes/:id`	CU	One delivery note.
Invoices	`/customer/invoices`	CU	Your invoices.
Invoice	`/customer/invoices/:id`	CU	One invoice.
Payments	`/customer/payments`	CU	Your outstanding balance. Payment history is not available in this prototype.
Profile	`/customer/profile`	CU	Your account details. Sign out from here.

18.11 Vehicle Management

Screen	Route	Who	What you do here
VMS Dashboard	`/vms`	AD, DP, QA	Fleet compliance overview: vehicles and drivers by document status.
Vehicles	`/vms/vehicles`	AD, DP, QA	Vehicles with their compliance state.
Driver Documents	`/vms/drivers`	AD, DP, QA	Drivers and their documents.
Document Expiry	`/vms/document-expiry`	AD, DP, QA	All vehicle and driver documents sorted by urgency: expired, today, 7 days, 30 days.

Screen	Route	Who	What you do here
VMS Reports	`/vms/reports`	AD, DP, QA	Fleet and compliance reports.
Compliance Rules	`/vms/settings`	AD	Choose which documents are required and when expiry blocks assignment.
New Vehicle	`/vms/vehicles/create`	AD	Add a vehicle (VMS).
Edit Vehicle	`/vms/vehicles/:id/edit`	AD	Change a vehicle (VMS).
Vehicle Documents	`/vms/vehicles/:id/documents`	AD, DP, QA	A vehicle's documents and expiry dates.
Vehicle Details	`/vms/vehicles/:id`	AD, DP, QA	One vehicle: details and compliance.
Driver Documents	`/vms/drivers/:id/documents`	AD, DP, QA	A driver's documents and expiry dates.

18.12 Documents

Screen	Route	Who	What you do here
Document Repository	`/documents`	AD, SE, SM, CS, DP, FI, WM, IE, QA	All documents you may see. Search by type, record or status.
Upload Documents	`/documents/upload`	AD, SE, SM, CS, DP, FI, WM, IE, QA	Upload a document: type, what it belongs to, the record, the file and key fields.
OCR Queue	`/documents/ocr-queue`	AD, SE, SM, CS, DP, FI, WM, IE, QA	Documents waiting for OCR or review.
Verified	`/documents/verified`	AD, SE, SM, CS, DP, FI, WM, IE, QA	Verified documents.
Rejected	`/documents/rejected`	AD, SE, SM, CS, DP, FI, WM, IE, QA	Rejected documents with the reason.
Document Types	`/documents/types`	AD	Document types and the fields each one captures.
Document Reports	`/documents/reports`	AD, CS, QA	Document counts by type, status and expiry.
New Document Type	`/documents/types/create`	AD	Add a document type.
Edit Document Type	`/documents/types/:id/edit`	AD	Change a document type.
OCR Review	`/documents/:id/review`	AD, SE, SM, CS, DP, FI, WM, IE, QA	Check the extracted fields against the file, correct them, then verify or reject.
Document	`/documents/:id`	AD, SE, SM, CS, DP, FI, WM, IE, QA	One document: file, fields, status and history.

18.13 Invoices

Screen	Route	Who	What you do here
Invoice Dashboard	`/invoices/dashboard`	AD, SM, FI	Invoice overview: issued value, orders delivered but not invoiced, recent invoices.
Invoices	`/invoices`	AD, SM, FI	All invoices. Search and open.
New Invoice	`/invoices/create`	AD, FI	Pick an order, choose template and options, check the live preview, then issue the invoice.
Invoice Templates	`/invoices/templates`	AD, FI	Invoice designs. Set the default template.
Invoice Reports	`/invoices/reports`	AD, FI	Invoice totals by status, customer and period.
New Template	`/invoices/templates/create`	AD, FI	Design a new invoice template.
Edit Template	`/invoices/templates/:id/edit`	AD, FI	Change an invoice template.
Template Preview	`/invoices/templates/:id/preview`	AD, SM, FI	See a template with sample data.
Print Invoice	`/invoices/:id/print`	AD, SM, FI	Print-ready invoice.
Invoice	`/invoices/:id`	AD, SM, FI	One invoice: lines, totals, status. Print, email or cancel.

18.14 Communication

Screen	Route	Who	What you do here
Communication Center	`/communication`	AD, SE, SM, CS, DP, FI	Communication hub: email, WhatsApp, automation and calls at a glance.
Email	`/communication/email`	AD, SE, SM, CS, DP, FI	Email home: counts and mode (demo or live).
WhatsApp	`/communication/whatsapp`	AD, SE, SM, CS, DP, FI	WhatsApp home: session state and recent messages.
Automated Notifications	`/communication/automation`	AD	Switch automation on or off and set, for each business event, the channel, recipients and templates. See the send history.
Call History	`/communication/calls`	AD, SE, SM, CS	Call overview: today’s calls, outcomes and follow-ups due.
Call Follow-ups	`/communication/calls/follow-ups`	AD, SE, SM, CS	Follow-ups by overdue, today and upcoming. Complete them here.
All Calls	`/communication/calls/list`	AD, SE, SM, CS	All calls you may see.

Screen	Route	Who	What you do here
Log a Call	`/communication/calls/create`	AD, SE, SM, CS	Log a call: customer, direction, status, outcome, duration, notes and follow-up.
Edit Call	`/communication/calls/:id/edit`	AD, SE, SM, CS	Change a call.
Call Details	`/communication/calls/:id`	AD, SE, SM, CS	One call: details, notes, follow-up and linked order.
Compose Email	`/communication/email/compose`	AD, SE, SM, CS, DP, FI	Write an email (or use a template) and send it. In demo mode it is recorded, not sent.
Inbox	`/communication/email/inbox`	AD, SM, CS	Received emails (demo mailbox unless a live gateway is connected).
Sent	`/communication/email/sent`	AD, SE, SM, CS, DP, FI	Emails you sent and their status.
Drafts	`/communication/email/drafts`	AD, SE, SM, CS, DP, FI	Saved email drafts.
History	`/communication/email/history`	AD, SE, SM, CS, DP, FI	Full email log.
Email Templates	`/communication/email/templates`	AD, SE, SM, CS, DP, FI	Email templates with variables.
New Template	`/communication/email/templates/create`	AD	Add an email template.
Edit Template	`/communication/email/templates/:id/edit`	AD	Change an email template.
Email Settings	`/communication/email/settings`	AD	Email server settings (used only through the gateway).
Compose	`/communication/whatsapp/compose`	AD, SE, SM, CS, DP, FI	Write and send a WhatsApp message. In demo mode it is recorded, not sent.
History	`/communication/whatsapp/history`	AD, SE, SM, CS, DP, FI	WhatsApp message log.
WhatsApp Templates	`/communication/whatsapp/templates`	AD, SE, SM, CS, DP, FI	WhatsApp templates.
New Template	`/communication/whatsapp/templates/create`	AD	Add a WhatsApp template.
Edit Template	`/communication/whatsapp/templates/:id/edit`	AD	Change a WhatsApp template.
OpenWA Settings	`/communication/whatsapp/settings`	AD	OpenWA session settings (QR, start / stop) through the gateway.

18.15 Administration

Screen	Route	Who	What you do here
User Management	`/admin/users`	AD	All user accounts with role, hub and status.
Role Management	`/admin/roles`	AD	All roles with their data scope and number of users.
Permission Matrix	`/admin/permissions`	AD	Matrix of permissions by role. Change grants for editable roles.
User Role Assignment	`/admin/role-assignment`	AD	Give users their roles and home hub.
Audit Log	`/admin/audit-logs`	AD	Who changed what and when (administration and security events).
Application Settings	`/admin/settings`	AD	Company-wide settings: localisation, numbering, security, approval rules, inventory, warehouse, cold chain.
Create User	`/admin/users/create`	AD	Create a user: name, login, role, hub and (for portal users) the linked customer or driver.
Edit User	`/admin/users/:id/edit`	AD	Change a user.
User Details	`/admin/users/:id`	AD	One user: details, roles, status; reset password, unlock, activate / deactivate.
Create Role	`/admin/roles/create`	AD	Create a custom role: name, scope, landing page and permissions.
Edit Role	`/admin/roles/:id/edit`	AD	Change a role.
Role Details	`/admin/roles/:id`	AD	One role: permissions and users.

18.16 Organization

Screen	Route	Who	What you do here
Company	`/admin/organization/company`	AD	Company profile used across the app and on documents.
Branches	`/admin/organization/branches`	AD	Branches.
Business Units	`/admin/organization/business-units`	AD	Business units.
Departments	`/admin/organization/departments`	AD	Departments.
Designations	`/admin/organization/designations`	AD	Job titles.
Employee Master	`/admin/organization/employees`	AD	Employee master linked to users.
Warehouse Master	`/admin/organization/warehouses`	AD	Warehouse master.

18.17 AI & Intelligence

Screen	Route	Who	What you do here
AI Overview	`/ai`	AD, SE, SM, CS, DP, FI, WM, IE, QA	AI overview: what the assistant, forecasting and insights can do.
AI Assistant	`/ai/assistant`	AD, SE, SM, CS, DP, FI, WM, IE, QA	Ask questions about your data in plain words and get tables and charts back.
Jarvis	`/ai/jarvis`	AD, SE, SM, CS, DP, FI, WM	Voice-style assistant (“Jarvis”) for quick questions and actions with confirmation.
Forecasting	`/ai/forecasting`	AD, SM, DP, FI, WM	Forecast overview and recent forecasts.
Predictive Inventory	`/ai/predictive-inventory`	AD, SM, DP, FI, WM	Predicted stock needs, risks and reorder suggestions.
Replenishment	`/ai/replenishment`	AD, SM, DP, FI, WM, IE	Purchase requests to suppliers and their status.
AI Insights	`/ai/insights`	AD, SM, CS, DP, FI, WM, QA	Automatically found issues and opportunities. Mark them reviewed or resolved.
Model Management	`/ai/models`	AD, SM, DP, FI, WM	Forecasting models: enable or disable, see accuracy.
AI Settings	`/ai/settings`	AD	Choose the AI provider and data-sharing level.
AI Activity	`/ai/activity`	AD, SE, SM, CS, DP, FI, WM, IE, QA	Log of what the assistant did for you.
New Forecast	`/ai/forecasting/new`	AD, SM	Create a forecast: what, where, how far ahead, which model.
Forecast History	`/ai/forecasting/history`	AD, SM, DP, FI, WM	Past forecasts.
Compare Forecasts	`/ai/forecasting/compare`	AD, SM, DP, FI, WM	Compare forecasts side by side.
Forecast Report	`/ai/forecasting/:id/print`	AD, SM, FI	Printable forecast report.
Forecast	`/ai/forecasting/:id`	AD, SM, DP, FI, WM	One forecast with chart and accuracy.
Reorder Recommendations	`/ai/predictive-inventory/reorder`	AD, SM, DP, FI, WM	Suggested reorders. Add to a replenishment draft or dismiss.
Inventory Risk Analysis	`/ai/predictive-inventory/risks`	AD, SM, DP, FI, WM	Stock-out, excess, slow and non-moving risks.
Demand Forecasting	`/ai/predictive-inventory/demand`	AD, SM, DP, FI, WM	Demand forecast by product and hub.
Planning Parameters	`/ai/predictive-inventory/settings`	AD, SM, DP, FI, WM	Planning parameters (Administrator).
Prediction Detail	`/ai/predictive-inventory/lines/:id`	AD, SM, DP, FI, WM	Prediction for one product at one hub.

Screen	Route	Who	What you do here
New Replenishment Draft	`/ai/replenishment/create`	AD, DP, WM, IE	Create a purchase request: supplier, hub, products and quantities.
Replenishment Draft	`/ai/replenishment/:id`	AD, SM, DP, FI, WM, IE	One purchase request: submit, approve, reject, convert, cancel, and receive goods.

18.18 Reports

Screen	Route	Who	What you do here
Reports Center	`/reports/center`	AD, SE, SM, CS, DP, FI, WM, IE, QA	All reports you may see. Search, favourite and open.
Report	`/reports/view/:id`	AD, SE, SM, CS, DP, FI, WM, IE, QA	Run a report: set dates and filters, view table and chart, export CSV or print.

18.19 My Account

Screen	Route	Who	What you do here
My Profile	`/profile`	All	Your profile.
Notification Center	`/notifications`	All	All your notifications. Mark them read and open the linked record.
Change Password	`/account/change-password`	All	Change your password (required after an administrator reset).
Preferences	`/settings`	All	Your preferences: theme and notifications.
Sign Out	`/logout`	All	Confirm sign out.

19 · Glossary

19.1 Business terms

Term	Meaning in Bellomassi
Approval checks	Six automatic checks shown to the Sales Manager: customer active, hub active, credit within limit, stock available, order total valid, products active.
Available stock	Total − reserved − damaged quantity at a hub.
Batch-controlled product	A product whose scrap requests must name a batch number. Batch tracking on stock is not implemented.
Closed with exceptions	Label of dispatch status PARTIALLY_DELIVERED: all stops closed, at least one failed or partial.
Cold chain	Keeping frozen (−25 to −18 °C) and chilled (0 to 4 °C) goods in range. Ranges are configured; live readings are not available.
Consolidation	Customer Service's five-point check of an approved order before it is released to dispatch.
Credit limit / balance	Maximum outstanding amount allowed for a customer / the outstanding amount recorded on the customer.
Credit warning	Flag on an order whose total would take the customer over the credit limit.
Customer app	Mobile-first part of the application for customers (`/customer/*`).
Data scope	Which records a role can see (all, own, linked customer, etc.).
Delivery	One stop on a dispatch, created per order when the dispatch is confirmed.
Delivery note	Proof-of-delivery document generated when a stop is completed (BM-DN-YYYY-NNNNN).
DEMO GPS	Simulated vehicle position used for tracking; no real GPS device.
Dispatch	One vehicle run from one hub on one date with a driver and a sequence of stops.
Dispatch confirmation	The Dispatcher's loading sign-off. Stock is deducted and the driver can start.
Driver app	Mobile-first part of the application for drivers (`/driver/*`).
Exception review	The Dispatcher's decision on a failed or partial delivery (return goods, re-queue).
FEFO	First-expiry-first-out picking strategy (a setting only).
Follow-up	A future action scheduled on a call record.
GRN	Goods receipt note — a stock receipt.
Hub	Location that holds stock and runs dispatches.
Invoice basis	Bill ordered quantities or delivered quantities (from the delivery note).
Landing page	First page a role sees after sign-in.
OTP	One-time 6-digit code the customer gives the driver to confirm the handover (simulated, shown in the customer app).
POD	Proof of delivery: receiver name, optional signature and photo, customer remarks.
Quarantine	Damaged quantity held at a hub (shown on the QA dashboard).

Term	Meaning in Bellomassi
Re-queue	Putting a failed order back to READY_FOR_DISPATCH so it can be planned again.
Ready for dispatch	Order released by Customer Service to the Dispatcher.
Reorder level	Available quantity at or below which a stock line is Low Stock.
Replenishment	Purchase request to a supplier (draft → approval → goods receipt).
Return for correction	Sales Manager sends an order back to the salesman with a reason; the salesman fixes and resubmits.
Scrap / write-off	Removing unusable stock after approval; posts a Scrap Write-off movement.
Sequence	The order in which a driver must deliver the stops of a dispatch.
Simulated	Works realistically in the demo without a real external system.
Supplier portal	Read-only dashboard for a supplier showing approved replenishment.
VIP customer	Customer flag; customer-app orders get High priority and VIP reports.
VMS	Vehicle Management System — vehicle and driver document compliance.
Zone	Group of pincodes served from a hub.

19.2 Technical terms

Term	Meaning
`can(user, key)`	Permission check used by routes and services (`src/config/permissions.js`).
Data version	`APP_DATA_VERSION` 8 — the LocalStorage schema version; older data is migrated on load.
Integration gateway	Optional backend proxy (`VITE_INTEGRATION_GATEWAY_URL`) for live email, WhatsApp, OCR and AI. Not configured in the demo.
LocalStorage key	Browser storage entry; all start with `bellomassi_` (chapter 01 §1.6).
`navigation.js`	Single list of 206 route entries driving menus, routes, breadcrumbs and access.
Permission key	Name of one capability, e.g. `orders.approve` (110 keys).
`ProtectedRoute`	Router guard that sends signed-out users to sign-in and unauthorised users to `/unauthorized` .
Service	Module in `src/services` that owns the business rules for one area and returns `{ ok, record }` or `{ ok: false, error }` .
`storageService`	Only module that reads / writes browser storage; JSON safety, in-memory fallback, transactions.
Timeline	Event history of an order (`bellomassi_order_timeline`) or dispatch (`bellomassi_dispatch_timeline`).
Transaction	Several LocalStorage keys written together (`storage.transaction`) so a step is all-or-nothing.

19.3 Document numbering

Record	Format / prefix
Order	BM-ORD-...
Dispatch	BM-DSP-...

Record	Format / prefix
Delivery note	BM-DN-YYYY-NNNNN
Invoice	BM-INV-YYYY-NNNNN (prefix, year and digits configurable)
Stock receipt	GRN-…
Stock transfer	TRF-…
Stock adjustment	ADJ-…
Scrap request	SCR-…
Replenishment request	RPL-…
Call	CALL-…

19.4 Role abbreviations

SA Super Admin · AD Administrator · SE Sales Executive (SALESMAN) · SM Sales Manager · CS Customer Service · DP Dispatcher · DR Driver · FI Finance Executive · WM Warehouse Manager · IE Inventory Executive · QA QA Manager · CU Customer · SU Supplier.

19.5 Status label legend

Implemented · Partially Implemented · Simulated · UI Only · Coming Soon · Not Implemented · Not Verified — see the README.